Concentrated Partial ML estimation for Dynamic Spatial Panel Data Probit Models with Fixed Effects

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July 18, 2023

Abstract

In this paper we propose three different concentrated partial maximum likelihood estimators (CPMLE) for a new specification of a spatial dynamic panel data probit (SDPD-probit) model, which allows to deal with cross-sectional dependence, time dependence and individual (spatial) or time fixed effects in a nonlinear setting. The first ML-based estimator is a panel version of the bivariate PMLE proposed by Wang et al. (2013) and Billé and Leorato (2020); the second one is the same estimator based on univariate (rather than bivariate) probabilites. We adjust the MLE and concentrate out the fixed effects following Carro (2007) and Fernández-Val (2009). Proper marginal effects for this new model specification are also defined. We provide extensive Monte Carlo simulations for the finite sample properties of those estimators, as well as their asymptotic properties using the increasing domain definition for the spatial component and under the assumption of near-epoch dependence. Finally, the third estimator is a feasible version of PMLE which make use of the coding technique, see Besag (1974) and Arbia (2014), and a block-diagonal structure of the variance-covariance matrix which can be used to overcome computational issues raised by very large datasets.

Keywords: Spatial Dynamic Panel Data Probit Models; Nonlinear Modeling; Concentrated Partial Maximum Likelihood; Bias Reduction; Discrete Choices.

JEL codes: C13, C33, C35, C51, C55, C63.

1 Introduction

The evaluation of spatial interaction models has received a great attention in the literature in the last decades since the pionering work by Ord (1975). In particular, the estimation of discrete choice panel data models has been challenged recently by the increasing availability of very large datasets in many economic fields, , such as in health economics, environmental economics and industrial economics to name only a few. In these, and in many other situations, the application of the state-of-the-art methodologies requires increasing computer resources both in terms of computer space and in terms of the speed of calculation. As a consequence, new model specifications and new estimation techniques are needed which compromise between accuracy and computer efficiency.

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Electronic copy available at: https://ssrn.com/abstract=4514415

In this paper we propose a spatial dynamic panel data probit (SDPD-probit) model with both space and time dependence and fixed effects that act in space. Following Pakel (2019), our proposed estimation strategies for this model will be based on the partial maximum likelihood estimator (PMLE). However, while Pakel (2019) estimates dynamic panel data models with different types of cross-sectional dependence in the error terms, in our case we extend the procedure to account for spatial effects directly to the dependent variables. The inclusion of a spatial autoregressive term on the dependent (latent) variable rather than on the error term is a source of additional complexity (both computational and methodological), in that it not only affects the conditional spatial covariance structure, but also changes the marginal effects of the regressors by allowing indirect neighboring effects. Because of this difference, while in Pakel (2019)'s framework the estimation of the spatial parameter can be considered of secondary importance, this is no longer true with our specification. Recently, Zhang et al. (2022) have proposed a nonlinear dynamic spatial panel data model. However, the two models do not overlap in different ways. While we include the spatial autoregressive component Wy_t^* in the latent linear equations, they consider the same component in the linear equations as Wy_t which is observed. W in Zhang et al. (2022) is allowed to be time-dependent and they also include a space-time lagged dependent variable, i.e. Wy_{t-1}. However, the model in Zhang et al. (2022) is not attributable to a spatial dynamic probit specification. Indeed, Zhang et al. (2022) define $y_{it} = F(\rho W_i y_t + \alpha y_{i,t-1} + \gamma W_i y_{t-1} + \beta' X_{it} + \mu_i + \text{error}_{it})$ for a continuous (and invertible) F, which clearly excludes $F(u) = \mathbf{1}\{u \geq 0\}$.

We propose three different estimation strategies to tackle the computational problems while increasing efficiency as much as possible. The first one is a panel version of the Partial Maximum Likelihood strategy proposed by Wang et al. (2013) and recently improved by Billé and Leorato (2020). We refer to this method as to the bivariate Concentrated Partial ML estimation (or bivariate CPMLE). The second strategy is a univariate version of the first one which we will refer to as the univariate CPMLE. Finally, the third strategy is based on the so-called coding technique, a method introduced in spatial statistics by Besag (1974) and extended by Arbia (2014) to the case of spatial linear regression models. We call this estimation method the bivariate Feasible Concentrated Partial ML (bivariate FCPMLE). The modification of the above mentioned estimators in considering time fixed effects rather than individual fixed effects is also quite straightforward.

In summary, the contributions of this paper to the literature on spatial models for binary dependent variables are the following:

(i) we propose an extension to panel data of the PMLE proposed in Billé and Leorato (2020), both in the bivariate and univariate versions;

(ii) we provide detailed analysis and sufficient conditions for the near-epoch dependence (NED) property of the data generating process and we study the asymptotic properties of the two preceding estimators;

(iii) we propose a new definition of long- and short-run direct and indirect effects;

(iv) we define an alternative version of the bias correction approach introduced in Pakel (2019),which is less computationally demanding;

(v) we present an extensive Monte Carlo analysis aimed at assessing the small sample performances of the CPMLEs; and

(vi) as a way to tackle the complexity of very large datasets, following the ideas of Besag (1974) and Arbia (2014), we also consider a computationally less demanding estimation strategy based on the use of a coding technique.

The remainder of the paper is organized as follows. Section 2 is devoted to the specification of the spatial dynamic panel data probit model which we will consider throughout the paper. Here we discuss the hypotheses underlying the model and we describe the associated marginal effects both in the short-term and in the long-term dynamics. We also introduce the idea of the coding technique which will be then used to develop the bivariate FCPMLE. Section 3 describes the three ML estimators and their asymptotic properties while Section 4 studies their finite sample performances via Monte Carlo simulations. Finally, Section 5 summarises the results and reports some concluding remarks.

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2 Model Specification

In this section we specify a spatial dynamic panel data probit (SDPD-probit) model through the assumption that there exists an underlining (latent) Gaussian multivariate spatio-temporal process that generates the binary observed data over time.

Specifically, let $\boldsymbol{y}_{Nt}$ be an N-dimensional stochastic vector of spatial binary variables located on a possibly unevenly spaced lattice $Z \subseteq \mathbb{R}^N$ at time t. The SDPD-probit model is then defined as

$$\begin{aligned} \boldsymbol{y}^*_{Nt} &= \alpha \boldsymbol{y}_{Nt-1} + \rho \boldsymbol{W}_N \boldsymbol{y}^*_{Nt} + \boldsymbol{X}_{Nt}\boldsymbol{\beta} + \boldsymbol{\mu}_N + \boldsymbol{\varepsilon}_{Nt}, \quad \boldsymbol{\varepsilon}_{Nt} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}) \\ \boldsymbol{y}_{Nt} &= \boldsymbol{1}_N (\boldsymbol{y}^*_{Nt} > \boldsymbol{0}), \quad \forall t \end{aligned} \tag{1}$$

where $\boldsymbol{y}^*_{Nt}$ is the stochastic vector of latent continuous variables at time t, $\boldsymbol{W}_N$ is the N-dimensional spatial weighting matrix of known constants and which we assume to be time-invariant, $\boldsymbol{X}_{Nt}$ is the N by k matrix of exogenous regressors at time t, $\boldsymbol{\mu}_N$ is a N-dimensional column vector of individual (spatial) fixed effects, ε_{Nt} is the N-dimensional vector of i.i.d. standard Normal innovations at time t and $\boldsymbol{1}_N (\boldsymbol{y}^*_{Nt} > \boldsymbol{0}) = (\boldsymbol{1}(y^*_{1t} > 0), \ldots, \boldsymbol{1}(y^*_{Nt} > 0))' \ \forall t$.

The dynamic nonlinear panel data model in equation (1) is both a spatial extension of the dynamic nonlinear panel data model studied, among others, by Carro (2007), and a dynamic panel data extension of the spatial nonlinear model studied by Billé and Leorato (2020). In particular, $\boldsymbol{y}^*_{Nt}$ reflects the unobserved utility functions in doing an action by an economic agent, see Manski et al. (1981) and McFadden (2001). The model assumes that the individual choice at time $t-1$, i.e. $\boldsymbol{y}_{Nt-1}$, directly affects the utility function of the same individual at time t, i.e. $\boldsymbol{y}^*_{Nt}$. Indeed, it is resonable to assume that an individual choice have an impact on his/her own utility function one period later. Furthermore, the model assumes spillover effects among the individuals' utility functions. This phenomena is of particular interest especially when there are for instance effects of information sharing among the economic agents. In doing so, the individuals' utility functions are affected each other through the use of the spatial weighting matrix $\boldsymbol{W}_N$.

Throughout this paper, we assume to deal with balanced panel data, an assumption particularly convenient in the case of subsampling (subsection 2.2) and likely to hold for many geographical data. In the following, we list all the necessary assumptions for stability and identifiability of the above model specification.

Lemma 1 *Let $\overline{\tau}$ denotes the spectral radius of the square N-dimensional $\boldsymbol{W}_N$ matrix, i.e.:*
$\overline{\tau}_{\boldsymbol{W}_N} = max\{|\omega_1|, ..., |\omega_N|\}$, where $\omega_1, ..., \omega_N$ are the eigenvalues of $\boldsymbol{W}_N$, respectively. Then, $\boldsymbol{A}_\rho := (\boldsymbol{I} - \rho \boldsymbol{W}_N)$ is nonsingular for all values of ρ in the interval $(-1/\overline{\tau}, 1/\overline{\tau})$.

Assumption 1 *(a) The spatial weights matrix $\boldsymbol{W}_N$ is a row-stochastic time-invariant square matrix with zero diagonal elements. (b) $\rho \in (-1/\overline{\tau}, 1/\overline{\tau})$.*

Assumption 2 *Matrices $\boldsymbol{W}_N$ and $\boldsymbol{A}_\rho^{-1}$ are uniformly bounded in both row and column sum norms.*

Assumption 3 *The disturbances ε_{it} are Gaussian and i.i.d. across i and t with zero mean and finite unitary variance, and their higher than fourth moments exist.*

Assumption 4 *The elements of $\boldsymbol{X}_{Nt}$ are uniformly bounded constants in N and t, $\boldsymbol{X}_{Nt}$ has a full column rank, and $lim_{T\to\infty} \sum_t (\boldsymbol{X}'_{Nt}\boldsymbol{X}_{Nt})/NT$ exists and it is nonsingular.*

Under the above-mentioned regularity conditions, the model in equation (1) can be written in reduced form as

$$\begin{aligned} \boldsymbol{y}^*_{Nt} &= \boldsymbol{A}_\rho^{-1}\alpha \boldsymbol{y}_{Nt-1} + \boldsymbol{A}_\rho^{-1}\boldsymbol{X}_{Nt}\boldsymbol{\beta} + \boldsymbol{A}_\rho^{-1}\boldsymbol{\mu}_N + \boldsymbol{A}_\rho^{-1}\boldsymbol{\varepsilon}_{Nt}, \\ \boldsymbol{y}_{Nt} &= \boldsymbol{1}_N (\boldsymbol{y}^*_{Nt} > \boldsymbol{0}), \quad \forall t \end{aligned} \tag{2}$$

where $\boldsymbol{\nu}_{Nt} = \boldsymbol{A}_\rho^{-1}\boldsymbol{\varepsilon}_{Nt}$, the covariance matrix is $\boldsymbol{\Sigma}_\nu = \mathbb{E}\left[\left(\boldsymbol{A}_\rho^{-1}\varepsilon_{Nt}\right)\left(\boldsymbol{A}_\rho^{-1}\varepsilon_{Nt}\right)'\right] = \boldsymbol{A}_\rho^{-1}(\boldsymbol{A}_\rho{}^{-1})'$, and again $\boldsymbol{\varepsilon}_{Nt} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. Note that the covariance matrix is time-invariant.

In this paper we estimate the set of parameters of interest $\boldsymbol{\theta} = (\alpha, \rho, \boldsymbol{\beta}')'$ of the model in equation (1) by making use of the definition of a partial MLE (PMLE) in section 3 with large datasets.

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2.1 Marginal Effects

In this section we intend to briefly describe the marginal effects derived from the model reported in equation (1).

Starting from its reduced form in equation (2), we obtain the expected value of y_i, for all $i = 1, \dots, N$, as follows

$$\mathbb{E}\left(y_{i,t} \mid \boldsymbol{y}_{N,t-1}, X_{N,t}\right) = \mathrm{P}\left(y_{i,t} = 1 \mid \boldsymbol{y}_{N,t-1}, X_{N,t}\right) = \Phi\left(\{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\{\boldsymbol{A}_\rho^{-1}\left(\alpha \boldsymbol{y}_{N,t-1} + \boldsymbol{X}_{N,t}\boldsymbol{\beta} + \mu_N\right)\}_i\right) \tag{3}$$

where $\boldsymbol{\Sigma}_\nu$ is the covariance matrix implied by the reduced form. In the above equation (3), and throughout all the paper, we also adopted the following notation: for any matrix A, $\{A\}_{ij}$ is the (i,j)th component, and $\{A\}_{i\cdot}$ indicates the ith row vector; if instead $\boldsymbol{z}$ is a vector, $\{\boldsymbol{z}\}_i$ denotes the i-th component.

Billé and Leorato (2020), among few others, proposed specific definitions of the marginal effects for static spatial (first-order) autoregressive models with (first-order) autoregressive disturbances. In the nonlinear panel data literature, most of the attention is focused on the estimation of the average marginal effects (AME) defined as the expected effect of a change in one of the regressors on the conditional mean of the dependent variable. The effect depends on the fixed effects and the common approach is to average them out, see for example Chernozhukov et al. (2013), Davezies et al. (2022), Dobronyi et al. (2021), Aguirregabiria and Carro (2021). Most of the cited works focus on discrete regressors (Chernozhukov et al., 2013) or more specifically on the effect of past choices (f.i. a change on the binary lagged dependent variable $y_{i,t-1}$) on the conditional probability of $y_{i,t}$, as in Dobronyi et al. (2021) and Aguirregabiria and Carro (2021). Aguirregabiria and Carro (2021) explicitly derive the $n-$periods forward AME.

Our approach here is different in many respects: (i) as a natural extension of the effects in Billé and Leorato (2020), we are interested in the impact of marginal changes of continuous regressors on the conditional probability of $y_{i,t} = 1$. This definition is somewhat nearer to equation (1) in Davezies et al. (2022); (ii) differently from Davezies et al. (2022), however, the fixed effects are not averaged out, since our definition is unit specific and thus conditional to both the xs and the fixed effects.

One peculiarity of spatial models is in fact the existence of regional spillovers, such that a change of the regressor k in unit/region j can have an effect on the dependent variable of unit $i \neq j$. It is therefore natural to think of a definition of marginal effects of a regressor that is time and unit ($i = j$ or $i \neq j$) specific. Indeed, in the spatial literature, the main interest is being able to disentangle the direct effect, namely the effect of a variable in one location on the response on the same location, and the indirect effects (of a variable in one location on the response in neighboring locations). These effects can further be transitionary (or contemporaneous) and permanent, as defined by Debarsy et al. (2012) for the linear models, depending on whether the marginal change of the kth regressor is contemporaneous or occurred in the past.

We will refer to *contemporaneous effects* at time t, when the derivative of the conditional mean of y_{it} is computed with respect to the regressor $x_{jh,t}$. In particular, if $i = j$ we have the direct contemporaneous effects, while for $i \neq j$ we obtain the indirect effects. The definition of these effects is a straighforward extension of the direct and indirect effects defined in Billé and Leorato (2020) for a (static) spatial probit model.

Let us denote, for convenience, $\boldsymbol{A}_\rho^{-1}\left(\alpha \boldsymbol{y}_{N,t-1} + \boldsymbol{X}_{N,t}\boldsymbol{\beta} + \boldsymbol{\mu}_N\right) = \boldsymbol{\zeta}_t = (\zeta_{1t}, \dots, \zeta_{Nt})'$. By considering the expected value in (3), we propose the following specification of the contemporaneous average marginal effects:

Definition 1 *(Average contemporaneous effect) The average contemporaneous effect of the regressor h for unit j to the $i-$th dependent variable, computed at the point $\boldsymbol{\zeta}_{i0} = (\zeta_{i10}, \dots, \zeta_{iT0})'$ is*

$$\begin{aligned} ACE_{ij;h}(\boldsymbol{\zeta}_{i0}) &:= \frac{1}{T}\sum_{t=0}^{T} \left.\frac{\partial \mathbb{E}\left(y_{i,t} \mid \boldsymbol{X}_{N,t}, y_{i,t-1}\right)}{\partial X'_{jh,t}}\right|_{\zeta_{it}=\zeta_{it0}} \\ &= \frac{1}{T}\sum_{t=0}^{T} \phi\left(\{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\zeta_{it0}\right)\{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\{\boldsymbol{A}_\rho^{-1}\}_{ij}\beta_h \end{aligned} \tag{4}$$

Given the dynamic nature of the model, we also want to define the effect that a change in the $h-$th regressor causes r times ahead and the long-run effect. This type of non

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contemporaneous effect yields important complications that prevented its use, even in the easier case of a dynamic probit model for non-spatial panel data.

In order to define the effect of $X_{jh,t}$ r periods ahead, we need the derivatives

$$\frac{\partial \mathbb{E}\left(y_{i,t+r} \mid \boldsymbol{X}_{N,t+r}, \boldsymbol{X}_{N,t+r-1}, \ldots\right)}{\partial X_{jth}}.$$

For the rest of this session we are going to omit the dimension N in some of the indices, by simply writing $\boldsymbol{y}_t = \boldsymbol{y}_{Nt}$ and $\boldsymbol{X}_t = \boldsymbol{X}_{Nt}$ to reduce the notational complexity. The dynamic part of the model implies that a change in the variable X_h at a time t is able to affect the dependent variable at future times. However, the fact that the dynamic component enters into the linear latent regression through the non-differentiable transformation $y_{iy} = \mathbf{1}\{y_{it^*\geq 0}\}$ of y^*_{it}, prevents us to use the arguments in Debarsy et al. (2012) for the definition of the r period ahead effects. In order to circumvent this problem, we propose a reasonable substitute based on an approximate latent process.

The idea is the following: in the latent reduced form equation for any y^*_{is}, we replace to the binary variable $y_{i,s-1} = 1\left\{y^*_{i,s-1} \geq 0\right\}$ its conditional mean, thus obtaining

$$y^*_{is} = \{A_\rho^{-1}\}_{i\cdot}\left(\alpha \boldsymbol{y}_{s-1} + \boldsymbol{X}_s\boldsymbol{\beta} + \boldsymbol{\mu}\right) + \nu_{is} \approx \{A_\rho^{-1}\}_{i\cdot}\left(\alpha \boldsymbol{\Phi}_{s-1} + \boldsymbol{X}_s\boldsymbol{\beta} + \boldsymbol{\mu}\right) + \nu_{is}$$

where $\boldsymbol{\Phi}_{s-1}$ is the $N-$dimensional vector with $j-$th component

$$\Phi_{j,s-1} = \Phi\left(\{\boldsymbol{\Sigma}_\nu\}_{jj}^{-1/2}\{\boldsymbol{A}_\rho^{-1}\}_{j\cdot}\left(\alpha \boldsymbol{y}_{s-2} + \boldsymbol{X}_{s-1}\boldsymbol{\beta} + \boldsymbol{\mu}\right)\right).$$

Since each component of $\boldsymbol{\Phi}_{s-1}$ depends on $y^*_{N,s-2}$ through $y_{N,s-2}$, we replace again the binary variable by its conditional mean, and continue like this recursively. Following this procedure, we get for each component of $\boldsymbol{\Phi}_{s-1}$:

$$\Phi_{j,s-1} \approx \Phi\left(\{\boldsymbol{\Sigma}_\nu\}_{jj}^{-1/2}\{\boldsymbol{A}_\rho^{-1}\}_{j\cdot}\left(\alpha \boldsymbol{\Phi}_{s-2} + \boldsymbol{X}_{s-1}\boldsymbol{\beta} + \boldsymbol{\mu}\right)\right),$$

where

$$\Phi_{j,s-2} \approx \Phi\left(\{\boldsymbol{\Sigma}_\nu\}_{jj}^{-1/2}\{\boldsymbol{A}_\rho^{-1}\}_{j\cdot}\left(\alpha \boldsymbol{\Phi}_{s-3} + \boldsymbol{X}_{s-2}\boldsymbol{\beta} + \boldsymbol{\mu}\right)\right),$$

and so on.

Therefore, we get the approximate expression:

$$\Pr\left(y^*_{i,s+r} \geq 0 \mid \boldsymbol{X}_{s+r}, \boldsymbol{X}_{s+r-1}, \ldots\right) \approx \Phi\left(\{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\{A_\rho^{-1}\}_{i\cdot}\left(\alpha \boldsymbol{\Phi}_{s+r-1} + \boldsymbol{X}_{s+r}\boldsymbol{\beta} + \boldsymbol{\mu}\right)\right)$$

We can now compute the effect $X_{js,h}$, r periods ahead by taking the derivatives:

$$\frac{\partial \mathbb{E}\left(y_{i,s+r} \mid \boldsymbol{X}_{s+r}, \boldsymbol{X}_{s+r-1}, \ldots\right)}{\partial X_{js,h}} \approx \frac{\partial \Phi\left(\{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\{A_\rho^{-1}\}_{i\cdot}\left(\alpha \boldsymbol{\Phi}_{s+r-1} + \boldsymbol{X}_{s+r}\boldsymbol{\beta} + \boldsymbol{\mu}\right)\right)}{\partial X_{js,h}}$$

where the right-hand-side is a recursive composition function of lagged $\boldsymbol{\Phi}_{s-k}$, $k = 1, \ldots, r-1$. In order to simplify the expression, let us denote

$$c_{ij} = \{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\{\boldsymbol{A}_\rho^{-1}\}_{ij} \quad \text{and} \quad \phi_{i,s} := \phi\left(\{\boldsymbol{\Sigma}_\nu\}_{ii}^{-1/2}\{A_\rho^{-1}\}_{i\cdot}\left(\alpha \boldsymbol{\Phi}_{s-1} + \boldsymbol{X}_s\boldsymbol{\beta} + \boldsymbol{\mu}\right)\right).$$

By noting that the vector $\boldsymbol{\Phi}_{s-1}$, that appears as argument of $\phi_{i,s}$, depends on $\boldsymbol{X}_{s-1}$, we can easily see that, if $r = 1$,

$$\frac{\partial \mathbb{E}\left(y_{i,s} \mid \boldsymbol{X}_s, \boldsymbol{X}_{s-1}, \ldots\right)}{\partial X_{j(s-1),h}} \approx \phi_{i,s}\,\alpha \sum_{j_1} c_{ij_1}\phi_{j_1,s-1}c_{j_1,j}\beta_h$$

Similarly, for $r \geq 2$,

$$\frac{\partial \mathbb{E}\left(y_{i,s} \mid \boldsymbol{X}_s, \boldsymbol{X}_{s-1}, \ldots\right)}{\partial X_{j(s-2),h}} \approx \phi_{i,s}\,\alpha \sum_{j_1} c_{ij_1}\phi_{j_1,s-1}\alpha \sum_{j_2} c_{j_1,j_2}\phi_{j_2,s-1}c_{j_2,j}\beta_h$$

$$\frac{\partial \mathbb{E}\left(y_{i,s} \mid \boldsymbol{X}_s, \boldsymbol{X}_{s-1}, \ldots\right)}{\partial X_{j(s-3)h}} \approx \phi_{i,s}\,\alpha \sum_{j_1} c_{ij_1}\phi_{j_1,s-1}\alpha \sum_{j_2} c_{j_1,j_2}\phi_{j_2,s-1}\alpha \sum_{j_3} c_{j_2,j_3}\phi_{j_3,s-2}c_{j_3,j}\beta_h$$

and so on.

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Further, we define the matrices $\boldsymbol{C}^{(s)}$, with $(i,j)-$th components $C_{ij}^{(s)} = \phi_{i,s}c_{ij}$. Then, for an arbitrary $r \geq 0$,

$$\begin{aligned}\frac{\partial \mathbb{E}\left(y_{i,s} \mid \boldsymbol{X}_s, \boldsymbol{X}_{s-1}, \ldots\right)}{\partial X_{j(s-r),h}} &\approx \beta_h \phi_{i,s}\alpha \sum_{j_1} c_{ij_1}\phi_{j_1,s-1}\alpha \sum_{j_2} c_{j_1,j_2}\phi_{j_2,s-1}\cdots \sum_{j_{r-1}} c_{j_{r-2},j_{r-1}}\phi_{j_{r-1},s-r+1}c_{j_{r-1},j} \\ &= \beta_h \alpha \sum_{j_1} C_{i,j_1}^{(s)}\alpha \sum_{j_2} C_{j_1,j_2}^{(s-1)}\cdots\alpha \sum_{j_{r-1}} C_{j_{r-1},j}^{(s-r+1)} \\ &= \beta_h \alpha^r \{\boldsymbol{C}^{(s:s-r+1)}\}_{ij}\end{aligned}$$

where $\boldsymbol{C}^{(s:s-k)} = \prod_{h=0}^{k}\boldsymbol{C}^{(s-h)}$ and $\{\boldsymbol{C}^{(s:s-k)}\}_{ij}$ is its $(i,j)-$th component.

It must be pointed out that the matrices $\boldsymbol{C}^s$ only change in time because of the elements $\phi_{js} = \phi(\{\boldsymbol{\Sigma}_\nu\}_{jj}\bar{\zeta}_{js})$, where $\bar{\zeta}_{js} = \{A_\rho^{-1}\}_{j\cdot}\left(\alpha\boldsymbol{\Phi}_{s-1} + \boldsymbol{X}_s\boldsymbol{\beta} + \boldsymbol{\mu}_j\right)$. Therefore, if the $N-$dimensional vectors $\bar{\boldsymbol{\zeta}}_s = \bar{\boldsymbol{\zeta}}$, $s \geq 1$ are constant through time[1] the above expression simplifies, because $\boldsymbol{C}^{(s:s-k+1)} = \boldsymbol{C}^k$ for all s, k where $\boldsymbol{C}$ is the matrix

$$\{\boldsymbol{C}\}_{ij} = \{\boldsymbol{\Sigma}_\nu\}_{ii}\{\boldsymbol{A}_\rho^{-1}\}_{ij}\phi(\{\boldsymbol{\Sigma}_\nu\}_{ii}\bar{\zeta}_i).$$

By summing up, with respect to r, to the infinite future, we obtain the desired definition of the cumulative effect of $X_{js,h}$.

Definition 2 *(Long-run marginal effects). The cumulative effect of $X_{js,h}$ on $\boldsymbol{y}_i$ at the points $\bar{\boldsymbol{\zeta}}_s, \bar{\boldsymbol{\zeta}}_{s+1}, \ldots$ is given by the $(i,j)-$component of the matrix:*

$$LRE_h(\bar{\boldsymbol{\zeta}}_s, \bar{\boldsymbol{\zeta}}_{s+1}, \ldots) = \frac{\partial \mathbb{E}\left(\boldsymbol{y}_i \mid \boldsymbol{X}\right)}{\partial X_{js,h}} = \sum_{r=0}^{\infty}\beta_h\alpha^r\boldsymbol{C}^{(s+r-1:s)}. \tag{5}$$

If in particular $\bar{\boldsymbol{\zeta}}_s = \bar{\boldsymbol{\zeta}}_{s-1} = \ldots = \bar{\boldsymbol{\zeta}}$, for all $s \geq 1$, and if $|\alpha| \leq \|\boldsymbol{C}\|$,

$$LRE_h(\bar{\boldsymbol{\zeta}}) = \beta_h\sum_{k=0}^{\infty}\alpha^k\boldsymbol{C}^k = (I - \alpha\boldsymbol{C})^{-1}\beta_h. \tag{6}$$

The last identity in (6) requires the following constraint for α: $\alpha\|\boldsymbol{C}\| < 1$. If we limit the sum in (6) a finite number of terms, we have the $T-$period-ahead (cumulative) impact. We note that, in the case of a linear model, equation (6) applies with the simplification $\{\boldsymbol{\Sigma}_\nu\}_{ii}\phi_i = 1$, that corresponds to $\boldsymbol{C} = \boldsymbol{A}_\rho^{-1}$. In this case, the finite-sum version of (6) is consistent with equation (23) in Debarsy et al. (2012).

It is also worth noting that, if $\rho = 0$ (no spatial interaction in included in the model), equations (5) and (6) can be used to define the long-run coefficient for a *standard* dynamic panel probit models. In this particular case, $c_{ij} = \mathbf{1}\{i = j\}$ while $\phi_{i,s} = \phi\left(\alpha\Phi_{i,s-1} + X_{is}^\top\boldsymbol{\beta} + \mu_i\right)$, thus $\boldsymbol{C}^{(s)} = \mathrm{diag}\{\phi_{1,s}, \ldots, \phi_{N,s}\}$ implies that equation (5) simplifies to a diagonal matrix with $i-$th diagonal term equal to:

$$\{LRE_h(\bar{\boldsymbol{\zeta}}_s, \bar{\boldsymbol{\zeta}}_{s+1}, \ldots)\}_{ii} = \sum_{r=0}^{\infty}\beta_h\alpha^r\prod_{s=1}^{r-1}\phi_{i,s}.$$

This in particular simplifies to

$$\{LRE_h(\boldsymbol{\zeta}_1)\}_{ii} = \beta_h\sum_{r=0}^{\infty}\alpha^r\phi\left(\alpha\Phi_{i0} + X_{i1}^\top\boldsymbol{\beta} + \mu_i\right)^r = \beta_h/\left(1 - \alpha\phi\left(\alpha\Phi_{i0} + X_{i1}^\top\boldsymbol{\beta} + \mu_i\right)\right)$$

if the $z-$values $\bar{\boldsymbol{\zeta}}_s = \bar{\boldsymbol{\zeta}}_1 = \alpha\boldsymbol{\Phi}_0 + \boldsymbol{X}_1^\top\boldsymbol{\beta} + \boldsymbol{\mu}$ are time invariant.

2.2 A Coding Technique

In order to tackle the computational burden involved in estimating the model in equation (1) on very large data sets, we propose to extend the use of the coding technique introduced by Besag (1974) and Arbia (2014). The technique consists in drawing a subsample of pairs of observations from the original dataset that can be considered independent such that the

[1]For $s = 1$, $\bar{\boldsymbol{\zeta}}_{j1} = \{A_\rho^{-1}\}_{j\cdot}\left(\alpha\boldsymbol{\Phi}_0 + \boldsymbol{X}_1\boldsymbol{\beta} + \boldsymbol{\mu}_j\right)$, where $\boldsymbol{\Phi}_0$ can be either the (unconditional) mean of the initial state of the dependent variable $\boldsymbol{y}_0$, or $\boldsymbol{\Phi}_0 = \boldsymbol{y}_0$.

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joint probability density function can be easily obtained as the product of the bivariate densities. While many plausible criteria could be used in the process of selecting the pairs[2], an obvious choice is the likely presence of spatial dependence between closest units. In this section we briefly describe the systematic sampling (SYS) design, see Madow and Madow (1944) and Madow (1949), as a way to select pairs of observations when no a priori information on the spatial dependence structure is available. Moreover, w.l.o.g. we focus the attention on sampling pairs of spatial observations located on a regular lattice grid.

Suppose we have N observations in space, which constitute the population size. The SYS design starts with the definition of the *sampling interval* as $k = N/q$, where q is the final number of pairs and, therefore, $n = 2q$ is the final sample size. With a simple random sampling scheme, a spatial unit r is drawn with equal probability within the sampling interval, i.e. w.l.o.g. from 1 to k, and we define this unit to be the "first" element of the "first" pair of observations. Starting from the randomly selected unit r, each "first" element of all the other pairs are sequentially selected through the fixed sampling interval k, while each "second" element of the pairs is selected as the nearest neighbor unit of the first elements. Specifically, the first set of units $u_{q_1,r}$ has the following elements

$$u_{q_1,r} = \{i : i = r + (h-1)\,k \leq N; h = 1, \ldots, n\} \tag{7}$$

while the second set of units $s_{q_2,r}$ are such that

$$u_{q_2,r} = \{j : d_{ij} = \min d_{ih}; j, h = 1, \ldots, n; j, h \neq i\} \tag{8}$$

and finally the sample of pairs is simply $u_{q,r} = u_{q_1,r} \cup u_{q_2,r}$. The result is shown in Figure 1.

3 Maximum Likelihood Estimation

In this section we consider two independent problems. In subsection 3.1, we propose a maximum likelihood estimation method for the dynamic spatial panel probit model, based on concentrating out the fixed effects and partial information. Section 3.2 then discusses the asymptotic properties under the NED assumption. In section 3.3, we apply a subsampling procedure to select cross-sectional pairs of units that are treated as pairwise independent.

3.1 Concentrated Partial MLE

It is well known in the literature that writing down the exact log-likelihood function of a spatial model, even with only cross–sectional data, requires the computation of a N-dimensional integral. To avoid doing so, one may proceed in replacing the log–likelihood with a partial log–likelihood, in which the log-likelihood is approximated by the sum of the contributions of groups of spatial units (here we limit ourselves to the cases of groups of size 1 and 2), see Wang et al. (2013). Therefore, in a partial log-likelihood approach, groups are treated as if they were independent.

Given the model in equation (1), the partial log-likelihood function can be written as

$$\ell\left(\boldsymbol{\theta}, \boldsymbol{\mu}_N\right) = \sum_{i=1}^{G} \ell_i\left(\boldsymbol{\theta}, \mu_i\right) = \sum_{i=1}^{G}\sum_{t=1}^{T} \log p_{it}\left(\boldsymbol{y}_{it}\right) \tag{9}$$

where $G = N/2$ can be the total number of pairs (in the bivariate case) or the total number of observations, $G = N$ (in the univariate case). Now the point is to differently define the conditional probabilities $p_{it}(\boldsymbol{d})$, according to the bivariate and the univariate cases, respectively. Defining $\boldsymbol{X}_{\rho,t} = \boldsymbol{A}_\rho^{-1}\boldsymbol{X}_{Nt}$ and $\boldsymbol{y}_{\rho,t-1} = \boldsymbol{A}_\rho^{-1}\boldsymbol{y}_{N,t-1}$, the bivariate probability for pair $i = (i_1, i_2)$ is given by

$$\begin{aligned} p_{it}\left(\boldsymbol{d}\right) &= \Pr\left(y_{i_1 t} = d_1, y_{i_2 t} = d_2 \mid \boldsymbol{X}_{Nt}, y_{i_1,t-1}, y_{i_2,t-1}\right) \\ &= \int_{\{\jmath_1 u > -\jmath_1 \zeta_{i_1}\}} \frac{1}{\sigma_{i_1}}\phi\left(\frac{u}{\sigma_{i_1}}\right)\Phi\left(\jmath_2 \frac{\zeta_{i_2} + u\frac{\sigma_{i_1,i_2}}{\sigma_{i_1}^2}}{(\sigma_{i_2}^2 - \sigma_{i_1,i_2}^2/\sigma_{i_1}^2)^{1/2}}\right) du \end{aligned} \tag{10}$$

[2]For instance Billé and Leorato (2020) solved the problem by using a Kullback–Liebler divergence approach.

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where $\jmath_j = 2\,(d_j - 0.5)$, $d_j = 0, 1$, $j = 1, 2$, σ_{ij} is the (i, j) −th component of $\boldsymbol{\Sigma}_\nu$, $\sigma_i^2 = \sigma_{ii}$, and $\boldsymbol{\zeta} = \boldsymbol{X}_{\rho,t}\boldsymbol{\beta} + \alpha \boldsymbol{y}_{\rho,t-1} + \boldsymbol{A}_\rho^{-1}\boldsymbol{\mu}$, while the univariate case by

$$p_{it}(d) = \Pr\left(y_{it} = d \mid \boldsymbol{X}_{Nt}, y_{i,t-1}\right) = \Phi\left(2\,(d - 0.5)\,\zeta_i/\sigma_i\right) \tag{11}$$

where $d = 0, 1$.

In optimizing the log-likelihood function in equation (9), we concentrated out the individual and time fixed effects. We called this estimator *Concentrated Partial Maximum Likelihood Estimator* (CPMLE). The procedure is based on repeating the following three steps till the condition $\max_i \left(\left|\tilde{\theta}_{i,l-1} - \tilde{\theta}_{i,l}\right|\right) > \varpi$ hold, where ϖ is a fixed threshold, and the number of iterations l are limited by fixed number L:

1. given an initial estimate $\tilde{\boldsymbol{\theta}}$, minimize $\ell\left(\tilde{\boldsymbol{\theta}}, \boldsymbol{\mu}_N\left(\boldsymbol{\theta}\right)\right)$ with respect to $\boldsymbol{\mu}_N$ and get $\hat{\mu}_N(\tilde{\boldsymbol{\theta}})$;
2. minimize $\ell\left(\boldsymbol{\theta}, \hat{\boldsymbol{\mu}}_N\left(\boldsymbol{\theta}\right)\right)$ with respect to $\boldsymbol{\theta}$ and get $\hat{\boldsymbol{\theta}}$
3. set $\hat{\boldsymbol{\theta}} = \tilde{\boldsymbol{\theta}}$ and repeat the procedure until $\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}\|_\infty < \varpi$ or until the maximum number of iterations.

An incidental parameter problem may arise within panel data models with fixed effects, depending on both the cross-sectional N and the time T dimension. In discrete choice models this problem would produce inconsistent estimates. Intuitively, when $T >> N$, the ratio N/T should converge to zero as both the dimensions increase, with different rates of convergence, i.e. the rate of divergence is faster for the time dimension than the one for the cross-sectional dimension, to avoid the incidental parameter of the individual (spatial) fixed effects.

Let the functions $s_{\theta,i}(\boldsymbol{\theta}, \mu_i)$ and $s_{\mu,i}(\boldsymbol{\theta}, \mu_i)$ denote the partial derivatives of $\ell_i(\boldsymbol{\theta}, \mu_i)$ with respect to $\boldsymbol{\theta}$ and μ_i respectively.

Computation of the score vector. Given $\boldsymbol{\theta}$, we need to compute the values μ_i that solve $s_{\mu,i}(\boldsymbol{\theta}, \hat{\mu}_i(\gamma, \rho)) = 0$. There is no close form solution to this equations, the solutions need to be found computationally. We focus here on the case of regional specific effects only. The i−th score is a single equation if $G = N$, while the FOC in μ_i is a system of two equations when $G = N/2$, because for each i, we have the two fixed effects μ_{i_1}, μ_{i_2}. In the first case, it is easy to compute

$$s_{\mu,i}(\boldsymbol{\theta}, \mu_i) = \sum_t \frac{1}{p_{it}(\boldsymbol{y}_{it})} \frac{\partial p_{it}(\boldsymbol{y}_{it})}{\partial \mu_i} = \sum_t \frac{1}{p_{it}(\boldsymbol{y}_{it})} \frac{1}{\sigma_i} \phi\left(\frac{\jmath_i \zeta_i}{\sigma_i}\right), \tag{12}$$

while in the second case, if we denote by $f_i(u, v)$ the density of a bivariate Gaussian distribution with mean zero and covariance matrix Σ_i equal to the 2-dimensional submatrix with rows and columns (i_1, i_2) from Σ_ν, we get

$$\begin{aligned} s_{\mu,i}(\gamma, \rho, \mu_i) &= \begin{pmatrix} \frac{\partial \ell_i}{\partial \mu_{i_1}} \\ \frac{\partial \ell_i}{\partial \mu_{i_2}} \end{pmatrix} = \sum_t \frac{1}{p_{it}(\boldsymbol{y}_{it})} \begin{pmatrix} \frac{\partial p_{it}(\boldsymbol{y}_{it})}{\partial \mu_{i_1}} \\ \frac{\partial p_{it}(\boldsymbol{y}_{it})}{\partial \mu_{i_2}} \end{pmatrix} \\ &= \sum_t \frac{1}{p_{it}(\boldsymbol{y}_{it})} \begin{pmatrix} \int_{S_{i_2}} \jmath_{i_1} f_i(\zeta_{i_1}, u) du \\ \\ \int_{S_{i_1}} \jmath_{i_2} f_i(u, \zeta_{i_2}) du \end{pmatrix} \end{aligned} \tag{13}$$

where the sets $S_j = \{\jmath_u > -\jmath_j \zeta_j\}$. Note that the score components for $\boldsymbol{\theta}' = (\alpha, \boldsymbol{\beta}', \rho)$ are the same as those computed in Billé and Leorato (2020) for $\boldsymbol{\beta}$ and ρ.

3.2 Asymptotic distribution under near epoch dependence

3.2.1 Near epoch dependence

In this section, we introduce the assumptions on the spatio-temporal dependence structure used to derive the asymptotic distribution of $\hat{\boldsymbol{\theta}}$. These same assumptions are sufficient for the derivation of the order of magnitude of the components of the remainder terms in the asymptotic expansions presented in section 3.2.2. Through the computation of the asymptotic bias, we are also able to propose a bias corrected estimator, that we denote by $\tilde{\boldsymbol{\theta}}$ (see equation 23).

We assume that the data generating process satisfies near-epoch dependence (NED), which is a weaker and more flexible assumption than α−mixing. NED property was first

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introduced in time series (see for example Davidson (1994) and references therein) and then extended to spatial processes by Jenish and Prucha (2012).

We first need to specify the notion of a NED spatial-temporal field on a possibly unevenly spaced lattice. We here give a brief description of the topological specification, for a more detailed introduction the reader can refer for example to Qu et al. (2017).

Let $\|Z\|_p = (E|Z|^p)^{1/p}$ be the L_p norm of a random field Z.

Definition 3 *Let Z be a random field $Z = \{Z_{l,L}, l \in D_L, L \geq 1\}$ with bounded L_p norm for some $p \geq 1$, defined on an increasing domain D_L, that is $|D_L| \overset{L\to\infty}{\to} \infty$. Let further $\epsilon = \{\epsilon_{l,L}, l \in D_L, L \geq 1\}$ be another random field and $c = \{c_{l,L}\}$ an array of finite positive constants. Then, the random field Z is called $L_p(c)-$near-epoch dependent on ϵ if there exists a sequence $\psi(s) \geq 0$, such that $\lim_{s\to\infty} \psi(s) = 0$ and*

$$\|Z_{l,L} - \mathbb{E}\left(Z_{l,L} \mid \mathscr{F}_l(s)\right)\|_p \leq c_{l,L}\psi(s),$$

where $\mathscr{F}_l(s)$ is the $\sigma-$algebra generated by all $\epsilon_{l',L}$ such that $\|l - l'\|_\infty \leq s$.

Intuitively, the field Z is L_p-NED on ϵ if it can be approximated arbitrarily well by a sufficiently large sequence of the ϵ.

In our framework, similarly to Qu et al. (2017), we have $Z_{l,L} = y_{it}$ and $\epsilon_{l,L} = \varepsilon_{it}$, so that l depends on both time and space indices and, as time passes, all unit locations shift vertically upward to an affine plane parallel to the Euclidean space of the cross-sectional locations ($\mathbb{R}^d$, with $d \geq 1$). Each location $l \in D_L$ then has a one-to-one correspondence with the space and time dimensions $l = l(i,t)$ and distance is given by

$$\|l(i,t) - l(i',t')\|_\infty = \max\{|t - t'|, \|l(i,0) - l(i',0)\|_\infty\}.$$

So, for fixed N regions and T periods, we have $L = NT$ locations in R^{d+1}, and D_L is a finite subset of R^{d+1} increasing with L. In our paper, we mainly focus on $d = 1$ (univariate partial ML) and $d = 2$ (bivariate partial ML, where each location index i identifies a couple of units (i_1, i_2))

The following results concern sufficient conditions ensuring preservation of NED property on transformations of the data.

Let $a_N(i,j)$ be the $(i,j)-$th component of the matrix $A_\rho^{-1} = (I - \rho W_N)^{-1}$.

Assumption 5 *For every $\rho \in (-1/\bar{\tau}, 1/\bar{\tau})$, and for all $i = 1, \ldots, N$, there exists a nonnegative sequence $\psi(s) \geq 0$ and a bounded constant $c_i(\rho) \leq 1$, such that*

$$\frac{\sum_{i':\|l(i,0)-l(i',0)\|_\infty > s} a_N^2(i,i')}{\sum_{j=1}^N a_N^2(i,j)} \leq c_i(\rho)\psi(s) \tag{14}$$

with $\lim_{s\to\infty} \psi(s) = 0$.

Assumption 5 is a low-level condition used to prove that the random field of binary variables $\{y_{it}\}$ is $L_2(c)$-NED. Intuitively, it requires the impact of non-neighboring regions to decrease fast enough with distance. We point out that this condition is reasonably weak. In fact, considering that the sum in the denominator of equation (14) is equal to the $i-$th component of the main diagonal of $A_\rho^{-1}(A_\rho')^{-1}$, if we assume that, for all $\rho \in (-1/\bar{\tau}, 1/\bar{\tau})$ and for all N, all elements of the main diagonal of $A_\rho^{-1}(A_\rho')^{-1}$ are bounded (which is a consequence of boundedness in row and column sum norm, assumption 2):

$$\lim_N \sup_i \sum_{j=1}^N a_N^2(i,j) < \infty.$$

This implies convergence of the series $\sum_{j=1}^\infty a_N^2(i,j)$, for all i. From Cauchy's convergence criterion, we then have that

$$\lim_N \lim_{s\to\infty} \sum_{i':\|l(i,0)-l(i',0)\|_\infty > s} a_N^2(i,i') = 0,$$

showing that Assumption 5 and 2 are strictly related, but none of them implies the other.

Although in a different framework, Assumption 3(ii) in Qu et al. (2017) produces a similar behavior to Assumption 5: the truncation of all weights associated to units whose distance surpasses a certain threshold guarantees that the components of all $(W_N)^k$ $(k \geq 1)$ corresponding to units farther than s eventually become zero as s increases.

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Theorem 1 *Under Assumptions 1 – 5, the process defined by the variables $Z_{l,L} = y_{it}$ is $L_2((c_0)^{1/4})-$NED on ε, with sequence $(\psi(s))^{1/4}$ and $c_0 = \sup_i c_i(\rho_0) \leq 1$.*

In order to derive the asymptotic expansions necessary to compute the asymptotic bias and prove asymptotic normality, we need to exploit preservation of the NED property under suitable transformations of the random field $\{y_{it}, l = l(i,t) \in D_L\}$.

For every vector $\kappa = (\kappa_0, \kappa_1, \ldots, \kappa_{k+2})$, with $|\kappa| = \sum_j \kappa_j \leq 5$, let us define the random field:

$$q_\kappa = \{q_{\kappa,l}(y_{it}), l \in D_L, l = l(i,t)\} = \left\{ q_{\kappa,l}(y_{it}; \boldsymbol{\theta}, \eta_l) := \frac{\partial^{|k|} \log \Pr\left(y_{it} \mid \boldsymbol{X}_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \eta_l\right)}{\partial^{\kappa_0}\eta_l \partial^{\kappa_1}\theta_1 \cdots \partial^{\kappa_{k+2}}\theta_{k+2}} \right\}, \tag{15}$$

where, $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_{k+2}) = (\rho, \alpha, \beta_1, \ldots, \beta_k)$.

Note that all the random fields q_κ have domain $\{0,1\}^q$, where $q = 1$ and $q = 2$ for the univariate and bivariate PMLE respectively.

As will be shown in next subsection, the terms q_κ, for different κ, enter the asymptotic expansion of the PMLE estimator.

We are going to invoke Proposition 2 in Jenish and Prucha (2012) on the preservation of the NED property under Lipschitz transformations to prove that all q_κ are L_2-NED.

First, it is easy to see that:

$$|q_{\kappa,l}(y) - q_{\kappa,l}(y')| \leq 2\max\{q_{\kappa,l}(y), q_{\kappa,l}(y')\}\|y - y'\|.$$

It is known, from Faà di Bruno's formula, that any $|\kappa|-$th order partial derivative of the composite function $\log \circ f = \log(f)$ is a linear combination of terms:

$$[1/f]^{m_1+\ldots+m_{|\kappa|}} \prod_{j=1}^{|\kappa|} (f^{(j)})^{m_j},$$

where the terms $f^{(j)}$ are $j-$th order (partial) derivatives[3]. Choosing $f(\mu_i, \boldsymbol{\theta}) = p_{it}(y \mid X_{Nt}, y_{i,t-1}, \eta_l, \boldsymbol{\mu}_i)$, one can see that, in order to bound $2\max\{q_{\kappa,l}(y), q_{\kappa,l}(y')\}$, it is enough to satisfy

$$\min_{y_{it} \in \{0,1\}^q} \Pr\left(y_{it} \mid \boldsymbol{X}_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \mu_i\right) \geq \delta > 0 \tag{16}$$

and all derivatives

$$\mathbb{E}\left| \frac{\partial^{|\kappa|} \Pr\left(y_{it} \mid \boldsymbol{X}_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \mu_i\right)}{\partial^{\kappa_0}\mu_i \partial^{\kappa_1}\theta_1 \cdots \partial^{\kappa_{k+2}}\theta_{k+2}} \right|^{10+5\epsilon} < \infty \tag{17}$$

for $|\kappa| \leq 5$, and for some $\epsilon > 0$. This motivates the following assumption.

Assumption 6 *For all η_l in a compact subset of $\mathbb{R}$, conditions (16) and (17) hold uniformly in $\boldsymbol{\theta}$:*

$$\liminf_{N,T} \inf_{i,t} \inf_{\theta \in \Theta} \min_{y_{it} \in \{0,1\}^q} \Pr\left(y_{it} \mid \boldsymbol{X}_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \mu_i\right) \geq \delta > 0$$

and

$$\sup_{N,T} \sup_{l(i,t) \in D_L} \mathbb{E} \sup_{\theta} \left| \frac{\partial^{|\kappa|} \Pr\left(y_{it} \mid \boldsymbol{X}_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \mu_i\right)}{\partial^{\kappa_1}\boldsymbol{\theta} \partial^{\kappa_2}\mu_i} \right|^{5(2+\epsilon)} < \infty$$

for all $|\kappa| \leq 5$ and for some $\epsilon > 0$.

The uniform conditions in Assumption 6 allows us to: (i) exploit the Lipschitz condition in y, that guarantees the preservation of the NED property of $\{q_{\kappa,l}\}$; (ii) prove a Lipschitz condition in the parameter, (Condition 1 in Jenish and Prucha (2009)), that implies stochastic equicontinuity (in probability) of the family of random functions defined by (15), that is, $\{q_{\kappa,l}(y_{i,t}; \boldsymbol{\theta}, \mu_i),\ l = l(i,t) \in D_L\}$.

Note that the first part of Assumption 6, is a necessary condition to have

$$\sup_{i,t} \left| \sum_{y=\{0,1\}^q} \frac{1}{\Pr\left(y_{it} = y \mid \boldsymbol{X}_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \eta_l\right)} \right| < \infty, \text{ for all } N, T.$$

The above boundedness assumption, if we allow the y_{it} to be bi-variate binary vectors ($q = 2$) and $T = 1$, corresponds to Assumption 5 in Billé and Leorato (2020).

Proposition 1 *Under Assumptions 1 – 6, and if $\psi(s) = o(s^{-\lambda})$, $\lambda > 0$, the random fields $q_\kappa = \{q_{\kappa,l} : l \in D_L,\ L \geq 1\}$ are $L_2(2c_0C)-$NED on ε, with sequence $\tilde{\psi}(s) = o(s^{-\lambda\epsilon/8(1+\epsilon)})$ and with $C = \max\{1, \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)\}$.*

[3] A more precise combinatoric formulation, following Hardy (2006), can be found in the proof of Proposition 1

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3.2.2 Asymptotic expansion for models with regional or time effects

Let us denote by $s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i)$, $s_{\mu\theta,i}(\boldsymbol{\theta},\mu_i)$ the second order partial derivatives of $\ell_i(\boldsymbol{\theta},\mu_i) = \sum_t \log \Pr\left(y_{it} \mid X_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \mu_i\right)$.

It is well known that the concentrated ML approach leads to asymptotically biased estimates for $\boldsymbol{\theta}$, because of the incidental parameter problem.

In this section we focus on finding an asymptotic expansion for the PMLE estimator $\hat{\boldsymbol{\theta}}$ that allows us to study its large sample behavior, in particular to prove its consistency, and asymptotic normality and to compute its asymptotic bias. The asymptotic bias is then used to propose an adjusted concentrated QML estimator. The adjustment is obtained by an ex-post correction of the estimator; a similar approach is followed in Fernández-Val (2009), or by Pakel (2019) among the others. Differently from Pakel (2019), we don't use a Jackknife bias correction but directly exploit the analytic form of the expected bias.

We exploit Carro's computations to derive the asymptotic expansion of the concentrated score, in order to write down an asymptotic expansion of the concentrated QML $\hat{\boldsymbol{\theta}}$ in the univariate partial QML case. We compute a similar expansion also for the bivariate case, using the more general results in McCullagh (1987).

Since we consider the asymptotic behavior as both T and N increase, we need to add an assumption on their rate of convergence.

In what follows, the index i (as in $s_{\theta i}(\boldsymbol{\theta}, \hat{\mu}_i(\boldsymbol{\theta}))$) refer to either the unit location i or to the ith couple $i = (i_1, i_2)$ depending on wether the partial ML is univariate or bivariate. We first need to introduce some more assumptions.

Assumption 7 *(i) For every $i = 1, \dots, n$, the parameter space for $(\boldsymbol{\theta}, \mu_i)$ is equal to $\Theta \times \Lambda$, where Θ and Λ are compact subsets of $\mathbb{R}^{k+2}$ and $\mathbb{R}$ respectively*

(ii) For every $i = 1, \dots, n$ and every $\boldsymbol{\theta}$, the partial log-likelihood $\sum_t \log \Pr(y_{it} \mid X_{Nt}, y_{i,t-1}, \boldsymbol{\theta}, \mu_i) = \ell_i(\boldsymbol{\theta}, \mu_i)$ has a maximum at $\hat{\mu}_i(\boldsymbol{\theta})$

(iii) For all i, there exists a function $\boldsymbol{\theta} \mapsto \bar{\mu}_i(\boldsymbol{\theta})$, such that, for every $\delta > 0$, and for all $T > 1$,

$$\inf_i \inf_\theta \left\{ \mathbb{E}\ell_i(\boldsymbol{\theta}, \bar{\mu}_i(\boldsymbol{\theta})) - \sup_{|\mu_i - \bar{\mu}_i(\theta)| > \delta} \mathbb{E}\ell_i(\boldsymbol{\theta}, \mu_i) \right\} > 0$$

and

$$\inf_i \left\{ \mathbb{E}\ell_i(\boldsymbol{\theta}_0, \bar{\mu}_i(\boldsymbol{\theta}_0)) - \sup_{|\boldsymbol{\theta} - \boldsymbol{\theta}_0| > \delta} \mathbb{E}\ell_i(\boldsymbol{\theta}, \bar{\mu}_i(\boldsymbol{\theta})) \right\} > 0$$

Assumption 8 *For all $\kappa = (\kappa_0, \dots, \kappa_{k+2})$, $|\kappa| \leq 5$, the following conditions hold: (i) $\inf_{N,T} \underline{\lambda}(J_{NT}^{(\kappa)}) > 0$ and $\sup_{N,T} \bar{\lambda}(J_{NT}^{(\kappa)}) < \infty$, where $J_{NT}^{(\kappa)}$ is the quantity:*

$$J_{NT}^{(\kappa)} := (NT)^{-1} \mathrm{Var}\left(\sum_i \sum_t q_{\kappa, l(i,t)}(y_t; \boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)) \right),$$

and where $\underline{\lambda}(A)$ and $\bar{\lambda}(A)$ denote the minimum and maximum eigenvalues of the p.d. matrix A. (ii) $q_{\kappa,l}(y_{it}; \boldsymbol{\theta})$ is a $L_2(\tilde{d})-$NED sequence on ε, with coefficients $\tilde{\psi}(r)$ that satisfy

$$\sum_{r=1}^{\infty} r^{d-1} \tilde{\psi}(r) < \infty$$

and such that

$$\sup_{N,T} \sup_{l \in D_L} \tilde{d}_{l,L} \leq K < \infty.$$

(iii) The matrix $H_{NT,0} := \mathbb{E}H_{NT}(\boldsymbol{\theta}_0)$, where H_{NT} is the Hessian matrix defined in (21), satisfies both $\inf_{N,T} \underline{\lambda}(H_{NT,0}) > 0$ and $\sup_{N,T} \bar{\lambda}(H_{NT,0}) < \infty$.

Assumption 7.(i) is a standard condition on the parameter space and need not be discussed further. Assumption 7.(ii)–(iii) gives identification condition, guaranteeing the existence of $\hat{\mu}_i$ for all i but also of its population counterpart $\bar{\mu}_i(\boldsymbol{\theta})$, that maximizes $\mathbb{E}\ell_i(\boldsymbol{\theta}, \mu_i)$ with respect to μ_i. It also yields $\bar{\mu}_i(\boldsymbol{\theta}_0) = \mu_{i0}$.

Assumption 8 is a technical condition that permits to use the rate of convergence and the asymptotic Gaussian approximation for some of the terms involved into the asymptotic expansion (by means of Theorem 2 of Jenish and Prucha (2012)), and this is used to assess their order of magnitude (in the O_p sense).

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Theorem 2 *Under assumptions 1 – 7,*

$$\Pr\left(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| > \delta\right) \leq o(N/T^2).$$

Theorem 3 *Under assumptions 1 – 8,*
(i) for all $i \leq N$,

$$|\hat{\mu}_i(\hat{\boldsymbol{\theta}}) - \mu_{i0}| \to_T 0 \ \ \textit{in prob.}$$

(ii) Further,

$$s_{\theta i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)) = s_{\theta i}(\boldsymbol{\theta}_0, \mu_{i0}) + B_i(\boldsymbol{\theta}_0) + T^{-1/2} a_i(\boldsymbol{\theta}_0) + O_p(T^{-1}), \tag{18}$$

where

$$B_i(\boldsymbol{\theta}) = -\frac{1}{2} \frac{\left(s_{\theta\mu\mu,i}(\boldsymbol{\theta}, \hat{\mu}_i(\boldsymbol{\theta})) + s_{\mu\mu\mu,i}(\boldsymbol{\theta}, \hat{\mu}_i(\boldsymbol{\theta})) \frac{\partial \hat{\mu}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)}{s_{\mu\mu,i}(\boldsymbol{\theta}, \hat{\mu}_i(\boldsymbol{\theta}))} + \frac{\partial}{\partial \mu_i} \left(\frac{\mathbb{E} s_{\theta\mu,i}(\boldsymbol{\theta}, \mu_i)}{\mathbb{E} s_{\mu\mu,i}(\boldsymbol{\theta}, \mu_i)} \right)_{|_{\mu_i = \hat{\mu}_i(\boldsymbol{\theta})}}$$

in the univariate case, while

$$\begin{aligned} B_i(\boldsymbol{\theta}) = &- \sum_{j,h=1}^{2} \frac{\mathbb{E} s_{\theta\mu_j} s_{\mu_h}}{\mathbb{E} s_{\mu_j \mu_h}} + \sum_{j,h,l,k}^{2} \frac{s_{\theta\mu_j} \mathbb{E} s_{\mu_h \mu_k} s_{\mu_l}}{\mathbb{E} s_{\mu_j \mu_h} \mathbb{E} s_{\mu_l \mu_k}} \\ &+ \frac{1}{2} \sum_{j,h,l} \sum_{r,s,u} \frac{s_{\theta\mu_j} \mathbb{E} s_{\mu_h \mu_l} \mathbb{E} s_{\mu_r \mu_s \mu_u}}{\mathbb{E} s_{\mu_j \mu_r} \mathbb{E} s_{\mu_h \mu_s} \mathbb{E} s_{\mu_l \mu_u}} \\ &- \frac{1}{2} \sum_{j,h=1}^{2} s_{\theta\mu_j \mu_h} \left[3 \left(\mathbb{E} s_{\mu_j \mu_h} \right)^{-1} + \left(\mathbb{E} s_{\mu_j \mu_{3-h}} \right)^{-1} \left(\mathbb{E} s_{\mu_{3-j} \mu_{3-h}} \right) \left(\mathbb{E} s_{\mu_{3-j} \mu_h} \right)^{-1} \right], \end{aligned} \tag{19}$$

in the bivariate case.

The first expansion is the same as in Carro, while the second is its bivariate extension.

In order to simplify the notation, we now focus on the univariate partial maximum likelihood. The bivariate case is analogous. By a standard Taylor expansion argument on the concentrated log-likelihood, we can write

$$\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\hat{\boldsymbol{\theta}}, \hat{\mu}_i(\hat{\boldsymbol{\theta}})) = \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)) + H_{NT} \sqrt{NT} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + O_p((NT)^{1/2} \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|^2) \tag{20}$$

where

$$H_{NT} = \left\{ \frac{1}{NT} \sum_i \frac{\partial s_{\theta,i}(\boldsymbol{\theta}, \hat{\mu}_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right\}_{|\boldsymbol{\theta} = \boldsymbol{\theta}_0}. \tag{21}$$

Since $\sum_{i=1}^{N} s_{\theta,i}(\hat{\boldsymbol{\theta}}, \hat{\mu}_i(\hat{\boldsymbol{\theta}})) = 0$ by definition, we have, using the equation (18):

$$H_{NT} \sqrt{NT} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)(1 + O_p(1/\sqrt{NT}) \|\sqrt{NT}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\|^2) = -\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)).$$

The right hand side is equal to

$$\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)) = \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \left(s_{\theta,i}(\boldsymbol{\theta}_0, \mu_{i0}) + B_i(\boldsymbol{\theta}_0) + T^{-1/2} a_i(\boldsymbol{\theta}_0) + O_p(T^{-1}) \right)$$

and is therefore a $O_p(1)$ (see Lemma 4 in Appendix). This implies $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| = O_p((NT)^{-1/2})$. Then,

$$H_{NT} \sqrt{NT} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \left(s_{\theta,i}(\boldsymbol{\theta}_0, \mu_{i0}) + B_i(\boldsymbol{\theta}_0) + T^{-1/2} a_i(\boldsymbol{\theta}_0) + O_p(T^{-1} + (NT)^{-1/2}) \right)$$

This implies that

$$H_{NT} \sqrt{NT} \left(\hat{\boldsymbol{\theta}} + H_{NT}^{-1} \frac{1}{NT} \sum_i B_i(\boldsymbol{\theta}_0) - \boldsymbol{\theta}_0 \right) = -\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\boldsymbol{\theta}_0, \mu_{i0}) + \sqrt{N/T^2} A_N + O_p\left(\sqrt{\frac{N}{T^3}} + \frac{1}{T} \right) \tag{22}$$

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satisfies

$$H_{NT}\sqrt{NT}\cdot\mathbb{E}\left(\hat{\boldsymbol{\theta}}+H_{NT}^{-1}\frac{1}{NT}\sum_i B_i(\boldsymbol{\theta}_0)-\boldsymbol{\theta}_0\right)=O_p\left(\sqrt{\frac{N}{T^3}}+\frac{1}{T}\right).$$

The above computations prove that $\hat{\boldsymbol{\theta}}$ has asymptotic bias: $H_{NT}^{-1}\frac{1}{\sqrt{NT}}\sum_i B_i(\boldsymbol{\theta}_0)$. Since each $B_i(\boldsymbol{\theta}_0)=O_p(1)$, and if the Hessian matrix H_{NT}^{-1} has a bounded limit, the asymptotic bias is of order $O_p(\sqrt{N/T})$ and can be negligible only if $N/T=o(1)$.

The asymptotic bias is clearly in line with the computations in Carro (2007), and can be generalized to the bivariate case by replacing the bivariate version of $B_i(\boldsymbol{\theta}_0)$ into the formula.

The computation of $B_i(\boldsymbol{\theta}_0)$ can be used to define a bias-corrected version of the estimator for $\boldsymbol{\theta}_0$ is different from both Carro (2007) and Pakel (2019): on the one hand, the correction is applied to the estimator directly, instead of the score function; on the other hand, equation (22) is exploited directly to compute the correction, without resorting to resampling methods.

An estimator defined as

$$\hat{\boldsymbol{\theta}}+H_{NT}^{-1}\frac{1}{NT}\sum_i B_i(\boldsymbol{\theta}_0)\quad\text{or}\qquad\hat{\boldsymbol{\theta}}+H_{NT}^{-1}\frac{1}{NT}\mathbb{E}\sum_i B_i(\boldsymbol{\theta}_0)$$

is expected to have a smaller order bias than the original concentrated partial ML estimator.

All the terms H_{NT} and $B_i(\boldsymbol{\theta}_0)$ depend on the true parameter, and need to be estimated, and thus a feasible version of the corrected estimator is obtained by

$$\tilde{\boldsymbol{\theta}}=\hat{\boldsymbol{\theta}}+\hat{H}_{NT}^{-1}\frac{1}{NT}\sum_i\hat{B}_i. \tag{23}$$

The definitions of $\hat{B}_i$ and $\hat{H}_{NT}$ in (23) are given in the following subsection.

3.2.3 Asymptotic distribution and inference

The asymptotic expansion proved in section 3.2.2 is here used to prove asymptotic normality of the PMLE estimator as well as of its bias-corrected version. Further, this section also deals with inference, by proposing consistent estimation of the variance-covariance matrix and the derivation of the feasible estimator $\tilde{\boldsymbol{\theta}}$.

Throughout all this section, we will assume the following rate of convergence for T and N: $\lim_{N\to\infty}\frac{T}{N}=c\in(0,\infty)$, namely, that $T=O(N)$.

Let us first define

$$J_{NT}=\frac{1}{NT}\operatorname{Var}\left(\sum_i s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))\right) \tag{24}$$

Theorem 4 *Let Assumptions 1–8 hold and let* $\mathbb{B}=\lim_{N,T}H_{NT,0}^{-1}\mathbb{E}\sum_i B_i(\boldsymbol{\theta}_0)/\sqrt{NT}<\infty$ *(componentwise). Then,*

$$\sqrt{NT}(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)\overset{N,T}{\to}\mathcal{N}(\mathbb{B},\mathbb{V}) \tag{25}$$

where $\mathbb{V}=\lim_{N,T}H_{NT,0}^{-1}J_{N,T}H_{NT,0}^{-1}$ *and* $H_{NT,0}^{-1}=\mathbb{E}_{\theta_0}H_{NT}^{-1}$.

Assume further that there exist a matrix $\hat{H}_{NT}$ *and* N *vectors* $\hat{B}_i$ *such that* $\hat{H}_{NT}$ *is nonsingular for all* N,T *and*

$$\|\hat{H}_{NT}-H_{NT}\|=o_p(1)$$

and

$$\sup_i\|\hat{B}_i-B_i(\boldsymbol{\theta}_0)\|=O_p(T^{-1/2}).$$

Then, if $N,T\to\infty$ *and* $N/T\to c$, $0<c<\infty$,

$$\sqrt{NT}(\tilde{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)\overset{N,T}{\to}\mathcal{N}(\mathbf{0},\mathbb{V}). \tag{26}$$

The estimation of H_{NT} and/or $H_{NT,0}$ through some consistent matrix $\hat{H}_{NT}$ is necessary for both bias correction and the variance-covariance matrix. Fortunately, the computation of $\hat{H}_{NT}$ does not pose particular problems, because the empirical Hessian matrix evaluated at $\hat{\boldsymbol{\theta}}$ is consistent for H_{NT} and $H_{NT,0}$. An alternative definition, based on *analytic derivation* of the Hessian is also proposed in Appendix D.

The remainder of this section is therefore devoted to the estimation of J_{NT} and of $B_i(\boldsymbol{\theta}_0)$.

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Theorem 5 *Under assumptions 1–8, and if $\hat{\boldsymbol{\theta}}^*$ is any estimator of $\boldsymbol{\theta}_0$ such that $\|\hat{\boldsymbol{\theta}}^* - \boldsymbol{\theta}_0\| = O_p(T^{-1/2})$, the asymptotic bias $B_i(\boldsymbol{\theta}_0)$ can be consistently estimated by:*

$$\hat{B}_i = -\frac{1}{2}\frac{s_{\theta\mu\mu,i}}{s_{\mu\mu,i}} + \frac{1}{2}\frac{s_{\theta\mu,i}s_{\mu\mu\mu,i}}{(s_{\mu\mu,i})^2} - \frac{\hat{\mathbb{E}}s_{\theta\mu,i}s_{\mu,i}}{s_{\mu\mu,i}} + \frac{s_{\theta\mu,i}\hat{\mathbb{E}}s_{\mu\mu,i}s_{\mu,i}}{(s_{\mu\mu,i})^2} \tag{27}$$

where all derivatives are computed at $(\hat{\boldsymbol{\theta}}^*, \hat{\mu}_i(\hat{\boldsymbol{\theta}}^*))$ and $\hat{\mathbb{E}}$ indicates the empirical version of the expected value $\mathbb{E}$.

Note that $\hat{\boldsymbol{\theta}}^*$ can be either $\hat{\boldsymbol{\theta}}$ or $\tilde{\boldsymbol{\theta}}$, and it actually only need an $\sqrt{T}-$consistency w.r.t. $\boldsymbol{\theta}_0$. The computation of the terms in (27) requires the definition of the partial derivatives, up to the third order, entering into the definition of B_i.

In the computation of the main terms of $\hat{B}_i$, expectations have to be taken and, in the case of cross-sectional dependence faced here, this adds considerable difficulties with respect to the framework in Carro (2007). Details on how to approximate and simplify all the terms in (27) can be found in the Appendix D.

For the estimation of J_{NT}, we propose an approach similar to Pakel (2019), based on a HAC estimator *à la* Newey and West (1987). It is first necessary to define

$$\hat{s}_{\theta,t} = N^{-1}\sum_{i=1}^{N} \frac{\partial \log \Pr(y_{it} \mid X_{Nt}, y_{i,t-1}\boldsymbol{\theta}, \mu_i)}{\partial\theta}\bigg|_{\theta=\hat{\theta}^*, \mu_i=\hat{\mu}_i(\hat{\theta}^*)}$$

and, for $j = 0, \ldots, T-1$:

$$\hat{\Omega}_j = T^{-1}\sum_{t=j+1}^{T} \hat{s}_{\theta,t}\hat{s}'_{\theta,t-j}$$

Differently from Pakel (2019), we consider the estimation of $\mathrm{Var}\left(\frac{1}{\sqrt{NT}}\sum_i s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)\right)$ directly and not of $\mathrm{Var}\left(\frac{1}{\sqrt{T}}\sum_i s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)\right)$, because the latter will lead to an indefinite limit in the Wald estimator. This requires a stronger condition on the convergence of the estimator $\hat{\boldsymbol{\theta}}$ used for the estimation of J_{NT}: instead of $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = O_p(T^{-1/2})$, we indeed need $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = O_p((NT)^{-1/2})$ (satisfied by the PMLE and the modified version in view of $T/N = O(1)$). We also need, as in similar proofs for consistency of Newey and West estimators, stronger conditions on the moments of the NED process.

Assumption 9 *$q_{\kappa,l}(y_{it};\boldsymbol{\theta})$ is a $L_2(\tilde{d})-$NED sequence on ε, and there exists a $\delta > 0$ such that the moments $\sup_\theta \sup_l \|q_{\kappa,l}(y_{it};\boldsymbol{\theta})\|_{2(2+\delta)} < \infty$. Moreover, the NED coefficients $\tilde{\psi}(r)$ satisfy*

$$\sum_{r=1}^{\infty} r^{d-1}[\tilde{\psi}(r)]^{\delta/(\delta+1)} < \infty.$$

We recall that a NED process with coefficient $\psi(r)$ is said to have size $-a$ if $\psi(r) \leq O(r^{-a})$. Then, Assumption 8(ii) implies that the process $q_{\kappa,l}$ has size $-a < -2$ because $\sum_r r^{d-1-a} < \infty$ necessarily implies $d - 1 - a < -1$, while assumption 9 requires a size smaller than $-d(\delta+1)/\delta$, where $d = 2$ in the univariate PMLE framework, while it is $d = 3$ in the bivariate case.

Theorem 6 *Let Assumptions 1–9, and let $w(m,j) = 1-[j/(m+1)]$, with $m = m(T)$, such that $\lim_T m(T) = \infty$ and $mT^{-1/3} \to 0$. Let also $\hat{\boldsymbol{\theta}}^*$ satisfy $\|\hat{\boldsymbol{\theta}}^* - \boldsymbol{\theta}_0\| = O_p((NT)^{-1/2})$, with $T/N \to c$ for some $0 < c < \infty$. Define*

$$\hat{J}_{NT} = \left[N\hat{\Omega}_0 + N\sum_{j=1}^{m(T)} w(m,j)\left(\hat{\Omega}_j + \hat{\Omega}'_j\right) - \frac{1}{NT}\left(\sum_i \hat{B}_i\right)\left(\sum_i \hat{B}_i\right)'\right],$$

where $\hat{B}_i$ is defined in (27). Then,

$$\left\| J_{NT} - \mathrm{Var}\left(\frac{1}{\sqrt{NT}}\sum_i s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0))\right)\right\| \to 0 \tag{28}$$

in probability.

Note that, while in Pakel (2019) and Newey and West (1987) $m = o(T^{-1/4})$, we prove consistency under the improved rate $o(T^{-1/3})$, that is likely to be the best possible rate. This is attained by adapting the arguments of Quah (1990) to the case of near-epoch dependence: the key point is to replace Davidov's and Peligrad's inequalities directly with summability properties of the NED sequence.

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3.3 Feasible CPMLE through Subsampling

The computation of the N–dimensional integral is intractable for large datasets, and asymptotic theory usually rely on the Normality assumption of the error terms. In this section we focus our attention on the case when the number of available cross-sectional observations is very large.

If we consider pairs of observations as described in subsection 2.2, it is quite reasonable to assume that observations corresponding to units distant (enough) in space can be approximately considered as uncorrelated, similarly to Arbia (2014), while units belonging to the same pair may display high correlation. To build up the procedure, we make use of the SYS design proposed in subsection 2.2 so as to obtain a subsample of observations of dimension $n << N$, where $n = 2q$ and q is the total number of pairs. The resulting estimator, called *Feasible Concentrated Partial Maximum Likelihood Estimator* (FCPMLE), is then a simplification of the CPMLE proposed in equation (9) with probabilites in equation (10). Here we focus only on the bivariate case.

Starting from model in equation (1), having a subsample n and a balanced panel, we specify separate latent models for every couple $i = 1, \dots, q$ over time $t = 1, \dots, T$, with the i-th couple being indexed by (i_1, i_2), as

$$\begin{aligned} \boldsymbol{y}_{it}^{*} &= \alpha \boldsymbol{y}_{it-1} + \rho \boldsymbol{W}_i \boldsymbol{y}_{it}^{*} + \boldsymbol{X}_{it}\boldsymbol{\beta} + \boldsymbol{\mu}_i + \boldsymbol{\varepsilon}_{it}, \quad \boldsymbol{\varepsilon}_{it} \sim \mathcal{N}\left(\boldsymbol{0}, \boldsymbol{I}_2\right) \\ \boldsymbol{y}_{it} &= \boldsymbol{1}_2\left(\boldsymbol{y}_{it}^{*} > \boldsymbol{0}\right), \end{aligned} \tag{29}$$

where $\boldsymbol{y}_{it}^{*} = \left(y_{i_1,t}, y_{i_2,t}\right)'$, $d = 0, 1$, $\boldsymbol{X}_{it} = \left(\boldsymbol{x}_{i_1,t}, \boldsymbol{x}_{i_2,t}\right)'$, and

$$\boldsymbol{W}_i = \begin{pmatrix} 0 & w_{i,12} \\ w_{i,21} & 0 \end{pmatrix}.$$

From the reduced form model we obtain the covariance matrix equal to

$$\boldsymbol{\Sigma}_{\nu,n} = \left(\boldsymbol{I} - \rho \boldsymbol{W}_n\right)^{-1}\left(\boldsymbol{I} - \rho \boldsymbol{W}_n\right)^{-1\prime} = diag\left(\boldsymbol{\Sigma}_{\nu,1}, \dots, \boldsymbol{\Sigma}_{\nu,i}, \dots, \boldsymbol{\Sigma}_{\nu,q}\right),$$

where $\boldsymbol{W}_n = diag\left(\boldsymbol{W}_1, \dots, \boldsymbol{W}_i, \dots, \boldsymbol{W}_q\right)$ is a block diagonal matrix, and $\boldsymbol{A}_\rho = \left(\boldsymbol{I} - \rho \boldsymbol{W}_n\right)$, and where

$$\boldsymbol{\Sigma}_{\nu,i} := \begin{pmatrix} \sigma_{i_1}^2 & \sigma_{i_1,i_2} \\ \sigma_{i_2,i_1} & \sigma_{i_2}^2 \end{pmatrix} := \frac{1}{(1 - \rho^2 w_{i,12} w_{i,21})^2} \begin{pmatrix} 1 + \rho^2 w_{i,12}^2 & \rho\left(w_{i,12} + w_{i,21}\right) \\ \rho\left(w_{i,12} + w_{i,21}\right) & 1 + \rho^2 w_{i,21}^2 \end{pmatrix}. \tag{30}$$

Under the above assumptions, the partial log-likelihood function can be written in a easier way

$$\ell\left(\boldsymbol{\theta}, \boldsymbol{\mu}_n\right) = \sum_{i=1}^{q} \ell_i\left(\boldsymbol{\theta}, \mu_i\right) = \sum_{i=1}^{q} \sum_{t=1}^{T} \log p_{it}\left(\boldsymbol{y}_{it}\right),$$

with probabilities computed as in equation (10), but with the following simplification

$$\begin{aligned} p_{it}\left(\boldsymbol{d}\right) &= \Pr\left(y_{i_1 t} = d_1, y_{i_2 t} = d_2 \mid \boldsymbol{X}_{nt}, y_{i_1,t-1}, y_{i_2,t-1}\right) \\ &= \int_{\{\jmath_1 u > -\jmath_1 \zeta_{i_1}\}} \frac{1}{\sigma_{i_1}} \phi\left(\frac{u}{\sigma_{i_1}}\right) \Phi\left(\jmath_2\left(\zeta_{i_2}\sqrt{1 + \rho^2 w_{i,21}^2} + u\frac{\rho(w_{i,12} + w_{i,21})}{\sqrt{1 + \rho^2 w_{i,21}^2}}\right)\right) du \end{aligned}$$

where $\jmath_j = 2\left(d_j - 0.5\right)$, $d_j = 0, 1$, $j = 1, 2$ and $\boldsymbol{\zeta} = \boldsymbol{X}_{\rho,t}\boldsymbol{\beta} + \alpha \boldsymbol{y}_{\rho,t-1} + \boldsymbol{A}_\rho^{-1}\boldsymbol{\mu}$, as in subsection 3.1. The reduced form model comes from equation (29) and from the i-th 2×2 block of the covariance matrix. This gives,

$$\begin{aligned} \zeta_{i_2} &= \frac{\rho w_{i_1} \boldsymbol{x}_{i_1,t}'\boldsymbol{\beta} + \boldsymbol{x}_{i_2,t}'\boldsymbol{\beta} + \alpha\rho w_{i_1} y_{i_1,t-1} + \alpha y_{i_2,t-1} + \left(\mu_{i_2} + \rho w_{i_1}\mu_{i_1}\right)}{1 - \rho^2 w_{i_1} w_{i_2}} \\ &= \frac{\rho w_{i_1} \boldsymbol{x}_{i_1,t}'\boldsymbol{\beta} + \boldsymbol{x}_{i_2,t}'\boldsymbol{\beta} + \alpha\rho w_{i_1} y_{i_1,t-1} + \alpha y_{i_2,t-1}}{1 - \rho^2 w_{i_1} w_{i_2}} + \tilde{\mu}_{i_2} \end{aligned}$$

where $\tilde{\mu}_{i_2} = \left(\mu_{i_2} + \rho w_{i_1}\mu_{i_1}\right) / \left(1 - \rho^2 w_{i_1} w_{i_2}\right)$, $\eta_{i_2} = \left(1 + \rho w_{i_1}\right) / \left(1 - \rho^2 w_{i_1} w_{i_2}\right)$. We then rename, for convenience, $\boldsymbol{\mu} = \tilde{\boldsymbol{\mu}}$.

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4 Finite Sample Properties

In this section we present the results of two different Monte Carlo experiments, aiming at comparing the statistical properties in finite samples of all the three estimators, and studying the behaviour of the FCPMLE when increasing the relative distance among the selected pairs by raising the domain.

All the DGPs are based on a fixed covariate $\boldsymbol{x}$ of dimension $(N \times T) \times 1$, whose distribution is Uniform, $\mathscr{U}(-1, 1)$, and the true coefficient is fixed to $\beta = 1$. The spatial autoregressive parameter ρ takes the value 0.6, while the dynamic coefficient is fixed to $\alpha = 0.5$. The nonnegative time-invariant and row-stochastic weight matrix $\boldsymbol{W}_N$ is built using a k-nearest neighbor criterion with $k = 11$, while $\varpi = 0.01$ (with maximum number of iterations $L = 20$). Finally, the cross-sectional dimension in equal to $N = \{100, 400\}$, the time dimension to $T = \{4, 10\}$ and the number of simulation runs is $1,000$ each. All the simulations were carried out through the R function `bobyqa` in the package *minqa* which is derivative-free method of minimization and it is able to find a global optimal solution even with rough surfaces of the likelihood function.

Recently, several studies have compared the finite sample performances of $\mathscr{M}$-estimators for dynamic discrete choice panel data models, see e.g. Arulampalam and Stewart (2009) and Akay (2012), and for static spatial discrete choice models, see e.g. Billé and Leorato (2020). In this section we allow for the evaluation of the proposed ML-based estimators within different scenarios, consindering time dependence, fixed effects and spatial (cross-sectional) dependence.

4.1 Comparison between the CPML estimators

The first set of MC simulations aims at comparing the finite sample properties of the three CPML estimators defined in section 3.1.

Table 1 reports the summary statistics of both the Bivariate and Univariate CPMLE estimators when $N = \{100, 400\}$ and $T = \{4, 10\}$. Regarding the bivariate CPML estimator, the estimates of the vector of parameters of interest $\boldsymbol{\theta} = (\alpha, \rho, \beta)'$ are very good in terms of both unbiasedness and consistency in finite samples. When $T = 4$ the slightly higher bias is likely due to the incidental parameter problem of the individual fixed effects, explained in section 3.1, and previously studied by Carro (2007). Billé and Leorato (2020) slightly underestimated the spatial parameter ρ, especially as the true value approached its upper limit, while this seems not to be the case in this context, probabily due to the adding time information. Results improve for all the parameters as the panel sample size increases, i.e. both from $T = 4$ to $T = 10$ with N fixed (increased time dimension) and from $N = 100$ to $N = 400$ with fixed T (increased cross-sectional dimension). The Monte Carlo distributions of the estimators of the parameters of interest $\boldsymbol{\theta} = (\alpha, \rho, \beta)'$ are approximately bell–shaped, especially those for ρ and α. Figures of the MC distributions are available upon request.

The same conclusion can be drawn for the univariate CPML estimator, with only a small increment in bias and RMSE. The table also reports the simulations with different sample sizes with the proposed bias reduction procedure. The bias reduction procedure seems to improve a bit the estimation of the correlation coefficient ρ, and in most of the cases also the estimation of the β and time coefficient α.

4.2 Feasible CPMLE

Table 2 reports the summary statistics of the Feasible (Bivariate) CPMLE estimator when $T = \{4, 10\}$, $N = \{2,500; 10,000\}$ and subsamples of cross-sectional dimension $n = \{100; 400\}$ are drawn for comparison with the previous cases. Results show very good estimator's performances in terms of the β coefficient, even better than the previous two estimators, but also an increased bias and RMSE for the autoregressive parameters. Regardless of the cross-sectional dimension N of the original samples, results are quite the same. Therefore, increasing distances between spatial units does not improve estimation results, but selecting a sufficient subsample size only matters.

Moreover, in Table 1 a small simulation has been carried out in order to evaluate how much of the bias of the Feasible (Bivariate) CPMLE estimator is due to the subsampling procedure or to the block diagonal structure of the $\boldsymbol{W}_N$ weighting matrix. The small simulation considers $N = 100$ and $T = 4$. As we can observe, the estimation results are

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quite the same in comparison with the cases $n = 100$ and $T = 4$ of Table 2. This means that the bias is mainly due to a model misspecification when imposing a block diagonal weighting matrix, rather than to the subsampling procedure.

Nevertheless, the results related to the FCPMLE are not promising when considering the model in equation (29) both in terms of the autocorrelation coefficent ρ and the dynamic coefficent α. For this reason, we conduct an additional simulation study based on the following model specification

$$\begin{aligned} \boldsymbol{y}_{it}^{*} &= \alpha \boldsymbol{y}_{it-1} + \boldsymbol{X}_{it}\boldsymbol{\beta} + \boldsymbol{\mu}_i + \xi_t \boldsymbol{\iota}_i + \boldsymbol{\nu}_{it}, \\ \boldsymbol{\nu}_{it} &= \lambda \boldsymbol{W}_i \boldsymbol{\nu}_{it} + \boldsymbol{\varepsilon}_{it}, \quad \boldsymbol{\varepsilon}_{it} \sim \mathcal{N}\left(\boldsymbol{0}, \boldsymbol{I}_2\right) \\ \boldsymbol{y}_{it} &= \boldsymbol{1}_2\left(\boldsymbol{y}_{it}^{*} > \boldsymbol{0}\right), \end{aligned} \tag{31}$$

where the autocorrelation is included in the error terms of the latent linear equation rather than in the latent dependent variable. Indeed, in many empirical applications the researcher can be interested only in autocorrelation in the error terms, justifying our last simulation study.

If the data generating process is given by (31), we have

$$\boldsymbol{\Sigma}_{\nu,n} = \left(\boldsymbol{I} - \lambda \boldsymbol{W}_n\right)^{-1}\left(\boldsymbol{I} - \lambda \boldsymbol{W}_n\right)^{-1'} = diag\left(\boldsymbol{\Sigma}_{\nu,1}, \ldots, \boldsymbol{\Sigma}_{\nu,i}, \ldots, \boldsymbol{\Sigma}_{\nu,q}\right),$$

where $\boldsymbol{W}_n = diag\left(\boldsymbol{W}_1, \ldots, \boldsymbol{W}_i, \ldots, \boldsymbol{W}_q\right)$ is a block diagonal matrix, and $\boldsymbol{A}_\lambda = \left(\boldsymbol{I} - \lambda \boldsymbol{W}_n\right)$, while $\boldsymbol{\Sigma}_{\nu,i}$ follows equation (30) but with λ instead of ρ.

The bi-variate probabilities are then defined as

$$\begin{aligned} p_{it}\left(\boldsymbol{d}\right) &= \Pr\left(y_{i_1 t} = d_1, y_{i_2 t} = d_2 \mid \boldsymbol{X}_{nt}, y_{i_1,t-1}, y_{i_2,t-1}\right) \\ &= \int_{\{\vartheta_1 u > -\vartheta_1 \vartheta \zeta_{i_1}\}} \frac{1}{\sigma_{i_1}} \phi\left(\frac{u}{\sigma_{i_1}}\right) \Phi\left(\vartheta_2\left(\zeta_{i_2}\sqrt{1+\lambda^2 w_{i,21}^2} + u\frac{\lambda(w_{i,12}+w_{i,21})}{\sqrt{1+\lambda^2 w_{i,21}^2}}\right)\right) du \end{aligned}$$

with the difference being the easier definition of $\boldsymbol{\zeta}_i$:

$$\begin{aligned} \zeta_{i_1} &= X_{i_1,t}'\boldsymbol{\beta} + \alpha y_{i_1,t-1} + \boldsymbol{\mu}_{i_1} + \xi_t \\ \zeta_{i_2} &= X_{i_2,t}'\boldsymbol{\beta} + \alpha y_{i_2,t-1} + \boldsymbol{\mu}_{i_2} + \xi_t \end{aligned}$$

Table 3 shows the summary statistics of the Bivariate FCPML estimates from model in equation (31). As we can observe, the estimation of α shows an improvement in terms of both unbiasedness and efficiency, whereas the autocorrelation parameter λ is heavily underestimated as much as ρ . The more the panel sample size increases the higher is the concentration of the α dynamic parameter around the true value, especially when the time dimension T is higher with fixed n. Slight differences can be found among the two cases with $N = 2,500$ and $N = 10,000$. In summary, the feasible estimator performs quite well with the subsampling procedure, showing robust estimates of the parameters of interest (α, β) in the presence of spatial error autocorrelation.

4.3 Comments on the Marginal Effects

This Section reports some comments on the Monte Carlo distributions of the short-term and long-term marginal effects defined in Section 2.1.

Figures 2, 3, 4, 5 show Gaussian Kernel densities for the (a) short term marginal effects and (b) long term marginal effects for different values of the cross-sectional dimension N and time dimension T. Purple and blue densities refer to the estimated marginal effects and true marginal effects, respectively. Purple and blue vertical lines reports their mean values, respectively. The number of MC replications equals to $1,000$ each.

Regardless of the N and T dimensions, the long-run effects seem to have larger variability around the mean values with respect to the short-term effects. The distributions are quite centerted around the true values, especially when considering the indirect effects for all the combinations of N and T. A sligh bias can be found in both the direct and total long-term effects, especially when $N = 100$.

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5 Conclusions

This paper aims at defining different strategies for the estimation of spatial dynamic panel data probit models in the presence of very large datasets. In particular, we proposed three different ML-based estimators and we study their asymptotic behaviour and their small sample properties through an extensive simulation exercise. All the three estimators perform well in terms of unbiasedness and efficiency and of their asymptotic behaviour.

More specifically, for the first estimator (the bivariate concentrated partial maximum likelihood estimators, or bivariate CPML), the estimates of the parameters of interest $\boldsymbol{\theta} = (\alpha, \rho, \beta)'$ are very good in terms of both unbiasedness and RMSE in finite samples. When $T = 4$ the slightly higher bias is likely due to the incidental parameter problem of the individual fixed effects. Results improve for all the parameters as the panel sample size increases, i.e. both the results improve for all the parameters as the panel sample size increases from $T = 4$ to $T = 10$ with N fixed (increased time dimension) and from $N = 100$ to $N = 400$ with fixed T (increased cross-sectional dimension). Similar conclusions can be drawn for the second estimator (the univariate concentrated partial maximum likelihood estimators, or univariate CPML), with only a small increase in the bias and in the RMSE. To tackle this problem, we introduce a bias-reduction procedure which, however, improves only marginally the quality of the estimation of the spatial correlation ρ, and in most of the cases also of the β and time coefficient α. For these two CPML estimators (both univariate and bivariate) we also provide a series of asymptotic results relying on the increasing domain case and under the assumption of near-epoch dependence.

In case we must deal with very large datasets the two preceding estimators may raise severe computational problems. For this reason, we also introduced a third estimator (the feasible concentrated partial maximum likelihood estimators, or FCPML) which is based on the analysis of sub-samples of data which are selected distant enough in space so as to be considered spatially independent. Also for this estimator results are very good for the β coefficient (which even outperforms the previous two estimators), but are worst in terms of an increased bias and RMSE for the spatial autoregressive and for the dynamic parameters ρ and α. Results do not depend on the cross-sectional dimension N of the original samples, therefore, increasing the distances between subsamples does not improve the estimation results, and it only matters selecting a sufficient large subsample size. A comparison with the other two partial ML approaches presented before shows that the larger bias is mainly due to the misspecification of the block diagonal weighting matrix, rather than to the subsampling procedure. Finally, when considering a spatial error version of the model specification, the estimation of α shows an improvement in terms of both bias and efficiency, whereas the autocorrelation parameter λ, as well as the parameter ρ, is heavily underestimated. In summary, in the case of spatial error autocorrelation, the feasible estimator performs quite well using the subsampling procedure, showing robust estimates of the parameters of interest (α, β).

Finally, we also introduce a novel way to measure short- and long-run marginal effects in a spatial dynamic probit panel data model, a result that fills an important gap for (spatial and non-spatial) dynamic probit models when the dynamic component is the time-lagged binary variable. Our results ultimately suggest the use of the univariate CPML approach, or the feasible FCPML estimator, when the bias in the estimation of non-regression parameters is irrelevant to the analysis which is often the case when using the spatial error model specification.

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A Tables

Bivariate CPMLE							
N	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	1.11188	1.10959	0.10418	0.15288	0.06519
100	10	$\rho = 0.6$	0.57604	0.58134	0.07890	0.08246	0.05441
100	10	$\alpha = 0.5$	0.52939	0.52078	0.08938	0.09409	0.05960
100	4	$\beta = 1$	1.21809	1.21255	0.16594	0.27404	0.11426
100	4	$\rho = 0.6$	0.55069	0.57114	0.14330	0.15154	0.09217
100	4	$\alpha = 0.5$	0.53857	0.52923	0.14681	0.15179	0.09786
400	10	$\beta = 1$	1.07138	1.06900	0.04670	0.08530	0.03265
400	10	$\rho = 0.6$	0.55783	0.55993	0.04219	0.05965	0.02839
400	10	$\alpha = 0.5$	0.54695	0.54673	0.04649	0.06608	0.03172
400	4	$\beta = 1$	1.16050	1.15962	0.07507	0.17719	0.05182
400	4	$\rho = 0.6$	0.56826	0.57364	0.06867	0.07565	0.04846
400	4	$\alpha = 0.5$	0.54098	0.54098	0.07993	0.08983	0.05190
Bivariate CPMLE (*W block diagonal*)							
N	T	True Value	Mean	Median	SD	RMSE	MAD
100	4	$\beta = 1$	1.08819	1.08527	0.13458	0.16090	0.09244
100	4	$\rho = 0.6$	0.12793	0.12822	0.05719	0.47552	0.03838
100	4	$\alpha = 0.5$	0.85406	0.85240	0.15565	0.38676	0.10396
Univariate CPMLE							
N	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	1.11522	1.11165	0.10850	0.15827	0.07066
100	10	$\rho = 0.6$	0.57098	0.57745	0.09584	0.10013	0.06519
100	10	$\alpha = 0.5$	0.53320	0.52919	0.10082	0.10615	0.07033
100	4	$\beta = 1$	1.22909	1.21718	0.18020	0.29147	0.12464
100	4	$\rho = 0.6$	0.54546	0.56510	0.17096	0.17945	0.12085
100	4	$\alpha = 0.5$	0.54084	0.52895	0.16050	0.16561	0.11078
400	10	$\beta = 1$	1.07133	1.06958	0.04746	0.08568	0.03324
400	10	$\rho = 0.6$	0.54834	0.54890	0.04780	0.07039	0.03318
400	10	$\alpha = 0.5$	0.55603	0.55663	0.05066	0.07554	0.03492
400	4	$\beta = 1$	1.16177	1.16032	0.07730	0.17929	0.05437
400	4	$\rho = 0.6$	0.56162	0.56787	0.08005	0.08877	0.05386
400	4	$\alpha = 0.5$	0.54617	0.54694	0.08818	0.09954	0.05781
Univariate CPMLE - Bias Reduction							
N	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	1.11032	1.10625	0.10895	0.15505	0.07114
100	10	$\rho = 0.6$	0.57217	0.57857	0.09543	0.09940	0.06504
100	10	$\alpha = 0.5$	0.53293	0.52902	0.10065	0.10590	0.07024
100	4	$\beta = 1$	1.21989	1.20858	0.18060	0.28455	0.12293
100	4	$\rho = 0.6$	0.54781	0.56729	0.16991	0.17774	0.11909
100	4	$\alpha = 0.5$	0.54105	0.52927	0.16046	0.16562	0.11139
400	10	$\beta = 1$	1.06680	1.06507	0.04769	0.08207	0.03346
400	10	$\rho = 0.6$	0.54946	0.54996	0.04762	0.06944	0.03291
400	10	$\alpha = 0.5$	0.55585	0.55636	0.05058	0.07535	0.03482
400	4	$\beta = 1$	1.15586	1.15409	0.07773	0.17417	0.05421
400	4	$\rho = 0.6$	0.56282	0.56900	0.07975	0.08799	0.05352
400	4	$\alpha = 0.5$	0.54658	0.54763	0.08800	0.09956	0.05783

Table 1: Summary statistics of the Bivariate and Univariate CPML estimates. The parameter vector $\boldsymbol{\theta} = (\beta, \rho, \alpha)' = (1, 0.6, 0.5)'$ is fixed, and the number of MC replications equals to $1,000$ each.

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Bivariate FCPMLE							
$N = 2,500$							
n	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	0.99953	0.99768	0.08477	0.08477	0.05731
100	10	$\rho = 0.6$	0.14516	0.14363	0.03616	0.45628	0.02341
100	10	$\alpha = 0.5$	0.94218	0.94025	0.07517	0.44852	0.05386
100	4	$\beta = 1$	1.07949	1.07356	0.13851	0.15970	0.09471
100	4	$\rho = 0.6$	0.12189	0.12068	0.05545	0.48131	0.03660
100	4	$\alpha = 0.5$	0.91552	0.91016	0.12951	0.43524	0.08892
400	10	$\beta = 1$	1.00137	1.00100	0.04320	0.04322	0.03001
400	10	$\rho = 0.6$	0.13587	0.13558	0.01815	0.46448	0.01277
400	10	$\alpha = 0.5$	0.87328	0.87212	0.03646	0.37505	0.02482
400	4	$\beta = 1$	1.09817	1.09473	0.06634	0.11849	0.04304
400	4	$\rho = 0.6$	0.12132	0.12058	0.02766	0.47948	0.01879
400	4	$\alpha = 0.5$	0.85809	0.85718	0.06396	0.36376	0.03875
$N = 10,000$							
n	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	0.98932	0.98896	0.08160	0.08229	0.05370
100	10	$\rho = 0.6$	0.14007	0.13966	0.03452	0.46122	0.02423
100	10	$\alpha = 0.5$	0.89841	0.89737	0.06392	0.40351	0.04307
100	4	$\beta = 1$	1.06880	1.07142	0.12541	0.14304	0.08418
100	4	$\rho = 0.6$	0.12934	0.12976	0.05491	0.47386	0.03636
100	4	$\alpha = 0.5$	0.88145	0.87663	0.11547	0.39854	0.07792
400	10	$\beta = 1$	1.00464	1.00409	0.04267	0.04292	0.02915
400	10	$\rho = 0.6$	0.12843	0.12883	0.02628	0.47230	0.01796
400	10	$\alpha = 0.5$	0.95180	0.95112	0.03798	0.45340	0.02402
400	4	$\beta = 1$	1.07607	1.07490	0.06511	0.10013	0.04581
400	4	$\rho = 0.6$	0.10462	0.10319	0.04166	0.49713	0.02795
400	4	$\alpha = 0.5$	0.92023	0.91795	0.06616	0.42540	0.04574

Table 2: Summary statistics of the Bivariate FCPML estimates from model in eqaution (29). The parameter vector $\boldsymbol{\theta} = (\beta, \rho, \alpha)' = (1, 0.6, 0.5)'$ is fixed, and the number of MC replications equals to $1,000$ each.

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Bivariate FCPMLE							
$N = 2,500$							
n	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	0.99875	0.99780	0.07778	0.07779	0.04884
100	10	$\lambda = 0.6$	0.10203	0.10134	0.04369	0.49989	0.02959
100	10	$\alpha = 0.5$	0.50685	0.50512	0.07130	0.07163	0.04711
100	4	$\beta = 1$	1.06737	1.06067	0.13314	0.14922	0.08537
100	4	$\lambda = 0.6$	0.10015	0.10031	0.07193	0.50500	0.04701
100	4	$\alpha = 0.5$	0.57264	0.57162	0.13173	0.15043	0.08940
400	10	$\beta = 1$	1.00814	1.00729	0.04121	0.04201	0.02733
400	10	$\lambda = 0.6$	0.10294	0.10263	0.02135	0.49752	0.01354
400	10	$\alpha = 0.5$	0.49382	0.49398	0.03627	0.03679	0.02381
400	4	$\beta = 1$	1.10837	1.10788	0.06499	0.12637	0.04499
400	4	$\lambda = 0.6$	0.10128	0.10230	0.03274	0.49979	0.02262
400	4	$\alpha = 0.5$	0.52240	0.52128	0.06574	0.06946	0.04542
$N = 10,000$							
n	T	True Value	Mean	Median	SD	RMSE	MAD
100	10	$\beta = 1$	0.99680	0.99276	0.07969	0.07975	0.05398
100	10	$\lambda = 0.6$	0.09988	0.10006	0.04143	0.50183	0.02758
100	10	$\alpha = 0.5$	0.49872	0.49683	0.06689	0.06690	0.04619
100	4	$\beta = 1$	1.07402	1.06889	0.12850	0.14829	0.08804
100	4	$\lambda = 0.6$	0.10352	0.10440	0.06560	0.50080	0.04296
100	4	$\alpha = 0.5$	0.51764	0.51526	0.12452	0.12576	0.07776
400	10	$\beta = 1$	1.01127	1.00869	0.03941	0.04099	0.02650
400	10	$\lambda = 0.6$	0.10207	0.10231	0.02108	0.49837	0.01496
400	10	$\alpha = 0.5$	0.50094	0.50108	0.03723	0.03724	0.02439
400	4	$\beta = 1$	1.08019	1.07983	0.06498	0.10322	0.04325
400	4	$\lambda = 0.6$	0.10157	0.10164	0.03432	0.49961	0.02322
400	4	$\alpha = 0.5$	0.52861	0.52759	0.06394	0.07005	0.04239

Table 3: Summary statistics of the Bivariate FCPML estimates from model in equation (31). The parameter vector $\boldsymbol{\theta} = (\beta, \lambda, \alpha)' = (1, 0.6, 0.5)'$ is fixed, and the number of MC replications equals to $1,000$ each.

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B Figures

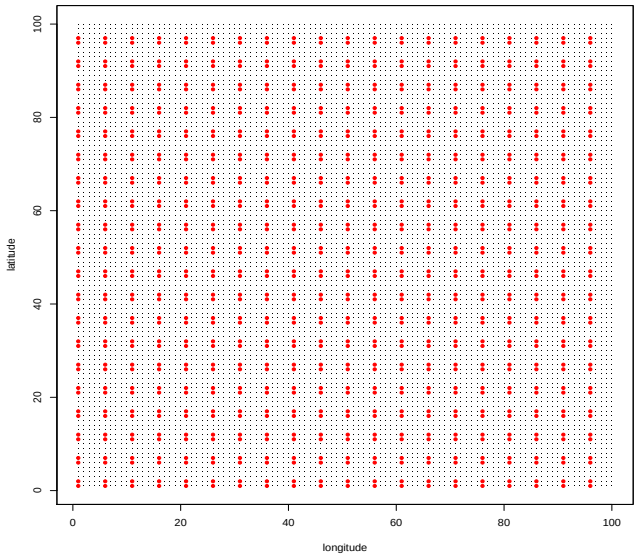


Figure 1: Systematic sampling of pairs of units from a regular square lattice grid.

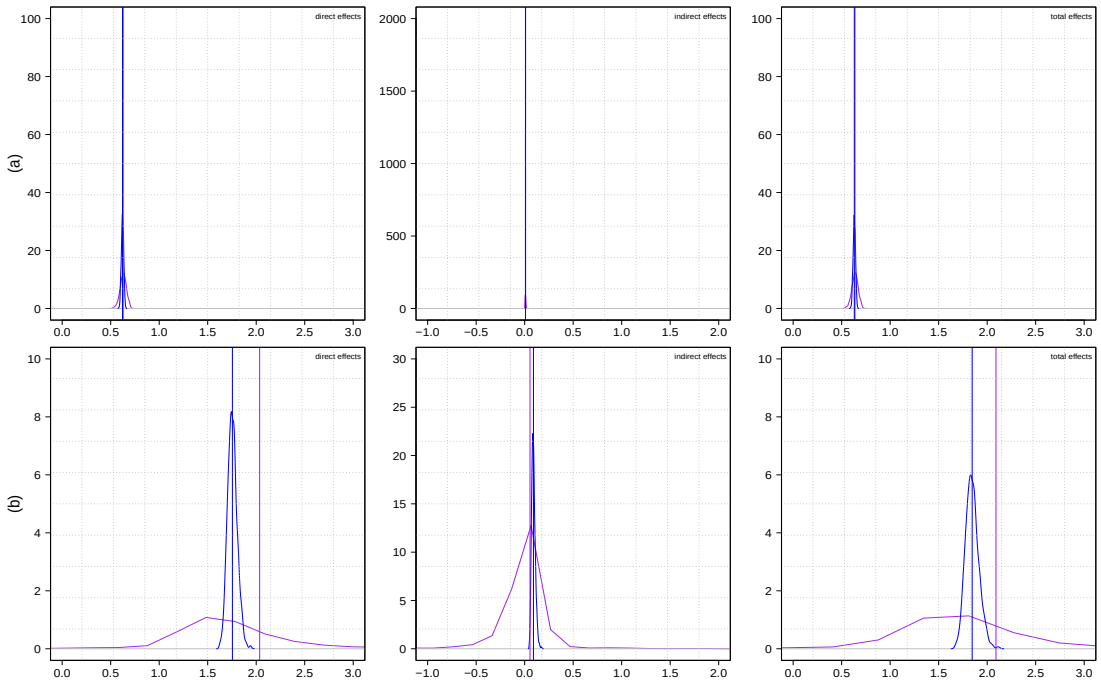


Figure 2: Gaussian Kernel densities for the (a) short term marginal effects and (b) long term marginal effects when $N = 100$ and $T = 4$. Purple and blue densities refer to the estimated marginal effects and true marginal effects, respectively. Purple and blue vertical lines reports their mean values, respectively. The number of MC replications equals to $1,000$ each.

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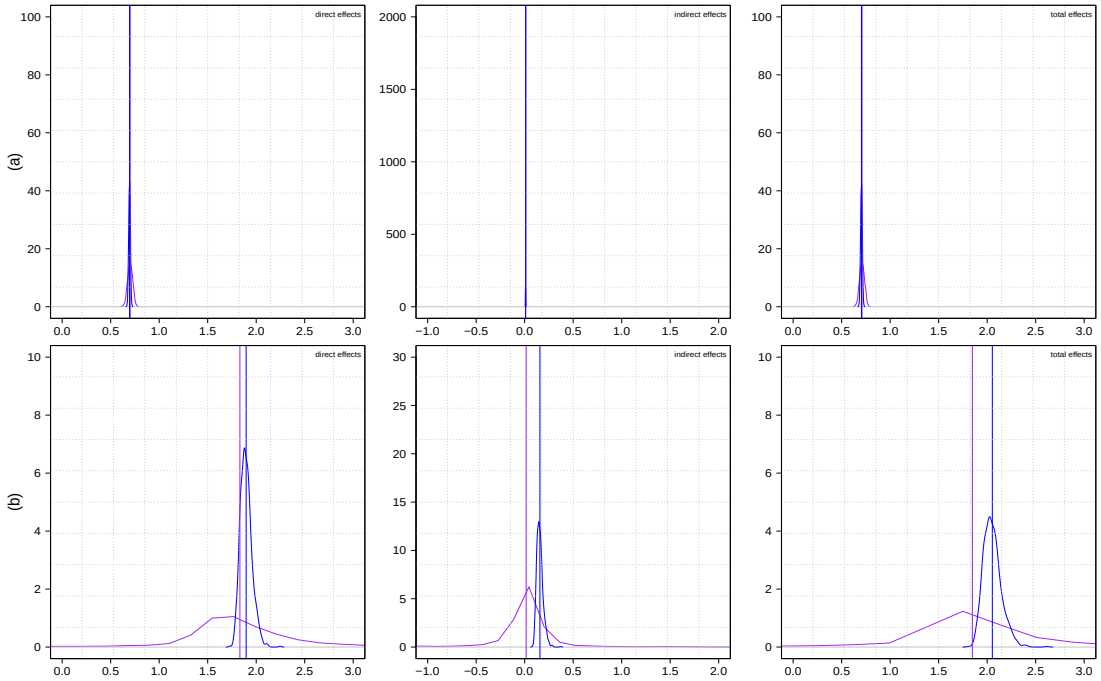


Figure 3: Gaussian Kernel densities for the (a) short term marginal effects and (b) long term marginal effects when $N = 100$ and $T = 10$. Purple and blue densities refer to the estimated marginal effects and true marginal effects, respectively. Purple and blue vertical lines reports their mean values, respectively. The number of MC replications equals to $1,000$ each.

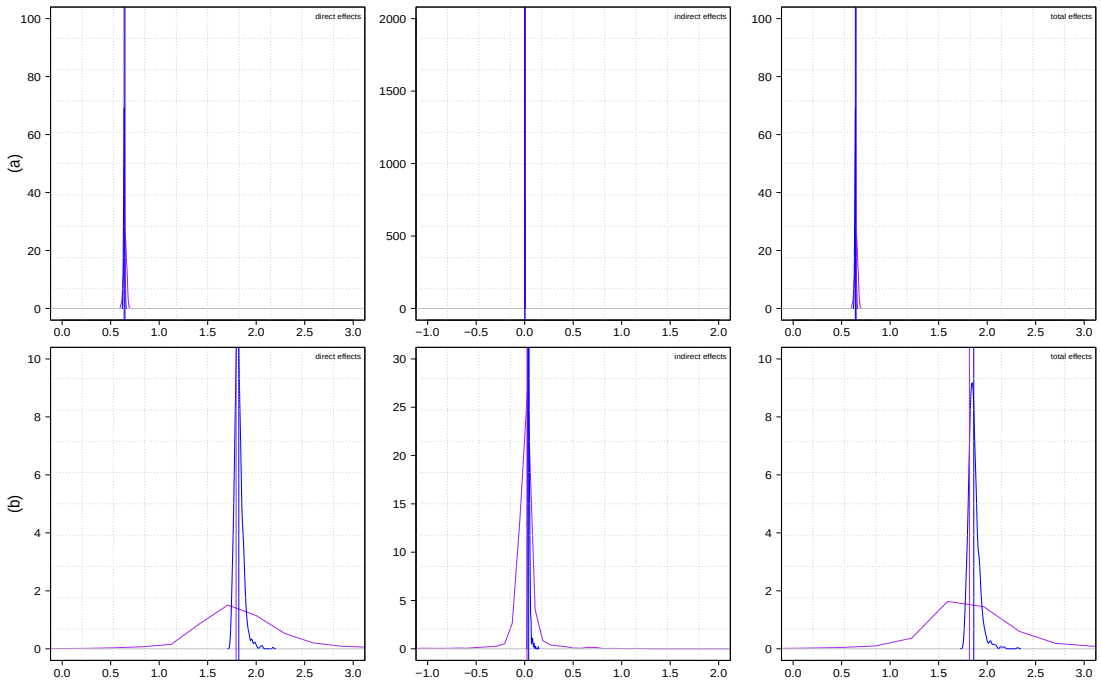


Figure 4: Gaussian Kernel densities for the (a) short term marginal effects and (b) long term marginal effects when $N = 400$ and $T = 4$. Purple and blue densities refer to the estimated marginal effects and true marginal effects, respectively. Purple and blue vertical lines reports their mean values, respectively. The number of MC replications equals to $1,000$ each.

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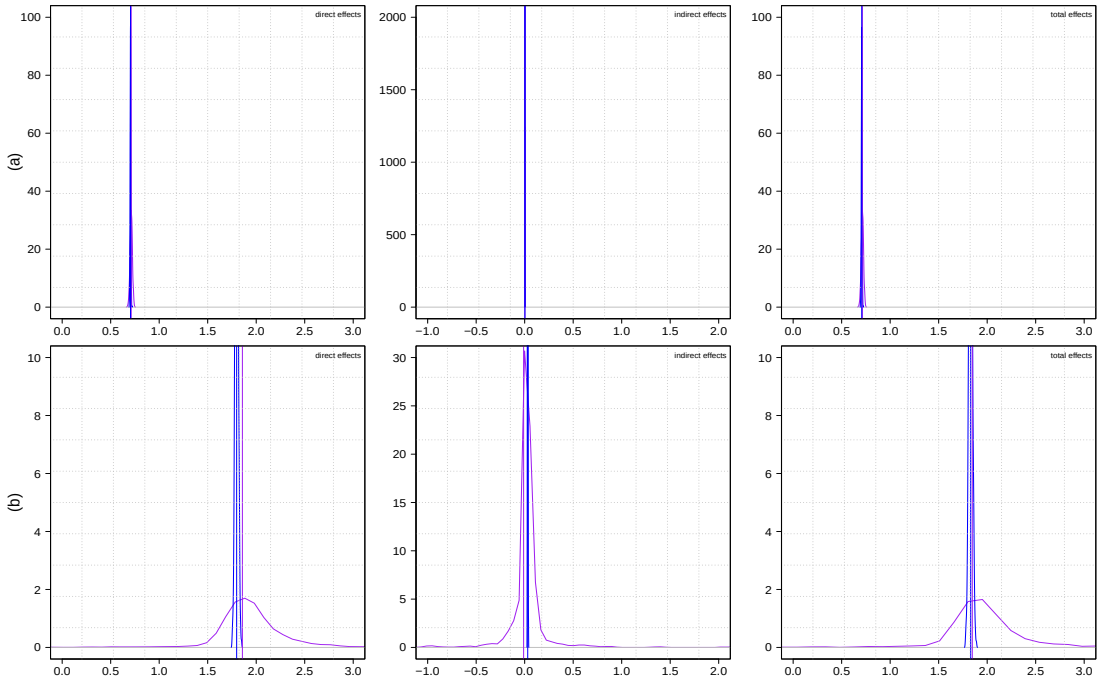


Figure 5: Gaussian Kernel densities for the (a) short term marginal effects and (b) long term marginal effects when $N = 400$ and $T = 10$. Purple and blue densities refer to the estimated marginal effects and true marginal effects, respectively. Purple and blue vertical lines reports their mean values, respectively. The number of MC replications equals to $1,000$ each.

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C Proofs and Auxiliary Lemmas

Proof of Theorem 1

We want to prove that $Z_{l,L}$ is a L_2-NED random field, where for every $l = l(i,t)$,

$$Z_{l,L} = y_{it} = 1\{\varepsilon_{\rho,it} \geq -X'_{\rho,it}\boldsymbol{\beta} - \mu_{\rho,i} - \alpha y_{\rho,\, i,(t-1)}\},$$

where $\boldsymbol{\varepsilon}_{\rho,t} = (I-\rho W)^{-1}\boldsymbol{\varepsilon}_t$, $\boldsymbol{X}_{\rho,t} = (I-\rho W)^{-1}\boldsymbol{X}_t$ $\boldsymbol{y}_{\rho,t-1} = (I-\rho W)^{-1}\boldsymbol{y}_{t-1}$, and so on.

Assuming the $\boldsymbol{X}_{NT}$ are exogenous, we can consider all expectations as conditional on the whole matrix $\boldsymbol{X}_{NT}$. The explicit reference to conditioning to the exogenous variables is however omitted for the ease of notation.

We need to prove that, for all s,

$$\mathbb{E}\left(Z_l.\mathbb{E}(Z_l \mid \mathscr{F}_l(s))\right)^2 \leq c_l^2\psi(s)^2.$$

Since y_{it} is a binary variable,

$$\begin{aligned}\mathbb{E}\left(Z_l.\mathbb{E}(Z_l \mid \mathscr{F}_l(s))\right)^2 &= \mathbb{E}(y_{it} - \mathbb{E}(y_{it} \mid \mathscr{F}_{l(i,t)}(s))^2 = \mathbb{E}\left(Var\left(y_{it} \mid \mathscr{F}_{l(i,t)}(s)\right)\right)\\ &= \mathbb{E}\left[\Pr\left(y_{it}=1 \mid X_{NT}, \mathscr{F}_{l(i,t)}(s)\right)\left(1-\Pr\left(y_{it}=1 \mid X_{NT}, \mathscr{F}_{l(i,t)}(s)\right)\right)\right]\end{aligned}$$

We can then focus on the probability

$$\Pr\left(y_{it}=1 \mid X_{NT}, \mathscr{F}_{l(i,t)}(s)\right) = \Pr\left(\varepsilon_{\rho,it} \geq -b_{it} - \alpha y_{\rho,i,(t-1)} \mid \mathscr{F}_l(s)\right),$$

where the term $b_{it} = X'_{\rho,it}\boldsymbol{\beta} + \mu_{\rho,i}$ includes all constants and exogenous terms.

Note that $\varepsilon_{\rho,it} = \sum_{j=1}^N a_N(i,j)\varepsilon_{jt}$ and $y_{\rho,i,(t-1)} = \sum_{j=1}^N a_N(i,j)y_{j,t-1}$, with the $a_N(i,j)$ as in Assumption 5. We can further decompose $\varepsilon_{\rho,it}$ as:

$$\sum_{j=1}^N a_N(i,j)\varepsilon_{jt} = \sum_{j:\|l(i,0)-l(j,0)\|>s} a_N(i,j)\varepsilon_{jt} + \sum_{j:\|l(i,0)-l(j,0)\|\leq s} a_N(i,j)\varepsilon_{jt}$$

and note that, conditional to $\mathscr{F}_{l(i,t)}(s)$, the first term only is random.

Note that the value of all the endogenous variables $y_{j,t-1}$ depends on all $\varepsilon_{j,t-1}$, belonging both to the group such that $\|l(j,t-1)-l(i,l)\| \leq s$ and $\|l(j,t-1)-l(i,l)\| > s$, therefore, it is not useful to decompose the endogenous variable in a similar way.

Assuming all $\varepsilon_{it} \sim \mathscr{N}(0,1)$, we have

$$\sum_{j:\|l(i,0)-l(j,0)\|>s} a_N(i,j)\varepsilon_{jt} \sim \mathscr{N}\left(0, \sum_{j:\|l(i,0)-l(j,0)\|>s} a_N(i,j)^2\right).$$

We denote $a_s = \sum_{j:\|l(i,0)-l(j,0)\|\leq s} a_N(i,j)^2$, $a = \sum_{j=1}^N a_N(i,j)^2$ and $\bar{a}_s = a - a_s$ (for convenience, all reference to the $i-$th row of A_ρ^{-1} and to the ρ coefficient is omitted) and write $b'_{it} = b_{it} + \sum_{j:\|l(i,0)-l(j,0)\|\leq s} a_N(i,j)\varepsilon_{jt}$.

We now marginalize w.r. to $y_{j,t-1}$, $j = 1,\ldots,N$:

$$\begin{aligned}&\mathbb{E}\left(Var\left(y_{it} \mid \mathscr{F}_{l(i,t)}(s)\right)\right)\\ &= \mathbb{E}\left[\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^N a_N(i,j)y_{j,t-1}}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\left(1-\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^N a_N(i,j)y_{j,t-1}}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\right)\right]\\ &= \mathbb{E}\sum_{y_1,\cdots,y_N}\Pr\left(\boldsymbol{y}_{t-1}=(y_1,\ldots,y_N)\right)\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^N a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\\ &\qquad\times\Pr\left(\varepsilon < -\frac{b'_{it}+\alpha\sum_{j=1}^N a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\\ &= \mathbb{E}\sum_{y_1,\cdots,y_N}\Pr\left(\boldsymbol{y}_{t-1}=(y_1,\ldots,y_N)\right)\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^N a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\\ &\qquad\times\Pr\left(\varepsilon > \frac{b'_{it}+\alpha\sum_{j=1}^N a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\end{aligned} \tag{32}$$

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Now, note that

$$\begin{aligned}
&\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\Pr\left(\varepsilon > \frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\\
&\quad\leq \min\left\{\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right),\Pr\left(\varepsilon > \frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\right\}\\
&\quad= \Pr\left(\varepsilon > \left|\frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\right|\right)
\end{aligned}$$

Then, we can bound the right-hand-side of (32):

$$\begin{aligned}
&\mathbb{E}\sum_{y_1,\cdots,y_N}\Pr\big(\boldsymbol{y}_{t-1}=(y_1,\ldots,y_N)\big)\Pr\left(\varepsilon \geq -\frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\\
&\qquad\qquad\times\Pr\left(\varepsilon > \frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\,\middle|\,\mathscr{F}_l(s)\right)\\
&\quad\leq \sum_{y_1,\cdots,y_N}\Pr\big(\boldsymbol{y}_{t-1}=(y_1,\ldots,y_N)\big)\mathbb{E}\Pr\left(\varepsilon > \left|\frac{b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j}{\sqrt{a-a_s}}\right|\right)\\
&\quad\leq \text{using the MGF inequality } P(Z>a)\leq e^{-a^2/2}\\
&\quad\leq \sum_{y_1,\cdots,y_N}\Pr\big(\boldsymbol{y}_{t-1}=(y_1,\ldots,y_N)\big)\mathbb{E}\left(\exp\left\{-\frac{(b'_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j)^2}{a-a_s}\right\}\right)
\end{aligned}$$

We can exploit the fact that $b'_{it}=\sum_{j:\|l(i,0)-l(j,0)\|\leq s}a_N(i,j)\varepsilon_{jt}+b_{it}$, and the expectation is made with respect to the r.v.

$$Z := \sum_{j:\|l(i,0)-l(j,0)\|\leq s}a_N(i,j)\varepsilon_{jt}\sim\mathscr{N}(0,a_s),$$

with the vector $\boldsymbol{y}=(y_1,\ldots,y_N)$ and the terms b_{it} being fixed. We rename for convenience $\gamma_z=b_{it}+\alpha\sum_{j=1}^{N}a_N(i,j)y_j$ and easily obtain:

$$\begin{aligned}
\mathbb{E}\exp\left\{-\frac{(Z-\gamma_z)^2}{2(a-a_s)}\right\} &= \int_{\mathbb{R}}\frac{1}{\sqrt{2\pi a_s}}e^{-\frac{(z-\gamma_z)^2}{2(a-a_s)}}e^{-z^2/a_s}dz\\
&= \int_{\mathbb{R}}\frac{1}{\sqrt{2\pi a_s}}\exp\left\{-\frac{a\,z^2}{2a_s(a-a_s)}-\frac{2\gamma_z z+\gamma_z^2}{2(a-a_s)}\right\}dz\\
&= e^{-\frac{\gamma_z^2}{2a}}\int_{\mathbb{R}}\frac{1}{\sqrt{2\pi a_s}}e^{-\frac{1}{2}\left(z\sqrt{\frac{a}{a_s(a-a_s)}}-\gamma_z\sqrt{\frac{a_s}{a(a-a_s)}}\right)^2}dz\\
&\quad\text{(applying the transformation } v=z\sqrt{a/a_s(a-a_s)})\\
&= e^{-\frac{\gamma_z^2}{2a}}\sqrt{\frac{a-a_s}{a}}\int_{\mathbb{R}}\frac{1}{\sqrt{2\pi}}e^{-\left(v-\gamma_z\sqrt{\frac{a_s}{a(a-a_s)}}\right)^2/2}dv\\
&= e^{-\frac{\gamma_z^2}{2a}}\sqrt{\frac{a-a_s}{a}}
\end{aligned}$$

because the integrand is the density of a $\mathscr{N}\left(\gamma_z\sqrt{a_s/a(a-a_s)},1\right)$ random variable. Finally, since $e^{-\frac{\gamma_z^2}{2a}}\leq 1$, we have obtained:

$$\mathbb{E}(y_{it}-\mathbb{E}(y_{it}\mid\mathscr{F}_{l(i,t)}(s))^2\leq\sum_{y_{j,t-1},j\leq N}\Pr\big(\boldsymbol{y}_{t-1}=(y_1,\ldots,y_N)\big)\sqrt{\frac{a-a_s}{a}}=\sqrt{\frac{a-a_s}{a}}.$$

From Assumption 5, we have therefore that

$$\left[\mathbb{E}(y_{it}-\mathbb{E}(y_{it}\mid\mathscr{F}_{l(i,t)}(s))^2\right]^{1/2}\leq(c_0\psi(s))^{1/4}.$$

□

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Proof of Proposition 1

We need to prove the claim of Section 3.2: conditions (16) and (17) imply

$$|q_{\kappa,l}(y) - q_{\kappa,l}(y')| \leq M_{k,l}(y, y')\|y - y'\|,$$

where $M_{\kappa,l}$ is such that

$$\mathbb{E}M_{\kappa,l}\left(y_{it}, \mathbb{E}(y_{it} \mid \mathscr{F}_l(s))\right)^{2+\epsilon} < \infty.$$

We will use the trivial bound

$$|q_{\kappa,l}(y) - q_{\kappa,l}(y')| \leq \left(|q_{\kappa,l}(y)| + \left|q_{\kappa,l}(y')\right|\right)\|y - y'\| \leq 2 \sup_{y_{it}=\{0,1\}} |q_{\kappa,l}(y)|\,\|y - y'\|$$

posing

$$M_{\kappa,l}(y, y') = 2 \sup_{y_{it}=\{0,1\}} |q_{\kappa,l}(y)|\,.$$

First, because of boundedness of y_{it}, we immediately have that $\|y - \mathbb{E}(y_{it} \mid \mathscr{F}_l(s))\|_r < \infty$ for all $r > 0$. Then, using the multivariate Faà di Bruno formula for the derivatives of $\log \Pr(y_{it} \mid X_{Nt}, \boldsymbol{\theta}, \mu_l)$ with respect to $(\boldsymbol{\theta}, \mu_l)$ (we refer to Hardy (2006) for the notation):

$$\begin{aligned}
&\mathbb{E}\left|M_{\kappa,l}\left(y_{it}, \mathbb{E}(y_{it} \mid \mathscr{F}_l(s))\right)\left(y - \mathbb{E}(y_{it} \mid \mathscr{F}_l(s))\right|^{2+\epsilon} \\
&\quad \leq \mathbb{E}\left|M_{\kappa,l}\left(y_{it}, \mathbb{E}(y_{it} \mid \mathscr{F}_l(s))\right)\right|^{2+\epsilon} \\
&\quad = \mathbb{E}\left(2 \sum_{\pi \in \Pi_{|\kappa|}} \prod_{B \in \pi} \left(\frac{\Pr^{(B)}(y_{it} \mid X_{NT}, y_{i,t-1}, \boldsymbol{\theta}, \mu_l)}{\Pr(y_{it} \mid X_{NT}, y_{i,t-1}, \boldsymbol{\theta}, \mu_l)}\right)\right)^{(2+\epsilon)} \\
&\quad \leq 2^{(2+\epsilon)(|\kappa|+1)} \max_{1 \leq B \leq \kappa} \mathbb{E}\left(\frac{\Pr^{(B)}(y_{it} \mid X_{NT}, y_{i,t-1}, \boldsymbol{\theta}, \mu_l)}{\Pr(y_{it} \mid X_{NT}, y_{i,t-1}, \boldsymbol{\theta}, \mu_l)}\right)^{|\kappa|(2+\epsilon)} \\
&\quad \leq 2^{(12+6\epsilon)} \max_{1 \leq B \leq \kappa} \mathbb{E}\left(\frac{\Pr^{(B)}(y_{it} \mid X_{NT}, y_{i,t-1}, \boldsymbol{\theta}, \mu_l)}{\Pr(y_{it} \mid X_{NT}, y_{i,t-1}, \boldsymbol{\theta}, \mu_l)}\right)^{10+5\epsilon} < \infty
\end{aligned}$$

Moreover, if Assumption 5 also holds, then $\|y - \mathbb{E}(y_{it} \mid \mathscr{F}_l(s))\|_2 < c_0^{1/4}\psi(s)^{1/4} = o\left(s^{-\lambda/4}\right)$ i.e. the process y_l is L_2-NED on ε with size $-\lambda/4$.

We have verified all assumptions of Theorem 17.16 in Davidson (1994) (or Proposition 3 in Jenish and Prucha (2012)), with $q = 2 = q/(q-1)$ and $r = 2+\epsilon$ and then can conclude that all sequences q_κ are L_2-NED on ε with size $-\lambda\epsilon/8(1+\epsilon)$ with scaling coefficients $c_0^{\epsilon/8(1+\epsilon)} \cdot C$, where

$$C = \max\{1, \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)\}$$

noting that, because of $\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^2 \leq \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)^{2/(2+\epsilon)}$, then

$$\begin{aligned}
&\left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^2\right)^{(2+\epsilon-2)/(2+2\epsilon)} \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)^{(2+\epsilon)/(2+2\epsilon)} \\
&\quad \leq \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)^{2\epsilon/[(2+2\epsilon)(2+\epsilon)]} \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)^{(2+\epsilon)/(2+2\epsilon)} \\
&\quad = \left(\mathbb{E}\left|2 \cdot q_{\kappa,l}(y_{it})\right|^{2+\epsilon}\right)^{1-\epsilon^2/[(2+\epsilon)(2+2\epsilon)]} \leq C.
\end{aligned}$$

□

Some Lemmas

Lemma 2 *If Assumption 6 holds, then, for all $\kappa = (\kappa_1, \kappa_2)$, $|\kappa| \leq 4$, the following Lipschitz condition is satisfied:*

$$|q_{\kappa,l}(y_{it}; \boldsymbol{\theta}, \mu_l) - q_{\kappa,l}(y_{it}; ; \boldsymbol{\theta}', \mu_l)| \leq B_{k,l}\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|$$

where $B_{k,l}$ is a random variable independent on $\boldsymbol{\theta}$. Further, for all κ with $|\kappa| \leq 4$, the random sequence $q_{\kappa,l}$ is stochastic equicontinuous, i.e.

$$\limsup_{N,T} \frac{1}{NT} \sum_{l \in D_L} \Pr\left\{\sup_{(\boldsymbol{\theta}, \boldsymbol{\theta}', \mu_l):\, \|\boldsymbol{\theta}' - \boldsymbol{\theta}\| \leq \delta} \left|q_{\kappa,l}(y_{it}; \boldsymbol{\theta}, \mu_l) - q_{\kappa,l}(y_{it}; \boldsymbol{\theta}', \mu_l)\right| > \epsilon\right\} \to 0 \quad \text{as } \delta \to 0. \tag{33}$$

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Proof of Lemma 2

Under Assumption 6, if we consider that $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_{k+2})$ and $\boldsymbol{\theta}' = (\theta'_1, \ldots, \theta'_{k+2})$ are $(k+2)-$dimensional parameter vectors (we omit the reference to μ_l in the function $q_{\kappa,l}$, because it remains constant):

$$\begin{aligned}\left|q_{\kappa,l}(y_{it};\boldsymbol{\theta}) - q_{\kappa,l}(y_{it};\boldsymbol{\theta}')\right| \leq{}& \left|q_{\kappa,l}(y_{it};\theta_1,\theta_2,\ldots,\theta_{k+2}) - q_{\kappa,l}(y_{it};\theta'_1,\theta_2,\ldots,\theta_{k+2})\right| \\ &+ \left|q_{\kappa,l}(y_{it};\theta'_1,\theta_2,\ldots,\theta_{k+2}) - q_{\kappa,l}(y_{it};\theta'_1,\theta'_2,\ldots,\theta_{k+2})\right| \\ &+ \ldots + \left|q_{\kappa,l}(y_{it};\theta'_1,\theta'_2,\ldots,\theta'_{k+1},\theta_{k+2}) - q_{\kappa,l}(y_{it};\theta'_1,\ldots,\theta_{k+1},\theta'_{k+2})\right| \\ \leq{}& \sup_{\boldsymbol{\theta}^*\in\Theta} \sum_{j=1}^{k+2} \left|\frac{\partial}{\partial\theta_j} q_{\kappa,l}(y_{it};\boldsymbol{\theta}^*)\right| \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|\end{aligned}$$

where

$$B_{\kappa,l} = \sup_{\boldsymbol{\theta}\in\Theta} \sum_{j=1}^{k+2} \left|\frac{\partial}{\partial\theta_j} q_{\kappa,l}(y_{it};\boldsymbol{\theta})\right|$$

is a random variable independent on $\boldsymbol{\theta}$, therefore, the first part of the lemma is proven.

Further, Assumption 6 also implies that each $B_\kappa = \{B_{\kappa,l}\}_l$, satisfies

$$\sup_{N,T}\sup_{l\in D_L} \mathbb{E}B^2_{\kappa,l} < \infty.$$

Then, we can apply Proposition 1 of Jenish and Prucha (2009), that implies stochastic equicontinuity in the sense specified by Definition 3(a) in Jenish and Prucha (2009).

□

Lemma 3 *Under assumptions 1 – 6, for all derivatives of $\ell_i(\boldsymbol{\theta},\mu_i)$ up to the fourth order, such as $s_{\mu\mu,i}, s_{\theta\mu\mu,i}, s_{\mu\mu\mu,i}$, satisfy a pointwise and uniform law of large numbers, that is (w.l.o.g., we write the result for $s_{\mu\mu,i}$), namely, for all μ_i,*

$$(i) \qquad \frac{1}{N}\sum_i \mathbb{E}\left|\frac{1}{T}s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i) - \frac{1}{T}\mathbb{E}s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i)\right| \to 0$$

$$(ii) \qquad \sup_\theta \mathbb{E}\left|\frac{1}{NT}\sum_i s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i) - \frac{1}{NT}\sum_i \mathbb{E}s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i)\right| \to 0$$

$$(iii) \qquad \limsup_{N,T}\sup_{\boldsymbol{\theta}'}\sup_{|\boldsymbol{\theta}-\boldsymbol{\theta}'|<\delta}\left|\frac{1}{NT}\sum_i \mathbb{E}s_{\mu\mu,i}(\boldsymbol{\theta}',\mu_i) - \frac{1}{NT}\sum_i \mathbb{E}s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i)\right| \overset{\delta\to 0}{\to} 0$$

Proof of Lemma 3

Assumptions 5 and 6 guarantee the L_2-ned property of the derivatives up to the 4th order, $q_{\kappa,l}$, $|\kappa| \leq 4$. Further, Assumption 6 is enough to guarantee not only stochastic equicontinuity, but also uniform integrability of $q_{\kappa,l}$, because of $\mathbb{E}|2q_{\kappa,l}|^{2+\varepsilon}(y_{it},\boldsymbol{\theta}) < \infty$ proven in Proposition 1:

$$\sup_{N,T}\sup_{l\in D_L} \mathbb{E}|q_{\kappa,l}(y_{it};\boldsymbol{\theta})|^{2+\epsilon} < \infty.$$

This is sufficient to apply Theorem 1 in Jenish and Prucha (2012):

$$\frac{1}{NT}\sum_{i,t}\left(q_{\kappa,l}(y_{it};\boldsymbol{\theta}) - \mathbb{E}\, q_{\kappa,l}(y_{it};\boldsymbol{\theta})\right) \to 0 \quad \text{in } L_1 \text{ norm}$$

that is equivalent to

$$\frac{1}{NT}\sum_{i}\left(s_{\cdot,i}(\boldsymbol{\theta},\mu_i(\boldsymbol{\theta})) - \mathbb{E}\, s_{\cdot,i}(\boldsymbol{\theta},\mu_i(\boldsymbol{\theta}))\right) \to 0 \quad \text{in } L_1 \text{ norm}$$

with $s_{\cdot,i}$ indicating one of the up to 4th order derivatives, such as $s_{\mu\mu,i}$, $s_{\theta\mu,i}$, and so on, and therefore implies (i).

We can further note, by repeating arguments similar to those in the proof of Proposition 1 that Assumption 6 is strong enough to imply not only uniform L_p-integrability of $q_{\kappa,l}$, but also of $\sup_{N,T}\sup_{l\in D_L} d_{l,L}$, for $p \geq 1$, where

$$d_{l,L} := \sup_{\boldsymbol{\theta}\in\Theta} |q_{\kappa,l}(y_{it},\boldsymbol{\theta})|,$$

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thus implying the *domination condition* of Jenish and Prucha (2009). Then, we can apply Theorem 2 of Jenish and Prucha (2009) concluding that

$$\sup_{\boldsymbol{\theta}} \left| (NT)^{-1} \sum_{l \in D_L} q_{\kappa,l}(y_{it};\boldsymbol{\theta}) - (NT)^{-1} \sum_{l \in D_L} \mathbb{E} q_{\kappa,l}(y_{it};\boldsymbol{\theta}) \right| \to_L 0 \quad \text{in prob.}$$

that is equivalent to claim (ii). Further, also from Theorem 2 of Jenish and Prucha (2009), uniform equicontinuity of the expectations $(NT)^{-1} \sum_{l \in D_L} \mathbb{E} q_{\kappa,l}(y_{it};\boldsymbol{\theta})$ (claim (iii)) follows.

Lemma 4 *If assumptions 1 – 8 are satisfied, then, for all* $|\kappa| \leq 4$*:*

$$\frac{1}{\sqrt{NT}} \sum_i \sum_t \left(q_{\kappa,l}(y_{it};\boldsymbol{\theta}_0,\mu_{i0}) - \mathbb{E} q_{\kappa,l}(y_{it};\boldsymbol{\theta}_0,\mu_{i0}) \right) \to \mathcal{N}(\mathbf{0}, \lim_{N,T} J_{NT}^{(\kappa)}). \tag{34}$$

Proof of Lemma 4 Assumption 8 (i) and (ii) corresponds to Assumption 4.b,4.c,4.d of Jenish and Prucha (2012). Assumption 4.a follows from Assumption 6, that yields the condition of integrability in $L_{2+\epsilon}$ of all $q_{\kappa,l}$, $|\kappa| \leq 4$. Finally, Assumption 3 of Jenish and Prucha (2012) is trivially satisfied because we assume $\{\varepsilon_l\}_{l \in D_L}$ to be an iid sequence.

Thus, Theorem 2 by Jenish and Prucha (2012) immediately gives (42).

□

Proof of Theorem 2

We first prove that

$$\Pr\left\{ \max_i \sup_{(\boldsymbol{\theta},\mu_i)} |\ell_i(\boldsymbol{\theta},\mu_i) - \mathbb{E}\ell_i(\boldsymbol{\theta},\mu_i)| \geq \eta \right\} = o(N/T^2), \tag{35}$$

following the same steps as Lemma 4 in Hahn and Kuersteiner (2011). A small difference with respect to their proof is that we define $\delta > 0$ such that $2\delta \max_{i,t} E \log p_{it}(y_{it}) < \eta/3$ and then find a partition $\Gamma_i, \ldots, \Gamma_j, \ldots, \Gamma_{N_\delta}$ of the parameter space, such that $\|(\boldsymbol{\theta},\mu) - (\boldsymbol{\theta}',\mu')\| < \delta/\sqrt{2}$, if both $(\boldsymbol{\theta},\mu) \in \Gamma_j$ and $(\boldsymbol{\theta}',\mu') \in \Gamma_j$. In this way, using Cauchy inequality,

$$\begin{aligned} \max\left\{ \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|, \|\mu - \mu'\| \right\} &\leq 2\frac{\|\boldsymbol{\theta} - \boldsymbol{\theta}'\| + \|\mu - \mu'\|}{2} < 2\sqrt{\frac{\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|^2}{2} + \frac{\|\mu - \mu'\|^2}{2}} \\ &= \sqrt{2}\|(\boldsymbol{\theta},\mu) - (\boldsymbol{\theta}',\mu')\| < \delta. \end{aligned}$$

Then, the steps in the proof of Theorem 3 of Hahn and Kuersteiner (2011) can be followed, taking into account that their $o(T^{-1})$ is replaced here by $o(N/T^2)$.

Proof of Theorem 3

(i)

$$|\hat{\mu}(\hat{\boldsymbol{\theta}}) - \mu_{i0}| \leq |\hat{\mu}(\hat{\boldsymbol{\theta}}) - \bar{\mu}(\hat{\boldsymbol{\theta}})| + |\bar{\mu}(\hat{\boldsymbol{\theta}}) - \mu_{i0}|.$$

The term $|\bar{\mu}(\hat{\boldsymbol{\theta}}) - \mu_{i0}| = |\bar{\mu}(\hat{\boldsymbol{\theta}}) - \bar{\mu}(\boldsymbol{\theta}_0)| = o_p(1)$ follows from Theorem 2. Similarly, the first term tends to zero in probability as a consequence of the uniform convergence (35) and of the well separetedness condition in Assumption 7.(ii), that together yield $\max_i \sup_\theta |\hat{\mu}_i(\boldsymbol{\theta}) - \bar{\mu}_i(\boldsymbol{\theta})| = o_p(1)$.

(ii) Since the asymptotic expansion and the derivation of the bias in the univariate case can be obtained by combining the computations of Carro (2007) and the asymptotic theorems for near epoch dependence, we give only the computations for the more interesting bivariate expansion.

Let $s_{\theta,0}, s_{\theta,\mu_i,\mu_j,0}$ (and so on), denote the partial derivatives computed at $(\boldsymbol{\theta}_0, \boldsymbol{\mu}_0) = (\boldsymbol{\theta}_0, \mu_{01}, \mu_{02})$. For notational convenience, we do not refer explicitly to the unit index in all the score functions (as in $s_{\mu,i}$ for the univariate case) but add an index to the component of the fixed effects' vector relative to which the derivative is taken (s_{θ,μ_i,μ_j}, for example).

We point out that Lemma 4 implies that the score $\frac{1}{\sqrt{T}} s_{\mu_i}(\boldsymbol{\theta}_0, \mu_{01}, \mu_{02})$ as well as the other derivatives (appropriately recentered), such as $\sqrt{T}\left(\frac{s_{\mu_j,\mu_l,0}}{T} - \mathbb{E}\frac{s_{\mu_j,\mu_l,0}}{T} \right)$, are asymptotically Gaussian and, therefore $O_p(1)$ terms.

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Using equation (7.10) in McCullagh (1987), we write then, for $i = 1, 2$:

$$\begin{aligned}\sqrt{T}(\hat{\mu}_i(\boldsymbol{\theta}_0) - \mu_{0i}) = &-\sum_{j=1}^{2} \frac{s_{\mu_j,0}/\sqrt{T}}{I_{ij}} + \frac{1}{\sqrt{T}} \sum_{j=1}^{2}\sum_{k=1}^{2}\sum_{l=1}^{2} \frac{Z_{jl}\, s_{\mu_k,0}/\sqrt{T}}{I_{ij} I_{lk}} \\ &+ \frac{1}{2\sqrt{T}} \sum_{j}\sum_{l}\sum_{r,s,u} s_{\mu_j,0} s_{\mu_l,0}/T\, I_{ir}^{-1} I_{js}^{-1} I_{lu}^{-1} I_{rsu} + O_p(T^{-1})\end{aligned} \tag{36}$$

where we set $Z_{jl} = \sqrt{T}\left(\frac{s_{\mu_j,\mu_l,0}}{T} - \mathbb{E}\frac{s_{\mu_j,\mu_l,0}}{T}\right)$ and $I_{jl} = \mathbb{E}\frac{s_{\mu_j,\mu_l,0}}{T}$, $I_{rsu} = \mathbb{E}\frac{s_{\mu_r,\mu_s,\mu_u,0}}{T}$, and so on. The remainder term is the product of T^{-1} and a combination of $O_p(1)$ quantities such as I_{rsu} and Z_{il}.

From pat (i), we have that $\|\hat{\mu}_i(\boldsymbol{\theta}_0) - \mu_{i0}\| \to_T 0$.

Next, we write the Taylor expansion of the score s_θ:

$$\begin{aligned}s_\theta(\boldsymbol{\theta}_0, \hat{\boldsymbol{\mu}}(\boldsymbol{\theta}_0)) = &\, s_{\theta_0} + (\hat{\mu}_1(\boldsymbol{\theta}_0) - \mu_{01}) s_{\theta\mu_1,0} + (\hat{\mu}_2(\boldsymbol{\theta}_0) - \mu_{02}) s_{\theta\mu_2,0} \\ &+ \frac{1}{2}\sum_{j=1}^{2}\sum_{k=1}^{2} (\hat{\mu}_j(\boldsymbol{\theta}_0) - \mu_{0j})(\hat{\mu}_k(\boldsymbol{\theta}_0) - \mu_{0k}) s_{\theta\mu_k\mu_j,0} + O_p(T^{-1/2}).\end{aligned} \tag{37}$$

We note that the terms entering $O_p(T^{-1/2}) = T^{-1/2} a_i(\boldsymbol{\theta}_0) + O_p(T^{-1})$, where the $a_i(\boldsymbol{\theta}_0)$ are determined by odd moments of asymptotically mean zero Gaussian random variables and have therefore mean zero (see for instance Carro (2007)), while the $O_p(T^{-1})$ does not disappear by taking expectations.

Using (36), we can write the following approximations:

$$\begin{aligned}(\hat{\mu}_i(\boldsymbol{\theta}_0) - \mu_{0i})^2 &= \frac{1}{T}\left(\sum_{j=1}^{2} \frac{s_{\mu_j,0}/\sqrt{T}}{I_{ij}}\right)^2 + O_p(T^{-3/2}) \\ &= \frac{1}{T}\sum_{j=1}^{2} I_{ij}^{-2} \frac{s_{\mu_j,0} s_{\mu_j,0}}{T} + \frac{2}{T} I_{i1} I_{i2} \frac{s_{\mu_1,0} s_{\mu_2,0}}{T} + O_p(T^{-3/2}) \\ &= -\frac{1}{T}\sum_{j=1}^{2} I_{ij}^{-2} I_{jj} + \frac{2}{T} I_{i1} I_{i2} I_{12} + O_p(T^{-3/2}) \\ &= \begin{cases} -\frac{3}{T} I_{11}^{-1} - \frac{1}{T} I_{12}^{-2} I_{22} & i = 1 \\ -\frac{3}{T} I_{22}^{-1} - \frac{1}{T} I_{12}^{-2} I_{11} & i = 2 \end{cases}\end{aligned} \tag{38}$$

where we used $\frac{s_{\mu_j,0} s_{\mu_j,0}}{T} = \mathbb{E}\frac{s_{\mu_j,0} s_{\mu_j,0}}{T} + O_p(T^{-1/2})$ and the information matrix identity, $\mathbb{E}\frac{s_{\mu_j,0} s_{\mu_j,0}}{T} = -I_{jj}$.

Similarly,

$$\begin{aligned}(\hat{\mu}_1(\boldsymbol{\theta}_0) - \mu_{01})(\hat{\mu}_2(\boldsymbol{\theta}_0) - \mu_{02}) &= \frac{1}{T}\sum_{j=1}^{2} I_{1j}^{-2} \frac{s_{\mu_j,0}}{\sqrt{T}} \sum_{k=1}^{2} I_{2k}^{-2} \frac{s_{\mu_k,0}}{\sqrt{T}} + O_p(T^{-3/2}) \\ &= \frac{1}{T}\left(-I_{11}^{-1} I_{12}^{-1} I_{11} - I_{11}^{-1} I_{22}^{-1} I_{12} - I_{12}^{-1} I_{12}^{-1} I_{12} - I_{12}^{-1} I_{22}^{-1} I_{22}\right) + O_p(T^{-3/2}) \\ &= -\frac{1}{T}\left(3 I_{12}^{-1} - I_{11}^{-1} I_{22}^{-1} I_{12}\right) + O_p(T^{-3/2}).\end{aligned} \tag{39}$$

Using similar approximations in (36), for $i = 1, 2$:

$$\begin{aligned}&\hat{\mu}_i(\boldsymbol{\theta}_0) - \mu_{0i} \\ &= -\sum_{j=1}^{2} I_{ij}^{-1} \frac{s_{\mu_j,0}}{T} + \frac{1}{T}\sum_{j=1}^{2}\sum_{k=1}^{2}\sum_{l=1}^{2} I_{ij}^{-1} I_{lk} \left(\frac{\mathbb{E} s_{\mu_j,\mu_l,0} s_{\mu_k,0}}{T} + O_p(T^{-1/2})\right) \\ &\quad + \frac{1}{2T}\sum_{j}\sum_{l}\sum_{r,s,u} (I_{jl} + O_p(T^{-1/2}))\, I_{ir}^{-1} I_{js}^{-1} I_{lu}^{-1} I_{rsu} + O_p(T^{-3/2})\end{aligned} \tag{40}$$

For notational convenience and following McCullagh (1987), we denote:

$$I^{ijl} = \sum_{r,s,u} I_{ir}^{-1} I_{js}^{-1} I_{lu}^{-1} I_{rsu}.$$

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Now, using equations (40), (38) and (39) into (41), we get:

$$
\begin{aligned}
&s_\theta(\boldsymbol{\theta}_0, \hat{\boldsymbol{\mu}}(\boldsymbol{\theta}_0)) - s_{\theta_0} \\
&= \sum_{i=1}^{2} \left(-\sum_{j=1}^{2} I_{ij}^{-1} \frac{s_{\mu_j,0}}{T} + \frac{1}{T} \sum_{k,j,l} \frac{\mathbb{E} s_{\mu_k \mu_j,0} s_{\mu_l,0}}{T} I_{ij}^{-1} I_{kl}^{-1} + \frac{1}{2T} \sum_{j,l} I^{ijl} I_{jl}^{-1} + O_p(T^{-3/2}) \right) \\
&\quad - \frac{1}{2T} \left[s_{\theta,\mu_1\mu_1,0}(3I_{11}^{-1} + I_{12}^{-2} I_{22}) + 2 s_{\theta,\mu_1\mu_2,0}(3I_{12}^{-1} + I_{11}^{-1} I_{12} I_{22}^{-1}) + s_{\theta,\mu_2\mu_2,0}(3I_{22}^{-1} + I_{12}^{-2} I_{11}) \right] \\
&\quad + T^{-1/2} a_i(\boldsymbol{\theta}_0) + O_p(T^{-1})
\end{aligned}
\tag{41}
$$

It is finally easy to see that the dominating term in the right hand side in (41) is equivalent to $B_i(\boldsymbol{\theta}_0)$.

Thus, we finally get:

$$
s_\theta(\boldsymbol{\theta}_0, \hat{\boldsymbol{\mu}}(\boldsymbol{\theta}_0)) = B_i(\boldsymbol{\theta}_0) + T^{-1/2} a_i(\boldsymbol{\theta}_0) + O_p(T^{-1}).
$$

□

Proof of Theorem 4

The assumptions 1–6 imply the asymptotic expansion in Theorem 3 and the equations (20) and (22).

We note that Assumption 8 (i) implies the existence and finiteness of the matrix J_{NT} defined in equation (24) in the sense specified in Corollary 1 of Jenish and Prucha (2012). Further, since $s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)) = \sum_t q_{\kappa,l(i,t)}(y_{it} \mid \boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0))$, from Assumption 8 (ii), and by applying Theorem 17.8 in Davidson (1994), we can conclude that also $s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0))$ is a NED sequence, with the same coefficients $\tilde{\psi}(r)$ and constants $\tilde{\tilde{d}}_i = \sup_t \tilde{d}_{l(i,t),L}$. Therefore, all the assumptions for the application of Corollary 1 of Theorem 2 of Jenish and Prucha (2012) are satisfied for the sequence $\{s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0))\}$, thus implying:

$$
\frac{1}{\sqrt{NT}} J_{NT} \sum_i \left(s_{\theta,i}(\boldsymbol{\theta}_0, \hat{\mu}_i(\boldsymbol{\theta}_0)) - B_i(\boldsymbol{\theta}_0) \right) \to \mathcal{N}(\mathbf{0}, I_{k+2}). \tag{42}
$$

Then, using (22) and (20) and exploiting uniform law of large numbers of Lemma 3,

$$
\begin{aligned}
&H_{NT}\sqrt{NT} \left(\hat{\boldsymbol{\theta}} - H_{NT}^{-1} \frac{1}{NT} \sum_i B_i(\boldsymbol{\theta}_0) - \boldsymbol{\theta}_0 \right) \\
&= (H_{NT,0} + o_p(1))\sqrt{NT} \left(\hat{\boldsymbol{\theta}} - (H_{NT,0} + o_p(1))^{-1} \frac{1}{NT} \sum_i B_i(\boldsymbol{\theta}_0) - \boldsymbol{\theta}_0 \right) \\
&= H_{NT,0}\sqrt{NT} \left(\hat{\boldsymbol{\theta}} - (H_{NT,0})^{-1} \frac{1}{NT} \sum_i B_i(\boldsymbol{\theta}_0) - \boldsymbol{\theta}_0 \right) + o_p(1) \\
&= H_{NT,0}\sqrt{NT} \left(\hat{\boldsymbol{\theta}} - (H_{NT,0})^{-1} \frac{1}{NT} \mathbb{E} \sum_i B_i(\boldsymbol{\theta}_0) - \boldsymbol{\theta}_0 \right) + o_p(1) \\
&= -\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\boldsymbol{\theta}_0, \mu_i(\boldsymbol{\theta}_0)) + O_p(\sqrt{N/T^3}).
\end{aligned}
$$

Therefore, using (42):

$$
\sqrt{NT} \left(\hat{\boldsymbol{\theta}} - H_{NT,0}^{-1} \frac{1}{NT} \sum_i \mathbb{E} B_i(\boldsymbol{\theta}_0) - \boldsymbol{\theta}_0 \right) \to \mathcal{N}(\mathbf{0}, \mathbb{V}),
$$

where $\mathbb{V} = \lim_{N,T} H_{NT,0}^{-1} J_{N,T} H_{NT,0}^{-1}$. This imples

$$
\sqrt{NT} \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \right) \to \mathcal{N}(\mathbb{B}, \mathbb{V}),
$$

where $\mathbb{B} = \lim_{N,T} H_{NT,0}^{-1} \mathbb{E} \sum_i B_i(\boldsymbol{\theta}_0)$ that exists finite by hypothesis.

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Further, since

$$\begin{aligned}\sqrt{NT}\left(\tilde{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\right) &= \sqrt{NT}\left(\hat{\boldsymbol{\theta}}-\hat{H}_{NT}^{-1}\frac{1}{NT}\sum_i \hat{B}_i-\boldsymbol{\theta}_0\right)\\ &= \left(\hat{\boldsymbol{\theta}}-H_{NT}^{-1}\frac{1}{NT}\sum_i \mathbb{E}B_i(\boldsymbol{\theta}_0)-\boldsymbol{\theta}_0\right)+(H_{NT}^{-1}-\hat{H}_{NT}^{-1})\frac{1}{\sqrt{NT}}\sum_i \mathbb{E}B_i(\boldsymbol{\theta}_0)\\ &\quad+\frac{1}{\sqrt{NT}}\hat{H}_{NT}^{-1}\sum_i\left(\mathbb{E}B_i(\boldsymbol{\theta}_0)-\hat{B}_i\right)\\ &= \left(\hat{\boldsymbol{\theta}}-H_{NT}^{-1}\frac{1}{NT}\sum_i \mathbb{E}B_i(\boldsymbol{\theta}_0)-\boldsymbol{\theta}_0\right)+A_{NT}+B_{NT}\end{aligned}$$

it is enough to prove that A_{NT} and B_{NT} are negligible to conclude the proof.

Indeed, this follows because $\|\hat{H}_{NT}-H_{NT}\|=o_p(1)$ and the existence of a bounded inverse $\hat{H}_{NT}^{-1}$ for all N,T, imply also $\|H_{NT}^{-1}-\hat{H}_{NT}^{-1}\|=o_p(1)$. Thus

$$A_{NT}=O_p(N/\sqrt{NT})o_p(1)=o_p(\sqrt{N/T})=o_p(1).$$

Similarly,

$$B_{NT}=O_p(1/\sqrt{NT})o_p(N)=o_p(1).$$

□

Proof of Theorem 5

We first rewrite the $i-$th contribution to the score correction, B_i, as (see Carro (2007) for details):

$$\begin{aligned}B_i(\boldsymbol{\theta}_0) &= \frac{1}{2}\frac{\left(s_{\theta_0\mu\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))+s_{\mu\mu\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))\frac{\partial\hat{\mu}(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}_0}\right)}{s_{\mu\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))}\\ &\quad-\frac{\partial}{\partial\mu_i}\left(\frac{\mathbb{E}s_{\theta_0\mu,i}(\boldsymbol{\theta},\mu_i)}{\mathbb{E}s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i)}\right)_{|_{\mu_i=\hat{\mu}_i(\boldsymbol{\theta}_0)}}\\ &= \frac{1}{2}\frac{s_{\theta_0\mu\mu,i}}{s_{\mu\mu,i}}-\frac{1}{2}\frac{s_{\theta_0\mu,i}}{s_{\mu\mu,i}}\frac{s_{\mu\mu\mu,i}}{s_{\mu\mu,i}}\\ &\quad-\frac{\mathbb{E}s_{\theta_0\mu\mu,i}\mathbb{E}s_{\mu\mu,i}+\mathbb{E}s_{\theta_0\mu,i}s_{\mu,i}\mathbb{E}s_{\mu\mu,i}-\mathbb{E}s_{\theta_0\mu,i}\mathbb{E}s_{\mu\mu\mu,i}-\mathbb{E}s_{\theta_0\mu,i}\mathbb{E}s_{\mu\mu,i}s_{\mu,i}}{\left(\mathbb{E}s_{\mu\mu,i}\right)^2}\\ &= \frac{1}{2}\frac{s_{\theta_0\mu\mu,i}}{s_{\mu\mu,i}}-\frac{1}{2}\frac{s_{\theta_0\mu,i}s_{\mu\mu\mu,i}}{(s_{\mu\mu,i})^2}-\frac{\mathbb{E}s_{\theta_0\mu\mu,i}}{\mathbb{E}s_{\mu\mu,i}}-\frac{\mathbb{E}s_{\theta_0\mu,i}s_{\mu,i}}{\mathbb{E}s_{\mu\mu,i}}+\frac{\mathbb{E}s_{\theta_0\mu,i}\mathbb{E}s_{\mu\mu\mu,i}}{\left(\mathbb{E}s_{\mu\mu,i}\right)^2}+\frac{\mathbb{E}s_{\theta_0\mu,i}\mathbb{E}s_{\mu\mu,i}s_{\mu,i}}{\left(\mathbb{E}s_{\mu\mu,i}\right)^2}\end{aligned}$$

where all functions in the right hand side are computed at $(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))$. Using the fact that $s_{\mu\mu,i}/T=\mathbb{E}s_{\mu\mu,i}/T+O_p(T^{-1/2})$ and that $\sqrt{T}(s_{\theta\mu\mu,i}/T-\mathbb{E}s_{\theta\mu\mu,i}/T)$ has asymptotically mean zero, B_i can be approximated by

$$\begin{aligned}B_i(\boldsymbol{\theta}_0) &= -\frac{1}{2}\frac{s_{\theta\mu\mu,i}}{s_{\mu\mu,i}}+\frac{1}{2}\frac{s_{\theta\mu,i}s_{\mu\mu\mu,i}}{(\mathbb{E}s_{\mu\mu,i})^2}-\frac{s_{\theta\mu,i}s_{\mu,i}}{\mathbb{E}s_{\mu\mu,i}}+\frac{s_{\theta\mu,i}\mathbb{E}s_{\mu\mu,i}s_{\mu,i}}{\left(\mathbb{E}s_{\mu\mu,i}\right)^2}+O_p(T^{-1/2})\\ &= -\frac{1}{2}\frac{s_{\theta\mu\mu,i}/T}{s_{\mu\mu,i}/T}+\frac{1}{2}\frac{\mathbb{E}s_{\theta\mu,i}\mathbb{E}s_{\mu\mu\mu,i}/T^2}{(\mathbb{E}s_{\mu\mu,i}/T)^2}-\frac{s_{\theta\mu,i}s_{\mu,i}/T}{\mathbb{E}s_{\mu\mu,i}/T}+\frac{s_{\theta\mu,i}/T\mathbb{E}s_{\mu\mu,i}s_{\mu,i}/T}{\left(\mathbb{E}s_{\mu\mu,i}/T\right)^2}+O_p(T^{-1/2})\end{aligned}\tag{43}$$

Numerators and denominators of the right-hand-side of the above display are all $O_p(1)$, because each $s_{\cdot,i}$ is the sum of T functions of the form $q_{\kappa,l}$.

Further, because of the Lipschitz condition induced on the derivatives $q_{\kappa,l}$ (and therefore on $s_{\theta\mu}, s_{\theta,\mu\mu}$ and so on) by Assumption 6, we have that

$$\|B_i(\boldsymbol{\theta}_0)-\hat{B}_i\|\leq C_B\|\boldsymbol{\theta}_0-\hat{\boldsymbol{\theta}}\|+O_p(T^{-1/2}),$$

where the constant C_B depends on the sum of Lipschitz coefficients of the derivatives $q_{\kappa,l}/T$, and is therefore an $O_p(1)$.

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To the reader's convenience, we include the details only for the first term of the difference $B_i(\boldsymbol{\theta}_0) - \hat{B}_i$, since all other terms can be proved to have the same rate of convergence using similar arguments:

$$\begin{aligned}
&\left\| -\frac{1}{2}\frac{s_{\theta\mu\mu,i}/T}{s_{\mu\mu,i}/T} + \frac{1}{2}\frac{s_{\theta\mu\mu,i}(\hat{\boldsymbol{\theta}},\hat{\mu}_i(\hat{\boldsymbol{\theta}}))/T}{s_{\mu\mu,i}(\hat{\boldsymbol{\theta}},\hat{\mu}_i(\hat{\boldsymbol{\theta}}))/T} \right\| \\
&\quad \leq \max_{\kappa:|\kappa|=3,\kappa_0=2} \sum_t \left\| \frac{1}{2}\frac{q_{\kappa,l(i,t)}(y_{it} \mid \boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))/T}{s_{\mu\mu,i}/T} - \frac{1}{2}\frac{q_{\kappa,l(i,t)}(y_{it} \mid \hat{\boldsymbol{\theta}},\hat{\mu}_i(\hat{\boldsymbol{\theta}}))/T}{s_{\mu\mu,i}/T + O_p(T^{-1/2})} \right\| \\
&\quad \leq \frac{1}{2} \max_{\kappa:|\kappa|=3,\kappa_0=2} \sum_t B_{\kappa,l(i,t)} \left\| \boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}} \right\| = O_p\left(\left\| \boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}} \right\| \right)
\end{aligned}$$

Thus, we have

$$\|B_i(\boldsymbol{\theta}_0) - \hat{B}_i\| = O_p(T^{-1/2}).$$

□

Proof of Theorem 6

Since

$$\begin{aligned}
\mathrm{Var}\left(\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) \right) &= \frac{1}{NT}\mathbb{E}\left(\sum_i \sum_j s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) s_{\theta,j}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))' \right) \\
&\quad - \frac{1}{NT}\sum_i \sum_j \mathbb{E}\left(s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) \right) \mathbb{E}\left(s_{\theta,j}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) \right)'
\end{aligned}$$

and given that, $\|B_i(\boldsymbol{\theta}_0) - \hat{B}_i\| = O_p((T)^{-1/2})$, we can write

$$\frac{1}{\sqrt{NT}}\sum_i \mathbb{E}\left(s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) \right) = \frac{1}{\sqrt{NT}}\sum_i B_i(\boldsymbol{\theta}_0) + O_p(T^{-1}) = \frac{1}{\sqrt{NT}}\sum_i \hat{B}_i + O_p(\sqrt{N}/T) + O_p(T^{-1}).$$

Thus, we can focus on the consistent estimation of the first term:

$$\frac{1}{NT}\mathbb{E}\left(\sum_i \sum_j s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) s_{\theta,j}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))' \right).$$

Now note that each vector $s_{\theta,i}$ has $j-$th component given by the sum

$$\sum_t q_{e_j,l(i,t)}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)),$$

where $\boldsymbol{e}_j$ is obtained by adding an initial zero term to the $j-$th basis vector, i.e. $\boldsymbol{e}_1 = (0,1,0,\ldots,0)$, …, $\boldsymbol{e}_{k+2} = (0,0,\ldots,1)$.

Similarly, using this notation, for all t,

$$s_{\theta,t} = \left\{ \frac{1}{N}\sum_i q_{e_j,l(i,t)}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) \right\}_{j=1,\ldots,k+2}$$

and $\frac{1}{NT}\mathbb{E}\left(\sum_i \sum_j s_{\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) s_{\theta,j}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))' \right) = \frac{N}{T}\sum_t \sum_{t'} \mathbb{E} s_{\theta,t} s_{\theta,t'}$. Whenever the derivatives are computed on the estimate $\hat{\boldsymbol{\theta}}^*$, we also define $\hat{s}_{\theta,t} = \left\{ \frac{1}{N}\sum_i q_{e_j,l(i,t)}(\hat{\boldsymbol{\theta}}^*,\hat{\mu}_i(\hat{\boldsymbol{\theta}}^*)) \right\}_{j=1,\ldots,k+2}$.

By using the same arguments as Newey and West (1987), we can limit the problem to focusing on linear transformations of the vectors $s_{\theta,t}$, for any arbitrary $\boldsymbol{c}$, such that $\|c\| < \infty$. Let

$$h_t = \boldsymbol{c}' s_{\theta,t} \quad \text{and} \quad \hat{h}_t = \boldsymbol{c}' \hat{s}_{\theta,t},$$

then the proof is completed once we are able to prove that: $|\hat{S}_{NT} - S_{NT}| \to 0$ in probability, where

$$\hat{S}_{NT} = \frac{N}{T}\left(\hat{h}_t^2 + 2\sum_{j=1}^{m} w(j,m) \sum_{t=j+1}^{T} \hat{h}_t \hat{h}_{t-j} \right)$$

and $S_{NT} = \frac{N}{T}\sum_t \sum_{t'} \mathbb{E} h_t h_{t'}$.

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Following the same computations as Newey and West (1987), we can decompose the difference into:

$$\begin{aligned}|\hat{S}_{NT} - S_{NT}| \leq & \left|\hat{S}_{NT} - \frac{N}{T}\left(h_t^2 + 2\sum_{j=1}^{m} w(j,m) \sum_{t=j+1}^{T} h_t h_{t-j}\right)\right| \\ & + \left|N\sum_{t=1}^{T}\left(h_t^2 - \mathbb{E}h_t^2\right)/T + \frac{2N}{T}\sum_{j=1}^{m} w(j,m) \sum_{t=j+1}^{T}\left(h_t h_{t-j} - \mathbb{E}h_t h_{t-j}\right)\right| \\ & + 2\sum_{j} |w(j,m) - 1| \sum_{t=j+1}^{T} |\mathbb{E}h_t h_{t-j}| \frac{N}{T} + 2\frac{N}{T}\sum_{j=1}^{m}\sum_{t=j+1}^{T} |\mathbb{E}h_t h_{t-j}| \\ = & A + B + C + D\end{aligned}$$

We'll prove that the 4 terms in the right had side are all $o_p(1)$.

$\underline{D \to 0}$. The key result that allows us to prove that $D = o_p(1)$ is summability of L_2-NED sequences. According to Theorem 17.7 in Davidson (1994), if a sequence X_t is L_2-NED with constants d and size -1, then the sequence:

$$\left\{\frac{|\mathbb{E}(X_t X_{t+j})|}{\|X_t\|_2 \|X_{t+j}\|_2}, j \in \mathbb{N}\right\}$$

is summable in t. This means that, for all t, the sum $\sum_{m=1}^{\infty} \frac{|\mathbb{E}(X_t X_{t+j})|}{\|X_t\|_2 \|X_{t+j}\|_2}$ is bounded.

In our case, we have $h_t = \boldsymbol{c}' s_{\theta,t}$ is a L_2-NED sequence, because each component of $s_{\theta,t}$ is the sample average of the N elements of an L_2-NED sub-sequence $\{q_{\kappa,l(i,t)}, i = 1, \ldots, N\}$ (Theorem 17.8 in Davidson (1994)). The coefficient of $s_{\theta,t}$ is $\tilde{\tilde{d}}_t \leq \sup_i \tilde{d}_{l(i,t),L}$, while the function $\tilde{\tilde{\psi}}(r) \leq \tilde{\psi}(r)$ and the size is clearly (at least) -1, because of Assumption 8. Another application of Theorem 17.8 implies that h_t is L_2-NED with coefficients $\tilde{\tilde{d}}\|c\|_\infty$ and the same size.

On the other hand, the same summability condition holds for the subsequence $\{q_{\kappa,l(i,t)}, i \geq 1\}$, that implies $\mathbb{E}\left(\sum_{i=1}^{N} q_{\kappa,l(i,t)}\right)^2 = O(N)$, thus

$$\|h_t\|_2^2 = N^{-2}\mathbb{E}\left(\sum_{i=1}^{N} \boldsymbol{c}'\{q_{\boldsymbol{e}_j,l(i,t)}\}_{j\leq k+2}\right)^2 \leq \|\boldsymbol{c}\|_\infty^2 O(N^{-1}),$$

for all t.

We can therefore conclude that

$$\begin{aligned}\frac{N}{T}\sum_{j=m+1}^{T-1}\sum_{t=j+1}^{T} |\mathbb{E}h_t h_{t-j}| &\leq \frac{N \sup_t \|h_t\|_2^2}{T}\sum_{t=1}^{T}\sum_{j=m+1}^{T-t} \frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2} \\ &\leq N \sup_t \|h_t\|_2^2 \sup_t \sum_{j=m+1}^{T-t} \frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}.\end{aligned}$$

For all t, the sum tends to 0 in probability as $T \to \infty$, because it is the remainder of a convergent series (in L_1), while $N \sup_t \|h_t\|_2^2 = O(1)$.

$\underline{C \to 0}$. The proof of this part follows similar steps to the ones used to prove that $D = o_p(1)$. We exploit in fact again the summability property discussed above:

$$\begin{aligned}\frac{N}{T}\sum_{j=1}^{m} |w(j,m) - 1| \sum_{t=j+1}^{T} |Eh_t h_{t-j}| &\leq \frac{N \sup_t \|h_t\|_2^2}{T}\sum_{j=1}^{m} \frac{j}{m+1}\sum_{t=1}^{T-j} \frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2} \\ = \frac{N \sup_t \|h_t\|_2^2}{T}&\left[\sum_{t=1}^{T-m-1}\sum_{j=1}^{m} \frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2} + \sum_{t=T-m}^{T-1}\sum_{j=1}^{T-t} \frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}\right]\end{aligned}$$

For the first sum, letting $m_0 = m_0(T)$, such that $\lim_T m_0 = \infty$, $\lim_T m_0/m = 0$, we

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have, using again summability,

$$T^{-1}\sum_{t=1}^{T-m-1}\sum_{j=1}^{m}\frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2} \leq \frac{T-m}{T}\sup_t\sum_{j=1}^{m}\frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}$$
$$\leq \frac{T-m}{T}\sup_t\left(\sum_{j=1}^{m_0}\frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}+\sum_{j=m_0+1}^{m}\frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}\right)$$
$$\leq \frac{T-m}{T}\left(\frac{m_0}{m+1}\sup_t\sum_{j=1}^{m_0}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}+\frac{m}{m+1}\sum_{j=m_0+1}^{m}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}\right)$$
$$\leq K_1\frac{T-m}{T}\frac{m_0}{m+1}+\frac{T-m}{T}\sum_{j=m_0+1}^{m}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}\to 0$$

because of $\sum_{j=1}^{\infty}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}=K_t<\infty$ for all t and by the Cauchy convergence criterion.

For the second sum,

$$T^{-1}\sum_{t=T-m}^{T-1}\sum_{j=1}^{T-t}\frac{j}{m+1}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2} \leq \sum_{t=T-m}^{T-1}\frac{T-t}{T(m+1)}\sum_{j=1}^{\infty}\frac{|\mathbb{E}(h_{t+j}h_t)|}{\|h_t\|_2\|h_{t+j}\|_2}$$
$$\leq \sum_{t'=1}^{m}\frac{t'}{T(m+1)}\sup_t K_t = O(\frac{m}{T})\to 0$$

Finally, by noting that $N\sup_t\|h_t\|_2^2=O(1)$ we can conclude that $C=o(1)$.

$\underline{B\to 0}$. We will prove that the two terms

$$\frac{N}{T}\sum_{t=1}^{T}\left|h_t^2-\mathbb{E}h_t^2\right|\to 0$$

and

$$\left|\frac{N}{T}\sum_{j=1}^{m}w(j,m)\sum_{t=j+1}^{T}h_th_{t-j}-\mathbb{E}h_th_{t-j}\right|\to 0$$

in probability.

Let us first denote $Z_{tj}=h_th_{t-j}-\mathbb{E}h_th_{t-j}$ for all j,t, so that we can focus on proving

$$\frac{N}{T}\sum_t Z_{t0}\to 0 \quad \text{and} \quad \left|\frac{N}{T}\sum_{j=1}^{m}w(j,m)\sum_{t=j+1}^{T}Z_{tj}\right|\to 0.$$

We note that, since for any $\kappa=(0,\boldsymbol{e}_j)$ $(j=1,\ldots,k+2)$

$$\sup_{i,t}\|q_{\kappa,l(i,t)}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))\|_{2(2+\delta)}<\infty,$$

we can invoke Theorem 17.16 in Davidson (1994), to prove that the product sequence $\{q_{\kappa_1,l}q_{\kappa_2,l'}\}$ is $L_2(d_q)-$NED, with size equal to $-a\delta/2(1+\delta)$, where $-a$ is the size of the NED sequence $\{q_\kappa\}$ (see in particular example 17.17 Davidson (1994), with $X_t=q_{\kappa_1,l}$, $Y_t=q_{\kappa_2,l'}$ and with $r=2+\delta$).

Then, by the preservation of the NED property with respect to linear combinations, we can define, for every κ_1,κ_2 and (j,t), the L_2-NED sequence

$$Q_{i,tj}^{\kappa_1,\kappa_2}=\sum_{i'=i+1}^{N}\frac{q_{\kappa_1,l(i,t)}q_{\kappa_1,l(i',t-j)}}{N}, \tag{44}$$

that has size $-a\delta/2(1+\delta)$ and coefficients equal to

$$d_i^{\mathcal{Q}}=C\tilde{d}_{l(i,t),L}\sup_{i\leq N}\tilde{d}_{l(i,t-j),L}\sup_l\|q_{\kappa_1,l}\|_{2(2+\delta)}\sup_l\|q_{\kappa_2,l}\|_{2(2+\delta)}.$$

Because of the assumption of the theorem, we have that

$$\sum_r[\tilde{\psi}(r)]^{\delta/(\delta+1)}r^{d-1}\leq\sum_r r^{-a\delta/(\delta+1)}r^{d-1}=\sum_r r^{d-1-a\delta/(\delta+1)}<\infty,$$

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implies $d < a\delta/(\delta+1)$. Therefore, $a > 2(\delta+1)/\delta$ in the univariate case and $a > 3(\delta+1)/\delta$ in the bivariate. Thus, $a\delta/2(\delta+1) > 1$ for $d \geq 2$. We then can invoke Theorem 17.7 of Davidson (1994) for summability of $Q_{i,tj}^{\kappa_1,\kappa_2}$:

$$\mathrm{Var}\left(\sum_{i=1}^{N} \frac{Q_{i,tj}^{\kappa_1,\kappa_2}}{\|Q_{i,tj}^{\kappa_1,\kappa_2}\|_2}\right) \leq \sum_{i=1}^{N}\sum_{i'=1}^{N} \mathbb{E}\left(\frac{Q_{i,tj}^{\kappa_1,\kappa_2} Q_{i',tj}^{\kappa_1,\kappa_2}}{\|Q_{i,tj}^{\kappa_1,\kappa_2}\|_2 \|Q_{i',tj}^{\kappa_1,\kappa_2}\|_2}\right) \leq O(N).$$

Further, since the derivatives $q_{\kappa,l}$ compose an L_2-NED sequence with size $a > 2$, we also have

$$\|Q_{i,tj}^{\kappa_1,\kappa_2}\|_2^2 = \mathbb{E}\left(\sum_{i'=i+1}^{N} \frac{q_{\kappa_1,l(i,t)} q_{\kappa_2,l(i',t-j)}}{N}\right)^2 = N^{-2}\mathbb{E}\left(\sum_{i'=1}^{N-i} q_{\kappa_1,l(i,t)} q_{\kappa_2,l(i+i',t-j)}\right)^2 = O(N^{-1})$$

for all $i = 1,\ldots,N$, again because of Theorem 17.7 of Davidson (1994).
This implies

$$\mathrm{Var}\left(\sum_{i=1}^{N} Q_{i,tj}^{\kappa_1,\kappa_2}\right) \leq \mathbb{E}\left(\sum_{i=1}^{N} Q_{i,tj}^{\kappa_1,\kappa_2}\right)^2 \leq O(1) \tag{45}$$

Now we remark that, since $h_t = \boldsymbol{c}' s_{\theta,t}$, we can bound, for every couple (j,t), $j = 0,\ldots,m$, $t = 1,\ldots,T$,

$$\begin{aligned}
(h_t h_{t-j})^2 = (\boldsymbol{c}' s_{\theta,t} s_{\theta,t-j}' \boldsymbol{c})^2 &= \left(\sum_{r_1=1}^{k+1}\sum_{r_2=1}^{k+1} c_{r_1} c_{r_2} \{s_{\theta,t}\}_{r_1} \{s_{\theta,t-j}\}_{r_2}\right)^2 \\
&\leq (k+2)^4 \sum_{r_1=1}^{k+1}\sum_{r_2=1}^{k+1} \frac{(c_{r_1} c_{r_2} \{s_{\theta,t}\}_{r_1} \{s_{\theta,t-j}\}_{r_2})^2}{(k+2)^2} \\
&\leq (k+2)^2 \|c\|_\infty^4 \sum_{r_1=1}^{k+1}\sum_{r_2=1}^{k+1} \left(\{s_{\theta,t}\}_{r_1} \{s_{\theta,t-j}\}_{r_2}\right)^2 .
\end{aligned}$$

The square of the product $\{s_{\theta,t}\}_{r_1}\{s_{\theta,t-j}\}_{r_2}$ of the r_1 and r_2 components of $s_{\theta,t}$ and $s_{\theta,t-j}$ is equal to

$$\left(\{s_{\theta,t}\}_{r_1} \{s_{\theta,t-j}\}_{r_2}\right)^2 = \left(N^{-2}\sum_{i=1}^{N}\sum_{i'=1}^{N} q_{\kappa_1,l(i,t)} q_{\kappa_2,l(i',t-j)}\right)^2 = \left(N^{-1}\sum_{i=1}^{N} Q_{i,tj}^{\kappa_1,\kappa_2}\right)^2$$

where κ_1 and κ_2 are the vectors defining the derivatives in the components r_1 and r_2 respectively of $s_{\theta,t}$ ($s_{\theta,t-j}$).
Thus, we have, for every $j = 0,1,\ldots,m$, $t = 1,\ldots,T$, using (45),

$$\begin{aligned}
\|h_t h_{t-j}\|_2 = \sqrt{\mathbb{E}(h_t h_{t-j})^2} &\leq (k+2)\|\boldsymbol{c}\|_\infty^2 \sup_{r_1,r_2} \frac{1}{N}\sqrt{\mathbb{E}\left(\sum_{i=1}^{N} Q_{i,tj}^{\kappa_1,\kappa_2}\right)^2} \\
&\leq \frac{C(k+2)\|\boldsymbol{c}\|_\infty^2}{N}
\end{aligned}$$

By applying the same results on preservation of the NED property, we have that, for all $j \geq 0$, the series Z_{tj} is L_2-NED with the same size as $Q_{i,jt}^{\kappa_1,\kappa_2}$ and with coefficient $K\|c\|_\infty \sup_i d_i^Q$.
We can once again invoke summability of Z_{tj}: there exists a constant $K < \infty$, such that, for all j,

$$\mathbb{E}\left(\sum_{t=1}^{T} \frac{Z_{tj}}{\|h_t h_{t-j}\|_2}\right) \leq \mathrm{Var}\left(\sum_{t=1}^{T} \frac{Z_{tj}}{\|Z_{tj}\|_2}\right) \leq KT$$

Then we can finally write, for any $\gamma > 0$

$$\begin{aligned}
\Pr\left(\sum_t Z_{t0}\frac{N}{T} > \gamma\right) &= \Pr\left(\sum_t \frac{Z_{t0}}{\|h_t^2\|_2}\frac{N\|h_t^2\|_2}{T} > \gamma\right) \\
&\leq (\text{Chebyshev}) \leq \mathbb{E}\left(\sum_{t=1}^{T} \frac{Z_{tj}}{\|Z_{tj}\|_2}\right)^2 \frac{N^2\|h_t^2\|_2^2}{T^2\gamma^2} \\
&\leq KT\frac{N^2C^2(k+2)^2\|\boldsymbol{c}\|_\infty^4}{T^2N^2\gamma^2} \to 0
\end{aligned}$$

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as $T \to \infty$.

Similarly, we can prove convergence in probability of $NT^{-1}\left|\sum_j w(j,m)\sum_{t=j+1}^{T} Z_{tj}\right|$. For any arbitrary $\gamma > 0$,

$$\begin{aligned}
\Pr\left(NT^{-1}\left|\sum_j w(j,m)\sum_{t=j+1}^{T} Z_{tj}\right| > \gamma\right) &\leq \sum_{j=1}^{m}\Pr\left(\left|\sum_{t=j+1}^{T} Z_{tj}\right| > \gamma T/(mNw(j,m))\right) \\
&\leq \sum_{j=1}^{m}\mathbb{E}\left(\sum_t \frac{Z_{tj}}{\|h_t h_{t-j}\|_2}\right)^2 \frac{m^2N^2\|h_t h_{t-j}\|_2^2 w(j,m)^2}{\gamma^2T^2} \\
&\leq \frac{m^2}{\gamma^2T^2}\frac{K'TN^2(k+2)\|\boldsymbol{c}\|_\infty^4}{N^2}\sum_{j=1}^{m}\frac{j^2}{(m+1)^2} \\
&= \frac{m^2}{\gamma^2T^2}K'T(k+2)\|\boldsymbol{c}\|_\infty^4\frac{m(m+1)(2m+1)}{6(m+1)^2} \to 0
\end{aligned}$$

if $m = o(T^{1/3})$.

$\underline{A \to 0}$. Now, we first note that, by applying a first order Taylor expansion, for any $j \geq 0$,

$$|\hat{h}_t\hat{h}_{t-j} - h_t h_{t-j}| = \left|\left(h_t^* h_{\theta,t-j}^* + h_{\theta,t}^* h_{t-j}^*\right)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\right|,$$

where h_t^* is equal to $\boldsymbol{c}'N^{-1}\sum_i\{q_{e_j,l(i,t)}(\boldsymbol{\theta}^*, \hat{\mu}_i(\boldsymbol{\theta}^*))\}_{j\leq k+2}$, $h_{\theta,t}^*$ is its derivative with respect to the $k+2$ dimensional vector $\boldsymbol{\theta}$, and $\boldsymbol{\theta}^*$ is a point in the ball centered at $\boldsymbol{\theta}_0$ with radius smaller than $\|\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}\|$.

Then $h_t^* h_{\theta,t-j}^*$ is a vector whose elements are combinations of products of the derivatives $q_{e_j,l(i,t)}$ and $q_{\kappa,l(i',t-j)}$, where $j = 1,\ldots,k+2$ and κ is allowed to vary within the $(k+2)^2$ possible vectors of the form $\kappa = e_j + e_{j'}$, $j, j' = 1,\ldots,k+2$ (all computed at $\boldsymbol{\theta}^*$):

$$\begin{aligned}
\|h_t^* h_{\theta,t-j}^*\|_2^2 &= \mathbb{E}\left(\boldsymbol{c}'\left\{N^{-1}\sum_i q_{e_j,l(i,t)}\right\}_{j\leq k+2}\left\{N^{-1}\sum_{i'}\sum_i q_{\kappa,l(i,t)}\right\}_{\kappa=e_j+e_{j'}}\boldsymbol{c}\right)^2 \\
&\leq \|\boldsymbol{c}\|_\infty^4\sum_{j=1}^{k+2}\sum_{\kappa=e_{j_1}+e_{j_2},j_1,j_2=1}^{k+2}\mathbb{E}\left(N^{-1}\sum_{i=1}^{N}Q_{i,tj}^{e_j,\kappa}\right)^2
\end{aligned}$$

where $Q_{i,tj}^{e_j,\kappa}$ is defined in (44). We can then exploit (45), to imply:

$$\|h_t^* h_{\theta,t-j}^*\|_2 \leq \|\boldsymbol{c}\|_\infty^2(k+2)^{3/2}O(N^{-1})$$

Then, for all $j = 0,\ldots,m$, there exists some finite constant K, such that, if N, T are large enough,

$$\begin{aligned}
\frac{N}{T}\sum_t \mathbb{E}|\hat{h}_t\hat{h}_{t-j} - h_t h_{t-j}| &\leq \frac{N}{T}\sum_t \mathbb{E}\left|\left(h_t^* h_{\theta,t-j}^* + h_{\theta,t}^* h_{t-j}^*\right)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\right| \\
&\leq \frac{N}{T}\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2\sum_t\|h_t^* h_{\theta,t-j}^*\|_2 \\
&\leq K\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 = O_p(1/\sqrt{NT})
\end{aligned}$$

Similarly,

$$\begin{aligned}
\sum_{j=1}^{m} w(j,m)\frac{N}{T}\sum_{t=j+1}^{T}|\hat{h}_t\hat{h}_{t-j} - h_t h_{t-j}| &\leq K\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2\sum_{j=1}^{m} w(j,m)\frac{(T-j)}{T} \\
&= O_p((NT)^{-1/2})\left[\frac{Tm(m+1)}{2T(m+1)} - \frac{m(m+1)(2m+1)}{6T(m+1)}\right] \\
&= O_p\left(\frac{m(3T-2m-1)}{T\sqrt{NT}}\right) \leq 3O_p(m/\sqrt{NT}) \to 0.
\end{aligned}$$

Therefore, the term A

$$A \leq \frac{N}{T}\sum_t\left|\hat{h}_t^2 - h_t^2\right| + \frac{2N}{T}\sum_{j=1}^{m} w(j,m)\sum_{t=j+1}^{T}\left|\hat{h}_t\hat{h}_{t-j} - h_t h_{t-j}\right|$$

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converges to zero in L_1, and thus in probability, and this completes the proof of the theorem.

□

D Computation of the asymptotic bias and of the Hessian matrix for the spatial probit model

In this section, we give some details on the computation of the terms entering the formula of $\hat{B}_i$ for the bias correction.

We also present an alternative to the purely computational approach for the estimation of the Hessian matrix, based on the approximation of the analytic expression of the Hessian.

The focus, as for Theorem 5, is on the univariate partial maximum likelihood approach.

Using the expression of $\hat{B}_i$ given in equation (27), we here derive the analytic expressions of viable approximations of all components in the case of a spatial probit model.

From (9) we clearly have, for all i:

$$s_{\mu,i}(\boldsymbol{\theta},\mu_i) = \frac{\partial}{\partial \mu_i}\ell_i\left(\boldsymbol{\theta},\mu_i\right) = \sum_{t=1}^{T}\frac{\frac{\partial}{\partial \mu_i}p_{it}\left(y_{it}\right)}{p_{it}\left(y_{it}\right)} \tag{46}$$

and

$$\begin{aligned} s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i) &= \frac{\partial}{\partial \mu_i}\sum_{t=1}^{T}\frac{\frac{\partial}{\partial \mu_i}p_{it}\left(y_{it}\right)}{p_{it}\left(y_{it}\right)} \\ &= \sum_{t=1}^{T}\left[\frac{\frac{\partial^2}{\partial \mu_i^2}p_{it}\left(y_{it}\right)}{p_{it}\left(y_{it}\right)} - \frac{\left(\frac{\partial}{\partial \mu_i}p_{it}\left(y_{it}\right)\right)^2}{p_{it}\left(y_{it}\right)^2}\right] \end{aligned} \tag{47}$$

Given the expression (12), we then get:

$$\begin{aligned} s_{\mu\mu,i}(\boldsymbol{\theta},\mu_i) &= \sum_t \frac{\partial}{\partial \mu_i}\frac{1}{p_{it}(y_{it})}\frac{\vartheta_i}{\sigma_i}\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right) \\ &= \sum_t \frac{-\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right)\frac{\vartheta_i\zeta_{it}}{\sigma_i}}{\sigma_i^2 p_{it}(y_{it})} - \frac{\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right)^2}{\sigma_i^2\, p_{it}^2(y_{it})}, \end{aligned} \tag{48}$$

where $\zeta_{it} = \{\boldsymbol{X}_{\rho,t}\boldsymbol{\beta}\}_i + \alpha y_{\rho,i,t-1} + \mu_i$ because of the reparametrization $\boldsymbol{\mu} = \boldsymbol{A}_\rho^{-1}\boldsymbol{\mu}$.

The term $s_{\mu\mu,i}$ can be conveniently estimated by replacing the terms in the sum by conditional expectations:

$$\begin{aligned} \frac{s_{\mu\mu,i}}{T} &\approx T^{-1}\mathbb{E}\left(\mathbb{E}\sum_t\left(\frac{-\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right)\frac{\vartheta_i\zeta_{it}}{\sigma_i}}{\sigma_i^2 p_{it}(y_{it})} - \frac{\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right)^2}{\sigma_i^2\, p_{it}^2(y_{it})}\Big|\boldsymbol{y}_{t-1},\boldsymbol{X}_t\right)\right) \\ &= T^{-1}\mathbb{E}\sum_t\left(\frac{-\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right)\frac{\vartheta_i\zeta_{it}}{\sigma_i}}{\sigma_i^2 p_{it}(y_{it})} - \frac{\phi\left(\frac{\vartheta_i\zeta_{it}}{\sigma_i}\right)^2}{\sigma_i^2\, p_{it}^2(y_{it})}\Big|\boldsymbol{y}_{t-1},\boldsymbol{X}_t\right) + O_p(T^{-1/2}) \\ &= T^{-1}\left[\sum_t \frac{-\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{\zeta_{it}}{\sigma_i}}{\sigma_i^2 p_{it}(1)}p_{it}(1) + \frac{-\phi\left(\frac{-\zeta_{it}}{\sigma_i}\right)\frac{-\zeta_{it}}{\sigma_i}}{\sigma_i^2 p_{it}(0)}p_{it}(0) - \left(\frac{\phi^2(\zeta_{it}/\sigma_i)}{\sigma_i^2 p_{it}(1)} + \frac{\phi^2(-\zeta_{it}/\sigma_i)}{\sigma_i^2 p_{it}(0)}\right) + O_p(T^{-1/2})\right] \\ &= -T^{-1}\sum_t \frac{\phi^2\left(\zeta_{it}/\sigma_i\right)}{\sigma_i^2}\left(\frac{1}{p_{it}(0)} + \frac{1}{p_{it}(1)}\right) + O_p(T^{-1/2}) \end{aligned} \tag{49}$$

With a similar computation, we obtain

$$T^{-1}s_{\theta\mu,i} \approx -T^{-1}\sum_t \frac{1}{p_{it}(1)(1-p_{it}(1))}\frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{\sigma_i}\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta} \tag{50}$$

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where

$$\frac{\partial \zeta_{it}/\sigma_i}{\partial \theta} = \begin{pmatrix} \frac{\partial \zeta_{it}/\sigma_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial \zeta_{it}/\sigma_i}{\partial \alpha} \\ \frac{\partial \zeta_{it}/\sigma_i}{\partial \rho} \end{pmatrix} = \frac{1}{\sigma_i} \begin{pmatrix} \boldsymbol{X}'_{\rho,it} \\ y_{\rho,it-1} \\ (\boldsymbol{A}_\rho^{-1}\boldsymbol{W})_i \left(\boldsymbol{X}_{\rho,t}\boldsymbol{\beta} + \alpha \boldsymbol{y}_{\rho,t-1}\right) - \frac{\zeta_{it}\partial\sigma_i/\partial\rho}{\sigma_i} \end{pmatrix} \tag{51}$$

and

$$\frac{\partial \sigma_i}{\partial \rho} = \frac{1}{2\sigma_i}\frac{\partial \sigma_i^2}{\partial \rho}$$

and

$$\frac{\partial \sigma_i^2}{\partial \rho} = -2\{\text{diag}(\boldsymbol{A}_\rho^{-1}\boldsymbol{W}\boldsymbol{\Sigma})\}_i$$

that is, the $i-$th term of the main diagonal of the matrix $-2\boldsymbol{A}_\rho^{-1}\boldsymbol{W}\boldsymbol{\Sigma}$.

By taking the derivative of $s_{\mu\mu,i}$ with respect to $\boldsymbol{\theta}$,

$$\begin{aligned} s_{\theta\mu\mu,i} = &\sum_t \phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{\jmath_i}{\sigma_i^2}\left[\left(\frac{\zeta_{it}^2}{\sigma_i^2}-1\right)\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta} + \frac{\zeta_{it}}{\sigma_i^3}\frac{\partial\sigma_i^2}{\partial\boldsymbol{\theta}}\right]\frac{1}{p_{it}(y_{it})} \\ &+ \sum_t 3\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)\left[\frac{\jmath_i^2\zeta_{it}}{\sigma_i}\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta} + \frac{\partial\sigma_i^2/\partial\boldsymbol{\theta}}{\sigma_i^2}\right]\frac{1}{\sigma_i^2 p_{it}^2(y_{it})} \\ &+ 2\sum_t \phi^3\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{\jmath_i\zeta_{it}}{\sigma_i^2}\frac{1}{p_{it}^3(y_{it})}\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta}, \end{aligned}$$

which leads to
$\frac{s_{\theta\mu\mu,i}}{T} \approx \sum_t \frac{1}{T\sigma_i^2}\frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{p_{it}(1)(1-p_{it}(1))}\left[\left(\frac{3\zeta_{it}}{\sigma_i}\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta} + \frac{\partial\sigma_i^2/\partial\boldsymbol{\theta}}{\sigma_i^2}\right) + 2\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{1-2p_{it}(1)}{p_{it}(1)(1-p_{it}(1))}\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta}\right]$.
Further,

$$\begin{aligned} s_{\mu\mu\mu,i} = &\sum_t \phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{\jmath_i\zeta_{it}^2}{\sigma_i^5 p_{it}(y_{it})} + 3\sum_t \phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{\zeta_{it}}{\sigma_i^4 p_{it}^2(y_{it})} \\ &+ 2\sum_t \phi^3\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{\jmath_i}{\sigma_i^3\, p_{it}^3(y_{it})}\frac{\partial\zeta_{it}/\sigma_i}{\partial\theta}, \end{aligned}$$

and

$$T^{-1}s_{\mu\mu\mu,i} \approx T^{-1}\sum_t \frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{\sigma_i^3\, p_{it}(1)(1-p_{it}(1))}\left[\frac{3\zeta_{it}}{\sigma_i} + 2\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\frac{1-2p_{it}(1)}{p_{it}(1)(1-p_{it}(1))}\right].$$

In the following, we need to refer explicitly to the time index in $\jmath_{it} = 2(y_{it}-1/2)$. We finally estimate the components in the last two terms of $B_i(\boldsymbol{\theta})$[4].

$$\begin{aligned} \hat{\mathbb{E}}s_{\mu\mu,i}s_{\mu,i} = &-\sum_t \frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{\sigma_i^3 p_{it}(1)(1-p_{it}(1))}\left[\frac{\zeta_{it}}{\sigma_i} + \frac{\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)(1-2p_{it}(1))}{p_{it}(1)(1-p_{it}(1))}\right] \\ &- 2\sum_t\sum_{t'>t}\frac{\jmath_{it'}\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\phi\left(\frac{\zeta_{it'}}{\sigma_i}\right)}{\sigma_i^3 p_{it}(y_{it})p_{it'}(y_{it'})}\left(\frac{\jmath_{it}\zeta_{it}}{\sigma_i} + \frac{\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)}{p_{it}\left(y_{it}\right)}\right) \end{aligned}$$

and

$$\begin{aligned} \hat{\mathbb{E}}s_{\theta\mu,i}s_{\mu,i} = &-\sum_t \frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{\sigma_i^2 p_{it}(1)(1-p_{it}(1))}\left[\frac{1}{2\sigma_i^2}\begin{pmatrix}\mathbf{0}' \\ 0 \\ \partial\sigma_1^2/\partial\rho\end{pmatrix} + \left(\frac{\zeta_{it}}{\sigma_i} + \frac{(1-2p_{it}(1))\phi\left(\zeta_{it}/\sigma_i\right)}{p_{it}(1)(1-p_{it}(1))}\right)\frac{\partial\zeta_i/\sigma_i}{\partial\boldsymbol{\theta}}\right] \\ &+ 2\sum_t\sum_{t'>t}\frac{\jmath_{it'}\jmath_{it}\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)\phi\left(\frac{\zeta_{it'}}{\sigma_i}\right)}{\sigma_i^2 p_{it}(y_{it})p_{it'}(y_{it'})}\left[\frac{1}{2\sigma_i^2}\begin{pmatrix}\mathbf{0}' \\ 0 \\ \partial\sigma_1^2/\partial\rho\end{pmatrix} + \left(\frac{\zeta_{it}}{\sigma_i} + \jmath_{it}\frac{\phi(\zeta_{it}/\sigma_i)}{p_{it}(1)}\right)\frac{\partial\zeta_i/\sigma_i}{\partial\boldsymbol{\theta}}\right]. \end{aligned}$$

The above equations give explicit expressions for the computation of all terms in (43). In particular, it must be pointed out that, if one considers the conditional expectation with respect to $y_{it'}$ both in $\mathbb{E}s_{\mu\mu,i}s_{\mu,i}$ and in $\mathbb{E}s_{\theta\mu,i}s_{\mu,i}$, the last terms are equal to zero and we get the simplifications:

[4]note that the products $s_{\mu,\mu,i}s_{\mu,i}$ or $s_{\theta,\mu,i}s_{\mu,i}$ would be equal to zero for all i because of $s_{\mu,i}(\boldsymbol{\theta}, \hat{\mu}_i(\boldsymbol{\theta})) = 0$

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$$\hat{\mathbb{E}}s_{\mu\mu,i}s_{\mu,i} = -\sum_t \frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{\sigma_i^3 p_{it}(1)(1-p_{it}(1))}\left[\frac{\zeta_{it}}{\sigma_i} + \frac{\phi\left(\frac{\zeta_{it}}{\sigma_i}\right)(1-2p_{it}(1)}{p_{it}(1)(1-p_{it}(1))}\right]$$

$$\hat{\mathbb{E}}s_{\theta\mu,i}s_{\mu,i} = -\sum_t \frac{\phi^2\left(\frac{\zeta_{it}}{\sigma_i}\right)}{\sigma_i^2 p_{it}(1)(1-p_{it}(1))}\left[\frac{1}{2\sigma_i^2}\begin{pmatrix}\mathbf{0}'\\0\\\partial\sigma_1^2/\partial\rho\end{pmatrix} + \left(\frac{\zeta_{it}}{\sigma_i} + \frac{(1-2p_{it}(1))\phi\left(\zeta_{it}/\sigma_i\right)}{p_{it}(1)(1-p_{it}(1))}\right)\frac{\partial\zeta_i/\sigma_i}{\partial\boldsymbol{\theta}}\right]$$

.

We can now sum up all the terms to compute $\hat{B}_i$:

$$\begin{aligned}\hat{B}_i &= -\frac{1}{2}\frac{s_{\theta\mu\mu,i}}{s_{\mu\mu,i}} + \frac{1}{2}\frac{s_{\theta\mu,i}s_{\mu\mu\mu,i}}{(s_{\mu\mu,i})^2} - \frac{\hat{\mathbb{E}}s_{\theta\mu,i}s_{\mu,i}}{s_{\mu\mu,i}} + \frac{s_{\theta\mu,i}\hat{\mathbb{E}}s_{\mu\mu,i}s_{\mu,i}}{(s_{\mu\mu,i})^2}\\ &= \hat{B}_{i1} + \hat{B}_{i2}\end{aligned}$$

where $\hat{B}_{i1} = -\frac{1}{2}\frac{s_{\theta\mu\mu,i}}{s_{\mu\mu,i}} - \frac{\hat{\mathbb{E}}s_{\theta\mu,i}s_{\mu,i}}{s_{\mu\mu,i}}$ and $\hat{B}_{i2} = \frac{1}{2}\frac{s_{\theta\mu,i}s_{\mu\mu\mu,i}}{(s_{\mu\mu,i})^2} + \frac{s_{\theta\mu,i}\hat{\mathbb{E}}s_{\mu\mu,i}s_{\mu,i}}{(s_{\mu\mu,i})^2}$.

By using all the expressions derived above, we can obtain, after some calculations

$$\hat{B}_{i1} = -\frac{T}{s_{\mu\mu,i}}\left[\frac{1}{T}s_{\theta\mu\mu,i} + \frac{1}{T}\hat{\mathbb{E}}s_{\theta\mu,i}s_{\mu,i}\right] \approx -\frac{1}{s_{\mu\mu,i}}\frac{1}{2}\sum_t \frac{\phi^2(\zeta_{it}/\sigma_i)}{\sigma_i^2 p_{it}(1)(1-p_{it}(1))}\frac{\zeta_{it}}{\sigma_i}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}$$

and

$$\hat{B}_{i2} = \frac{s_{\theta\mu,i}/T}{(s_{\mu\mu,i}/T)^2}\left[\frac{1}{T}s_{\mu\mu\mu,i} + \frac{1}{T}\hat{\mathbb{E}}s_{\mu\mu,i}s_{\mu,i}\right] \approx \frac{s_{\theta\mu,i}/T}{(s_{\mu\mu,i}/T)^2}\left[\frac{1}{2T}\sum_t \frac{\phi^2(\zeta_{it}/\sigma_i)}{\sigma_i^3 p_{it}(1)(1-p_{it}(1))}\frac{\zeta_{it}}{\sigma_i}\right]$$

so that

$$\hat{B}_i \approx \frac{1}{s_{\mu\mu,i}}\frac{1}{2}\sum_{t=1}^{T}\frac{\phi^2(\zeta_{it}/\sigma_i)}{\sigma_i^2 p_{it}(1)(1-p_{it}(1))}\frac{\zeta_{it}}{\sigma_i}\left[\frac{s_{\theta\mu,i}}{s_{\mu\mu,i}}\frac{1}{\sigma_i} - \frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\right] \tag{52}$$

Now, in order to compute an estimate of H_{NT}^{-1}, we can consider that

$$\begin{aligned}H_{NT} &= \left\{\frac{1}{NT}\sum_i \frac{\partial s_{\theta,i}(\boldsymbol{\theta},\hat{\mu}_i(\boldsymbol{\theta}))}{\partial\boldsymbol{\theta}}\right\}_{|\boldsymbol{\theta}=\boldsymbol{\theta}_0}\\ &= \frac{1}{NT}\sum_i s_{\theta\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) + \frac{1}{NT}\sum_i \frac{\partial\hat{\mu}_i(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}_{|\boldsymbol{\theta}=\boldsymbol{\theta}_0} s_{\theta\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))'\\ &= \frac{1}{NT}\sum_i s_{\theta\theta,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0)) - \frac{1}{NT}\sum_i \frac{s_{\theta\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))}{s_{\mu\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))}s_{\theta\mu,i}(\boldsymbol{\theta}_0,\hat{\mu}_i(\boldsymbol{\theta}_0))'.\end{aligned}$$

The terms $s_{\mu\mu,i}$ and $s_{\mu\theta,i}$ can be approximated using the formulas (49) and (50). The matrices $s_{\theta\theta,i}$ are approximated by:

$$\begin{aligned}\frac{s_{\theta\theta,i}(\hat{\boldsymbol{\theta}},\hat{\mu}(\hat{\boldsymbol{\theta}}))}{T} &= \frac{1}{T}\sum_{t=2}^{T}\frac{\phi'\left(\jmath_{it}\zeta_{it}/\sigma_i\right)}{p_{it}(y_{it})}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}'}\\ &\quad + \frac{1}{T}\sum_{t=2}^{T}\frac{\phi\left(\jmath_{it}\zeta_{it}/\sigma_i\right)}{p_{it}(y_{it})}\cdot\jmath_{it}\frac{\partial^2\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\\ &\quad - \frac{1}{T}\sum_{t=2}^{T}\frac{\phi^2\left(\jmath_{it}\zeta_{it}/\sigma_i\right)}{p_{it}^2(y_{it})}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}'}\\ &= \frac{1}{T}\sum_{t=2}^{T}\jmath_{it}\frac{\phi\left(\jmath_{it}\zeta_{it}/\sigma_i\right)}{p_{it}(y_{it})}\times\\ &\quad\times\left[\frac{\partial^2\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'} - \frac{\zeta_{it}}{\sigma_i}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}'} - \jmath_{it}\frac{\phi\left(\jmath_{it}\zeta_{it}/\sigma_i\right)}{p_{it}(y_{it})}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}'}\right].\end{aligned}$$

By applying again the conditional expectation of y_{it} for all t, we easily get:

$$\begin{aligned}\frac{s_{\theta\theta,i}(\hat{\boldsymbol{\theta}},\hat{\mu}(\hat{\boldsymbol{\theta}}))}{T} &\approx -\frac{1}{T}\sum_{t=2}^{T}\mathbb{E}\left(s_{it}^2\frac{\phi^2\left(\zeta_{it}/\sigma_i\right)}{p_{it}^2(y_{it})}\frac{\zeta_{it}}{\sigma_i}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}'} \mid \boldsymbol{X},\boldsymbol{y}_{t-1}\right)\\ &= -\frac{1}{T}\sum_{t=2}^{T}\frac{\phi^2\left(\zeta_{it}/\sigma_i\right)}{p_{it}(1)(1-p_{it}(1))}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}}\frac{\partial\zeta_{it}/\sigma_i}{\partial\boldsymbol{\theta}'}.\end{aligned}$$

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